SAP Extended Warehouse Management (SAP EWM)

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SAP Extended Warehouse Management (SAP EWM) | 9.5 FPS02

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Delivery Processing

Purpose

Extended Warehouse Management (EWM) has its own **Delivery Processing** area. This is a significant part of the logistics chain. Delivery processing offers functions for executing goods receipt and goods issue activities, as well as for executing posting changes and internal stock transfers. The following sections describe the available functions, as well as the integration with EWM.

For an overview of the available user transactions, see User Interfaces in Delivery Processing .

Corresponding to the goods receipt and goods issue, a distinction is made in delivery processing between the inbound delivery process and the outbound delivery process.

Inbound Delivery Process

In the inbound delivery process, delivery processing supports the following scenarios:

- Deliveries from the supplier, from any ship-from party, or from any ship-from location.
- Returns from the customer, from any ship-from party, or from any ship-from location.

From the purchase order, the inbound delivery process covers the notification, the inbound delivery, the goods receipt posting of the ordered goods, and the subsequent putaway.

Information about planning is saved in the inbound delivery, the status is monitored, and all the data collected during the inbound delivery is kept. On creation of the inbound delivery, activities such as the putaway or scheduling are triggered and the data generated by the inbound delivery process is recorded.

Outbound Delivery Process

In the outbound delivery process, delivery processing supports the following scenarios:

- Deliveries to the customer, to the warehouse, to any ship-to party, or to any ship-to location.
- Returns to the supplier, to any ship-to party, or to any ship-to location.

The outbound delivery process covers all shipping activities such as picking, packing, transportation and goods issue.

Information about shipping planning is saved in the inbound delivery, the status of the shipping activities is monitored, and all the data collected during the shipping process is recorded. On creation of the outbound delivery, shipping activities such as picking or shipment scheduling are triggered and the data generated by the shipping processing is recorded.

Features

In delivery processing, documents are provided which represent the delivery processes in the system. Delivery processing also provides a series of services and functions to configure how the documents are to be processed, and to define the business features of the documents in a way that best suits your business process.

The following services and functions are provided by delivery processing:

- Status management
- Quantity offsetting

- Text management
- Field control
- Validation
- Connection to the Post Processing Framework (PPF)
- Change documents
- Archiving

Documents in Delivery Processing

Use

A document contains all the data required to document and control a business process. Various documents are used in delivery processing to control the processes.

The inbound delivery process is represented in the system by the following documents:

- Inbound delivery notification
- Inbound delivery

The outbound delivery process is represented in the system by the following documents:

- Outbound delivery Request
- Outbound delivery order
- Outbound Delivery

The posting change process is represented in the system by the following documents:

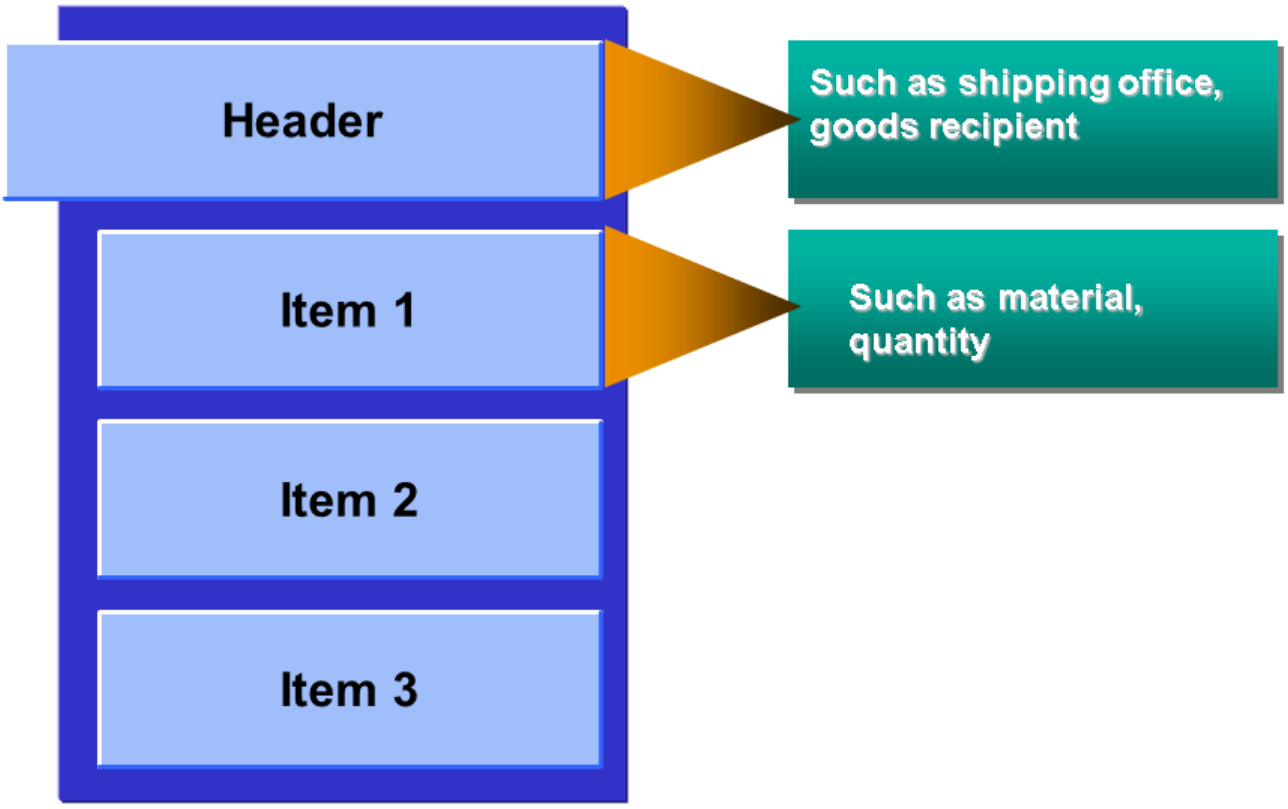
- Posting change request
- Posting change

The internal stock transfer process is mapped in the system using the Internal Stock Transfer document.

Every document provides various actions that are used to control the individual subsequent processes. The execution of the actions depends on the settings in the Post Processing Framework (PPF) .

Structure

A document comprises a document header and any number of document items. The following figure shows the structure of a delivery document.



Document Header

The document header contains general data and data that is relevant for the following documents:

- Inbound delivery
- Outbound delivery
- Posting change
- Internal stock transfer

This data is valid for the entire document. This data includes:

- Goods receiving office or shipping office
- Scheduling data (for example, shipping or delivery date)
- Weights and volumes for the complete inbound/outbound delivery
- Ship-from party, sold-to party and goods recipient numbers

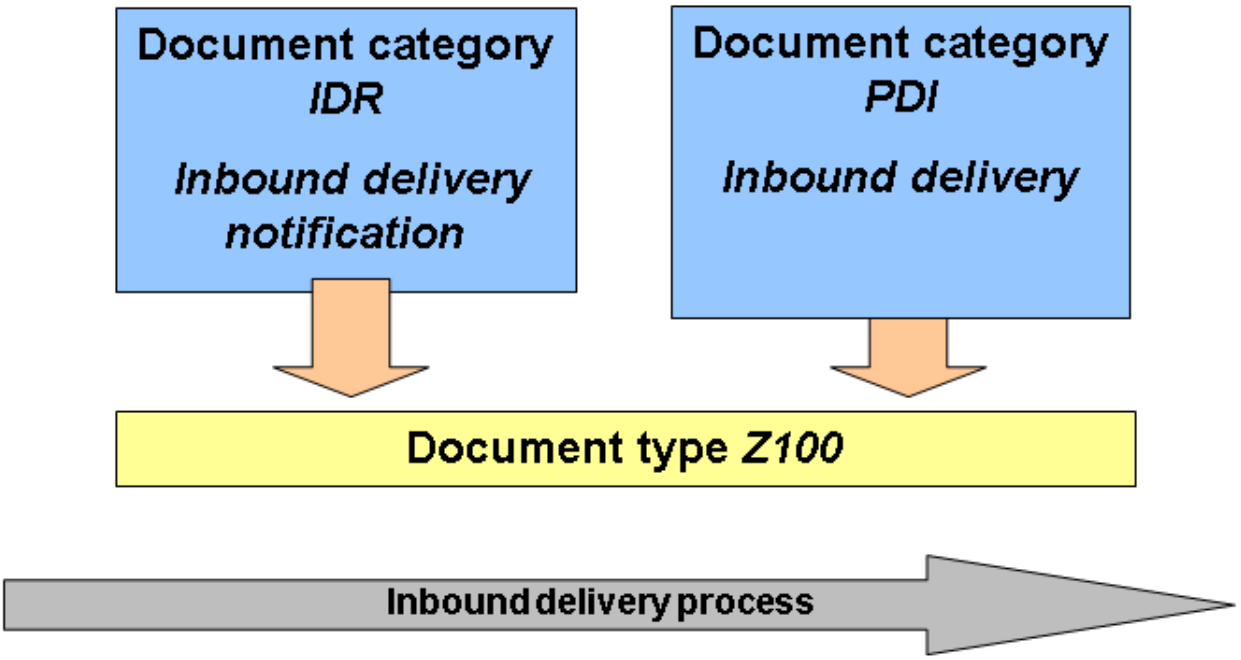
The document header is defined in the system by the document category and the document type.

The document category classifies the different documents that can be processed by the system in delivery processing. It is defined by the system and cannot be changed. The following document categories exist in delivery processing:

- Inbound delivery notification
- Inbound delivery
- Outbound delivery request
- Outbound delivery order
- Outbound delivery
- Posting change request
- Posting change
- Internal Stock Transfer

The document type classifies the documents with regard to the complete delivery process. The document type defines the business aspects of a document.

The following figure shows a possible relationship between the document category and the document type in the inbound delivery process. In this case, only one document type (Z100) is used for the complete inbound delivery process.



Using the document type determination, you can derive several document types from a document category and therefore define certain delivery processes.

In Customizing for Delivery Processing , you can define your own document types and document type determination.

Document Items

In the document items, you find the data relevant to the individual items. It includes:

- Material number
- Delivery quantity
- Location specifications (goods receiving point, unloading point)
- Putaway date or picking date
- Weights and volumes of the individual items
- Tolerances for underdelivery or overdelivery

The document items are defined in the system by the item category and by the item type.

The item category classifies the different items that can be processed by the system for delivery processing. It is defined by the system and cannot be changed.

The following item categories exist in delivery processing:

- Normal delivery item
- Return item
- Text item
- Packaging item
- Value item (in documents of the outbound delivery only)

The item type classifies the items for a document type with regard to the complete delivery process. The item type defines the business aspects of the document items. As for the document header, for the document items, you can also configure your business process via the item type and the item type determination in Customizing for Delivery Processing.

Service Profiles

Service profiles define how a document is processed in the system as well as its business scope.

Two types of service profile exist:

- System profile
- Customer profile (for example, status profile, text profile and incompleteness profile)

The system profile is not configurable and is linked to the document categories and item categories.

The customer profiles are linked to the document types and to the item types. In Customizing for Delivery Processing, you can define more customer profiles and therefore adjust the documents to suit your business process.

In delivery processing, you can configure the following functions using the customer profiles:

- Status management
- Quantity offsetting
- Text management
- Field control
- Incompleteness check
- Actions in the Post Processing Framework

- Partner processing
- Process codes
- Reference document types
- Date types

SAP delivers predefined status profiles for the standard processes in delivery processing.

General Settings

The general settings for the documents in delivery process are not controlled by the service profiles.

You can define the following general settings in Customizing for Delivery Processing:

- Number range intervals
- Delivery priority
- Shipping conditions
- Incoterms

> **ℹ Note**
> You can find more information in Customizing for Delivery Processing .

Inbound Delivery Notification

Definition

The inbound delivery notification is a document containing all the relevant logistics data in the inbound delivery process right from the origin of the inbound delivery process (shipping notification, delivery note or purchase order, for example).

The shipping notification from the supplier is saved in the inbound delivery notification.

Use

You use the inbound delivery notification to save the logistics data transferred data from a reference document in order to create an inbound delivery .

Using the inbound delivery notification, you can also process or reject incoming changes to an existing inbound delivery.

Structure

For a description of the structure, see Documents in Delivery Processing .

Integration

The execution of the actions depends on the settings in the Post Processing Framework (PPF) .

The following configuration exists in the standard system:

- The delivery process receives a message from a reference document with all the relevant logistics data. An inbound delivery notification is created automatically as a result of this message. All the information contained in the message is saved in the inbound delivery notification.

  An inbound delivery is created automatically from the inbound delivery notification.

- Changing an existing inbound delivery:

  If a change to an existing inbound delivery is rejected, an inactive inbound delivery notification (inactive version) is created in addition to the existing active inbound delivery notification. You can process the data in the inactive inbound delivery notification and you can create a new inbound delivery by activating the request manually.

Inbound Delivery

Definition

The inbound delivery is a document containing all the data required for triggering and monitoring the complete inbound delivery process. This process starts on receipt of the goods in the yard and ends on transferal of the goods at the final putaway, production or outbound delivery.

Use

The inbound delivery is used as a working object in the inbound delivery process.

You use this document in delivery processing when executing the following actions:

- Registering the delivery in the yard
- Unloading the delivery
- Canceling the "unloading of the delivery"
- Placing the delivery into stock (putaway)
- Canceling the "delivery putaway"
- Adjusting the delivery quantity to the quantity posted in the goods receipt (in case of overdelivery or underdelivery.
- Adjusting the delivery quantity and the transferred quantity

  In this case, the delivery quantity is adjusted to the quantity posted in the goods receipt. The quantity in the inbound delivery notification is also adjusted and a new inbound delivery is created.

- Posting a goods movement
- Canceling a goods movement
- Creating items
- Deleting items

You can save, validate and delete an inbound delivery.

Structure

For a description of the structure, see Documents in Delivery Processing .

Integration

The inbound delivery is created after activating an inbound delivery notification . The inbound delivery adopts all the relevant data from the inbound delivery notification.

The execution of the actions depends on the settings in the Post Processing Framework (PPF) .

Outbound Delivery Request

Definition

The outbound delivery request is a document containing all the relevant logistics data in the outbound delivery process right from the origin of the outbound delivery process (sales order, for example).

Use

You use the outbound delivery request to save the transferred data from a reference document in order to create an outbound delivery order .

Using the outbound delivery request, you can also process or reject incoming changes to an existing outbound delivery order.

Structure

For a description of the structure for an outbound delivery request, see Documents in Delivery Processing .

Integration

The execution of the actions depends on the settings in the Post Processing Framework (PPF) .

The following configuration exists in the standard system:

- The delivery process receives a message from a reference document with all the relevant logistics data. An outbound delivery request is created automatically as a result of this message. The delivery process saves the information from the message in the outbound delivery request. The message itself is not saved.

  An outbound delivery order is created automatically from the outbound delivery request.

- Changing an existing outbound delivery order:

  If a change to an existing outbound delivery order is rejected, an inactive outbound delivery request (inactive version) is created in addition to the existing active outbound delivery request. You can process the data in the inactive outbound delivery request and you can create a new outbound delivery order by activating the request manually.

Outbound Delivery Order

Definition

The outbound delivery order is a document containing all the data required for triggering and monitoring the complete outbound delivery process. This process starts with the first planning activities for the outbound delivery and continues until the finished goods have been loaded and sent.

Use

You either use the outbound delivery order to preview the planning of pending warehouse activities or to execute the actions required to create the delivery.

You can use the outbound delivery order to execute the following actions:

- Creating the delivery and therefore creating an outbound delivery
- Picking the delivery
- Canceling the "delivery picking"
- Loading the delivery
- Canceling the "delivery loading"
- Setting the status **Leave Yard**
- Adjusting the delivery quantity to the picked quantity
- Posting a goods movement
- Canceling a goods movement
- Creating items
- Deleting items

You can save, validate, delete and archive an outbound delivery order.

Structure

For a description of the structure for an outbound delivery order, see Documents in Delivery Processing .

Integration

The execution of the actions depends on the settings in the Post Processing Framework (PPF) .

The following configuration exists in the standard system:

- The outbound delivery order is created automatically after activating the outbound delivery request . Here, all the relevant data is copied from the outbound delivery request.

  One or several outbound deliveries are created from the outbound delivery order.

- The outbound delivery order has the status **Locked (Preceding Document)** when only used as a preview for planning warehouse activities. This status is determined via the corresponding outbound delivery request from the reference document. In this case, you cannot create a delivery.

Outbound Delivery

Definition

The outbound delivery is a document representing the goods to be delivered together to a goods recipient.

Use

The outbound delivery is used as the basis for printing the delivery note or for sending a shipping notification.

You can execute the following actions using this document:

- Posting a goods movement
- Canceling a goods movement
- Setting the status **Leave Yard**

You can save, process, delete and archive an outbound delivery.

Structure

For a description of the structure for an outbound delivery, see Documents in Delivery Processing .

Integration

You create an outbound delivery when you select the action **Create Outbound Delivery** in the outbound delivery order .

The execution of the actions depends on the settings in the Post Processing Framework (PPF) .

Maintenance of Proof of Delivery

Use

For monitoring and analysis of outbound delivery items, you can use this function to create or delete proof-of-delivery data, which is imported from SAP ERP and enriched in SAP Extended Warehouse Management by warehouse activity data.

Integration

SAP ERP replicates outbound deliveries indicated as relevant for proof of delivery (PoD) to SAP EWM for further processing.

The BI system extracts the data sets on PoD relevant outbound deliveries for enabling KPI reporting in SAP EWM.

Prerequisites

- You have limited the number of users who are allowed to process the data import by having assigned authorizations for the authorization object `/SCWM/PIMP`. For more information, see Reducing Authorization Checks in Applications.
- You have implemented SAP Note 1625878 in SAP ERP for setting up PoD relevancy, which is also used to support the stock transfer order process.
- You have not archived outbound deliveries in EWM, thus enabling import of PoD entries at item level.
- Items in outbound deliveries are selected as relevant for proof of delivery in SAP EWM.
- Confirmed PoD differences exist in ERP for the corresponding outbound deliveries in ERP.

Features

You can use this function in the background to regularly create proof-of-delivery (PoD) data.

PoD Data Creation

- Selecting outbound delivery items

  The system selects PoD relevant outbound delivery items according to criteria set by the user.

- Importing data

  The system imports PoD differences reported in SAP ERP.

  Before importing data from SAP ERP, you must define a specific RFC destination. In Customizing, see **Assign RFC Destinations to Logical Systems** under **Extended Warehouse Management** > **Interfaces** > **ERP Integration** > **General Settings** > **Assign RFC Destinations to Logical Systems.**.

  By implementing the BAdI `/SCWM/EX_POD_SELECT`, you can replace the standard PoD data import with your own PoD data import.

- Enriching data

  The system determines warehouse activities and tasks performed on selected outbound delivery items and updates the PoD data.

  By implementing the BAdI `/SCWM/EX_POD_ENRICH`, you can replace the standard determination of relevant warehouse tasks with your own determination.

PoD Data Deletion

The system selects and deletes relevant PoD data according to criteria set by the user.

The deletion of PoD data is not linked to delivery archiving because an analysis should be possible even if the delivery is archived.

PoD Data Test

The system creates an application log but does not make any changes to the database.

To activate the application log, on the SAP Easy Access screen choose **SCM Extended Warehouse Management** > **Extended Warehouse Management** > **Settings** > **Activate Application Log.**.

PoD Data in BI

For transfer of PoD transaction data from EWM into BI, use data source `0WM_POD`.

You can use the extracted PoD data with, for example, extracted data on outbound delivery items (data source `0WM_DLVI_OUT`) to build key performance indicators, such as **Ratio Outbound Delivery Errors** and monitor them in the warehouse cockpit. To build the key performance indicator, you must make sure that the PoD data in BI corresponds to the outbound delivery.

See Data Sources for more information.

Activities

To maintain proof-of-delivery data for relevant outbound deliveries, on the **SAP Easy Access** screen choose **Extended Warehouse Management** > **Interfaces** > **ERP Integration** > **Maintain Proof-of-Delivery Data.**

The system processes creation or deletion of proof-of-delivery data based on the selection criteria of at least warehouse number and a goods issue date interval.

After execution, the system displays a message log, which you can use to analyze the changes made. Depending on the application log settings, the message log is also saved to the database. Using transaction code `SLG1`, you can display on the **Analyze Application Log** screen the message log for the object `/SCWM/WME` and sub-object `PoD`.

Example

If you want to use this function to create PoD data in the background on a regular basis, such as weekly, then create a variant with a selection variable for the goods issue data.

For example, give customers 7 days to report incorrect deliveries and if you confirm the PoD based delivery in the same timeframe, you can use the variable current date - xxx, current date + yyy. Enter `14` as xxx value and `7-` as yyy value to import the confirmed PoD data from the previous week.

More Information

For more information, see:

- Proof of Delivery in EWM
- Proof of delivery in SAP ERP, in SAP Library for SAP ERP under **SAP ERP** > **SAP ERP Central Component** > **Sales and Distribution (SD)** > **Shipping (LE-SHP)** > **Proof of Delivery**.

Posting Change Request

Definition

A posting change request is a document into which the system copies all relevant logistics data from the initiator of this process for a posting change process triggered externally.

Use

You use the posting change request to save the logistics data transferred from a reference document in order to create a posting change . For more information about the posting change process, see Warehouse Request for Posting Changes .

Structure

For a description of the structure for a posting change request, see Documents in Delivery Processing .

Integration

The execution of the action depends on the settings in the Post Processing Framework (PPF) .

The following configuration exists in the standard system:

- The delivery process receives a message from a reference document with all the relevant logistics data. A posting change request is created automatically as a result of this message. All the information contained in the message is saved in the posting change request.

  A posting change is created automatically from the posting change request.

- Changing an existing inactive posting change request:

You can edit the data in the inactive posting change request, and generate a posting change by activating it manually (see: Using Inactive and Active Document Versions). You can also assign an inactive posting change request manually.

Posting Change

Definition

The posting change is a document containing all the data required for triggering and monitoring the complete posting change process.

Use

The posting change is used as a working object in the posting change process.

You use this document in delivery processing to perform planned posting changes that either have no reference to a posting change request or for which a posting change request acts as a data basis. For more information about the posting change process, see Warehouse Request of the Posting Change Type .

Structure

For a description of the structure, see Documents in Delivery Processing .

Integration

The posting change is generated after activating a posting change request. The posting change copies all the relevant data from the posting change request.

The execution of the actions depends on the Post Processing Framework (PPF) Settings .

Internal Stock Transfer

Definition

An internal stock transfer is a document containing all the data required for planning and monitoring warehouse-internal product movements.

Use

The warehouse-internal stock transfer serves as a working object in the stock transfer process for internal product movements. For more information about internal stock transfers, see Warehouse Request of the Internal Stock Transfer Type .

Structure

For a description of the structure for a delivery document, see Documents in Delivery Processing .

Integration

The execution of the actions depends on the settings in the Post Processing Framework (PPF) .

Change Documents in Delivery Processing

Use

You can use change documents to access information on changes made to a delivery document.

You cannot re-apply old statuses that are documented in the change documents.

Features

As soon as a delivery document is changed and saved by a user, a change document is created in delivery processing.

The change document contains the following information:

- which changes were made to a delivery document
- which user made the changes
- when the changes were made

Activities

In the maintenance transaction of the corresponding delivery document, choose **Display Change Document** (see Menu Paths and Paths in Customizing for Delivery Processing)

You receive a list with all the change documents for the selected delivery document.

Warehouse Request

Definition

The warehouse request is a document that enables the processing of warehouse activities for a specific product. The warehouse activities for a product include the following:

- Picking
- Putaway
- Posting change
- Stock transfer (within warehouse)
- Scrapping

The following types of warehouse requests exist:

- Inbound delivery
- Outbound delivery order
- Posting change
- Internal stock transfer

In addition to these working documents, the following predecessor documents exist:

- Inbound delivery notification
- Outbound delivery request
- Posting change request

Use

You use this document in Extended Warehouse Management (EWM) to:

- Process specific product quantities within a warehouse for your goods receipt process or goods issue process (for example, for putaway or picking).
- Create posting changes
- Process internal goods movements
- Process scrappings

EWM uses warehouse requests to form waves . EWM works with both inactive and active document versions .

It creates warehouse tasks to fulfill the warehouse request.

You can create, process, delete, or archive warehouse requests .

Structure

A warehouse request consists of

- Document categories with different document types. You can define the document types in Customizing for delivery processing. For more information about the paths in the Implementation Guide (IMG), see Menu Paths and Customizing Paths for Delivery Processing .
- Item categories with different item types. You can define the item types in Customizing for delivery processing. For more information about the paths in the Implementation Guide (IMG), see Menu Paths and Customizing Paths for Delivery Processing .
- Services that EWM can use. These include:
    - Text management
    - Quantity offsetting
    - Status management
    - Field control
    - Connection to the Post Processing Framework (PPF)
    - Incompleteness check

For more information about the structure, see Documents in Delivery Processing .

Integration

Systems outside of EWM send messages for a warehouse request to EWM . For a goods receipt process, this is an SAP ERP system. For a goods issue process, this is a sales order system.

ℹ Note

For more information about Customizing for integration between SAP ERP and EWM, see the Implementation Guide (IMG) for Supply Chain Management under Extended Warehouse Management > Interfaces > ERP Integration.

Using Inactive and Active Document Versions

Purpose

Extended Warehouse Management (EWM) can accept or reject change messages for inbound deliveries from SAP ERP. EWM uses inactive and active document versions for this. If warehouse activities (follow-on activities) have already begun forinbound deliveries, EWM rejects change messages from SAP ERP. For example, EWM has already created a warehouse task (WT) for an inbound delivery. In doing so, EWM has also already set the status type Warehouse Activities to the status value Partially Completed . In the case of outbound deliveries, EWM can only accept a reduction of the quantity from SAP ERP. If warehouse activities (follow-on activities) have already begun for inbound deliveries, EWM rejects the message from SAP ERP.

Prerequisites

- You have determined the document type for the Post Processing Framework (PPF) action for activating an inbound delivery notification or the outbound delivery request 3 Immediate Processing . You do this in Customizing for the PPF in the action profile in EWM. To do so, in Customizing for Extended Warehouse Management , choose Cross-Process Settings > Delivery Processing > Actions > Change Actions and Conditions > Define Action Profile and Actions . Select the action profile for an inbound delivery notification (/SCDL/IDR) or for an outbound delivery request (SCWM/ODR). Choose Action Definition in the dialog structure and go from there to the Action Definition tab page in the Action Settings screen area for processing time 3 Immediate Processing .

- In Customizing, you have assigned the defined action profile to a document type.

  For more information, see the Implementation Guide (IMG) for Extended Warehouse Management under Goods Receipt Process > Inbound Delivery > Define Document Types for Inbound Delivery Process Using Wizard, ,or under Goods Issue Process > Outbound Delivery > Define Document Types for Outbound Delivery Process Using Wizard .

- In the case of inbound deliveries, the status type Warehouse Activities has the status value Not Started in the document. On the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Inbound Delivery > Maintain Inbound Delivery. At header level, check whether the status type Warehouse Activities has the status value Not Started .

- The following conditions have been fulfilled in the case of outbound deliveries:

  - The status type Warehouse Activities has the status value Not Started in the document. On the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Outbound Delivery > Maintain Outbound Delivery Order or Maintain Outbound Delivery. At item level, check whether the status type Warehouse Activities has the status value Not Started for the respective delivery item.

  - The status type Locked (Overall Status) has the status value No in the document. On the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Outbound Delivery > Maintain Outbound Delivery Order or Maintain Outbound Delivery. At item level, check whether the status type Locked (Overall Status) has the status value No .

  - Neither you or the system have assigned a wave to the delivery item in the document.

Process

Creating New Documents in EWM

1. You create a new document (for example, an inbound delivery) in SAP ERP and you save it.
2. SAP ERP informs EWM of this and EWM creates a corresponding inactive document, for example, an inbound delivery notification IDN .
3. Depending on your Customizing settings for the PPF, the following is either performed by you or by EWM.
   a. If you determined document type 3 Immediate Processing in the PPF Customizing settings in the action profile in EWM, EWM automatically activates the inactive document. As a result of this activation, EWM creates the active successor document. For example, EWM automatically activates an inbound delivery notification IDN and creates an active inbound delivery ID .

      If EWM cannot perform the PPF action due to an error, EWM creates an inactive document, in other words, an inactive inbound delivery notification or an inactive outbound delivery request. You can manually activate and save these. To do so, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing and then one of the following options:

      - Inbound Delivery > Maintain Inbound Delivery Notification and the Activate pushbutton.
      - Outbound Delivery > Maintain Outbound Delivery Request and the Activate pushbutton.

   b. If you have **not** determined document type 3 Immediate Processing in the PPF Customizing settings in the action profile in EWM, EWM always creates an inactive document. You can manually activate and save these. To do so, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing and then one of the following options:

      - Inbound Delivery > Maintain Inbound Delivery Notification and the Activate pushbutton.
      - Outbound Delivery > Maintain Outbound Delivery Request and the Activate pushbutton.

Accepting Inbound Delivery Changes to SAP ERP Documents in EWM

1. You change an SAP ERP inbound delivery, for example, you change the quantity and save this change. SAP ERP creates a new document with a temporary document number.
2. SAP ERP notifies EWM to replace the inactive inbound delivery.
3. EWM creates a corresponding inactive document for the existing document. In other words, EWM creates an inactive inbound delivery notification for an existing active inbound delivery notification. The inactive document has the same document number as the existing active document. The active inbound delivery notification IDN corresponds to the active inbound delivery ID in EWM that contains the unchanged quantities.
4. EWM carries out one of the following options:
   a. If EWM has not yet carried out any changes for the active document (that is, the status type Warehouse Activities still has the status value Not Started), EWM accepts the message from SAP ERP.

      Depending on your Customizing settings for the PPF, the inactive document is either activated by you or by EWM. EWM copies the changes to the active document. For example, EWM activates an inactive inbound delivery notification with a quantity change from SAP ERP. EWM copies the quantity changes to the inbound delivery, that is, the inactive inbound delivery notification overwrites the active inbound delivery notification.

   b. If EWM has already carried out changes for the inactive document, or it cannot carry out the quantity change from SAP ERP, EWM rejects the change message from SAP ERP. EWM does not copy the changes to the active document. For example, EWM has already created a warehouse task for an inbound delivery and has set the status value

Partially Started for the status type Warehouse Activities . If you now make another change to the inbound delivery in SAP ERP, EWM does not copy this change to the active inbound delivery.

5. EWM informs SAP ERP of the rejection or acceptance.

6. SAP ERP deletes the temporarily created SAP ERP document number again.

Accepting Outbound Delivery Changes to SAP ERP Documents in EWM

1. In SAP Customer Relationship Management (SAP CRM), the quantity of an order is reduced and SAP CRM sends this quantity change to SAP ERP.

2. SAP ERP then sends the quantity change to EWM.

> **i Note**
> In contrast to inbound delivery cases, EWM does not create a corresponding inactive document (in other words, an inactive outbound delivery request) for the existing active document in the case of outbound deliveries.

3. EWM carries out one of the following options:

   a. If EWM has not yet carried out any changes for the active document (that is, the status type Warehouse Activities still has the status value Not Started), and the other prerequisites for outbound delivery cases apply (see 'Prerequisites'), EWM accepts the message from SAP ERP. EWM copies the quantity reduction to the active document.

   b. If EWM has already carried out changes for the active document, or it cannot carry out the quantity reduction from SAP ERP, EWM rejects the message from SAP ERP. EWM does not copy the quantity reduction to the active document.

4. EWM informs SAP ERP of the rejection or acceptance.

5. If necessary, SAP ERP changes its outbound delivery in accordance with the message.

> **i Note**
> For more information about document control and processing checked deliveries in SAP ERP, see the SAP Library under Documentation > SAP ERP Central Component > SAP ERP Central Component > Logistics > General Logistics (LO) > Service Parts Management:
>
> - > Architecture and Communication > Document Control
> - > Goods Issue Process > Processing Checked Deliveries.

Inbound Delivery Notification

Outbound Delivery Order

Warehouse Req. - Type Inbound Del. or Outbound Del. Order

Definition

This warehouse request defines a worklist for Extended Warehouse Management (EWM) and thereby determines whether goods are to be put away or removed from stock.

Use

The warehouse request of the type inbound delivery or outbound delivery order serves in EWM as the basis for performing putaway or stock removal activities. For example, EWM uses a warehouse request to create warehouse tasks for putaway or picking.

EWM also posts goods movements for goods receipt or goods issue based on the warehouse request.

EWM uses the inbound delivery notification or outbound delivery request, in other words, the messages from SAP ERP, to create a warehouse request for an inbound delivery or an outbound delivery order .

Structure

For a description of the structure of a warehouse request, see Warehouse Request .

Integration

EWM receives inbound delivery notifications or outbound delivery requests from other systems. Based on these messages, EWM can automatically generate the corresponding warehouse requests of the type inbound delivery or outbound delivery order. However, you can also create these warehouse requests manually. In the case of outbound deliveries, manual creation is limited to warehouse requests for the scrapping process.

See also:

Generation of Warehouse Request of the Inbound Delivery Type , Generation of Warehouse Request of Outbound Delivery Order Type , and Performing Scrapping .

The priority points are calculated in **SAP Advanced Planning and Optimization** . The inbound delivery only displays these priority points.

See also:

Determination of Priority for Inbound Deliveries

Generation of Warehouse Request of the Inbound Delivery Type

Use

Extended Warehouse Management (EWM) uses this function to create a warehouse request for your goods receipt process (inbound delivery notification).

Integration

When EWM creates a warehouse task for a warehouse request, it updates the document flow for the warehouse request of the inbound delivery type.

When you generate a warehouse request of the inbound delivery type, you can automatically generate value-added service orders (VAS orders) for the warehouse request item .

Prerequisites

The warehouse process type in the warehouse request must be defined.

For more information about defining and determining the warehouse process type, see the Implementation Guide (IMG) for SAP Supply Chain Management (SAP SCM) under Extended Warehouse Management > Cross-Process Settings > Warehouse Task > Define Warehouse Process Type and Determine Warehouse Process Type .

Features

The common functions that are performed automatically for both the inbound delivery and the outbound delivery order when a warehouse request is generated or changed (both inbound and outbound) enable you to perform automatic packing.

EWM automatically creates handling units in accordance with the packing instruction. It uses the condition technique to do this.

The additional functions that are performed automatically for the inbound delivery when a warehouse request is created or changed enable you to do the following:

- If packaging specifications are not available for packing products during goods receipt, EWM can automatically trigger a WebFlow or SAP Alert Framework to notify someone. For this to occur, you must have defined the WebFlow or SAP Alert Framework accordingly.

  For more information, see the IMG for SAP SCM under SCM Basis > Pack > Packaging Specification

  For more information about the WebFlow Engine, see the SAP Library under SAP Web Application Server > Business Management (BC-BMT) > WebFlow Engine (BC-BMT-WFM).

  For more information about Alert Management, see the SAP Library under SAP Web Application Server > Basis-Services/Communications Interface (BC-SRV) > Business Workplace and Services (BC-SRV-OFC) > Alert Management (BC-SRV-GBT-ALM).

- Determination of a counting for each delivery item

- Check of the handling unit type for notified handling units

  During the creation of an inbound delivery with the corresponding handling units, delivery processing checks whether the handling unit type is permissible for the warehouse in question.

  For more information, see the IMG for SAP SCM under Extended Warehouse Management > Goods Receipt Process > Slotting > Influence Parameters > HU Types > Define HU Types per Warehouse Number

- Creation of batches

  Before you create batches, you must assign a batch class of the class type 023 to the product subject to a batch management requirement in SAP ERP. You have also assigned the object characteristics LOBM_HERKL (country of origin) and LOBM_VFDAT (shelf life expiration date, minimum shelf life) to the batch class. A batch of the product must also exist already in EWM.

  If the batch is known in SAP ERP, the SAP ERP system sends this information to EWM. EWM automatically provides this in the delivery.

  If a batch is unknown in EWM, you can specify this batch manually in the inbound delivery.

  For more information, see Connection of Delivery Processing to Batch Management.

- Goods receipt posting

  When EWM automatically generates a delivery based on a warehouse request from SAP ERP, you can carry out partial goods receipt postings using the warehouse task confirmation. When you confirm a warehouse task, EWM automatically executes a corresponding partial goods receipt posting. It reports this to SAP ERP.

  You can use the delivery UI to either post the goods receipt for individual handling units only, or you can post it completely.

Activities

At the same time as generating an inbound delivery, SAP ERP sends a message for an inbound delivery notification or posting change request to EWM.

EWM automatically generates a warehouse request.

During the automatic creation and saving of the warehouse request, EWM triggers the following:

- Mapping of delivery data from SAP ERP to the EWM delivery data
- Data enrichment using data from the functions described under **Features**
- Determination of warehouse process type, which controls the processing of a warehouse task in EWM, such as whether EWM is to immediately confirm a warehouse task
- Determination of the goods movement bin

For more information, see Warehouse Order Creation and Creation of Warehouse Tasks for Warehouse Request .

EWM copies most of the settings that you make in Customizing for process control to the delivery documents and displays them on the user interface for the outbound delivery order. To do this, on the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Outbound Delivery** > **Maintain Outbound Delivery Order** . Display an outbound delivery order in the form view. Here you can find the settings from Customizing, such as in the field **Corr.Deliv.**, **Scrapping** or **Pickup**.

Generation of Warehouse Request of Outbound Delivery Order Type

Use

SAP Extended Warehouse Management (SAP EWM) uses this function to generate a warehouse request for your goods issue process (outbound delivery request).

Integration

When you generate a warehouse request of the type outbound delivery order, you can automatically generate value-added service orders (VAS orders) for the warehouse request item. For more information, see Autom. Generation of a VAS Order for the Warehouse Request Item.

When you generate a warehouse request of the type outbound delivery order, SAP EWM can perform route determination in accordance with your Customizing. For more information about route determination, see Route Determination.

Prerequisites

The warehouse process type in the warehouse request must be defined.

For more information about defining and determining the warehouse process type, see Customizing for **SAP Supply Chain Management (SAP SCM)** under **Extended Warehouse Management** > **Cross-Process Settings** > **Warehouse Task** > **Define Warehouse Process Type** and **Determine Warehouse Process Type**.

Features

The **additional functions** that are performed automatically for the outbound delivery order when a warehouse request is generated or changed enable you to do the following:

- Rough determination of the picking location

  The system determines the source storage bins for a delivery based on the stock removal strategies. However, it does not make quantity reservations for the products. You can use the activity area that is assigned to the storage bin as a selection criterion.

  ### i Note

  A delivery contains three delivery items of 10 tires each. The system determines the same source storage bin for each delivery item. It contains 10 tires only. Therefore, the total quantity of 30 tires does not exist at the source storage bin. For fixed storage bins, the system organizes the total quantity using replenishment strategies.

  For more information about Customizing for stock removal strategies, see Customizing for **SAP SCM** under **Extended Warehouse Management** > **Goods Issue Process** > **Strategies**.

- Determination of which products can be packed together for a shipment
- Determination of staging area
- Determination of door assignment
- Determination of warehouse process type

  For more information, see Customizing for **SAP SCM** under **Extended Warehouse Management** > **Cross-Process Settings** > **Warehouse Task** > **Determine Warehouse Process Type**.

- Route determination
- Determination of staging area group
- Adding packaging items

  If, for example, you use recyclable packing materials, you can manually add delivery items for the packing material to an outbound delivery order and then post goods issue again for this packing material (see also Processing of Packaging Items).

  On the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Maintain Outbound Delivery Order**. Select an outbound delivery order, and on the **Items** tab page, choose **Create**. Use the input help to select a packaging item type.

- You can control processes for outbound delivery orders as part of the configuration. For example, you can create an invoice before goods issue in the outbound delivery process. You can also do this on the basis of other delivery parameters, for example, the export relevance or the route that is determined. As a prerequisite, you have defined process profiles for the delivery document header and for the delivery document items in Customizing for SAP EWM. For more information, see Customizing for **Extended Warehouse Management** under **Cross-Process Settings** > **Delivery Processing** > **Process Management and Control**.

- You can enter the account assignment for direct outbound delivery orders on the user interface for the outbound delivery order in the detail view of the item on the **Account Assignments** tab page.

  ### Example

  A department within your company requests consumable material from your warehouse. Inventory management is no longer required for this consumable material. You update the consumable material, for example, for a cost center or project, by entering the account assignment on the user interface for the outbound delivery order.

Activities

- At the same time as generating the outbound delivery, SAP ERP sends a message for an outbound delivery request or posting change request to SAP EWM.
- SAP EWM automatically generates a warehouse request using the Post Processing Framework (PPF).
- During the automatic creation and saving of the warehouse request, SAP EWM triggers the following:
  - Mapping of delivery data from SAP ERP to the SAP EWM delivery data
  - Data enrichment using data from the functions described under **Features**
  - Wave assignment, depending on your Customizing
  - Determination of warehouse process type, which controls the processing of a warehouse task in SAP EWM, such as whether SAP EWM is to immediately confirm a warehouse task
- SAP EWM creates the warehouse tasks at the same time as releasing the wave. In accordance with the warehouse tasks, SAP EWM creates warehouse orders to put together work packages for the individual warehouse employees.

  For more information, see Warehouse Order Creation and Creation of Warehouse Tasks for Warehouse Request.
- SAP EWM copies most of the settings that you make in Customizing for process control to the delivery documents and displays them on the user interface for the outbound delivery order. To do this, on the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Outbound Delivery** > **Maintain Outbound Delivery Order**. Display an outbound delivery order in the form view. Here you can find the settings from Customizing, for example, in the **Corr.Deliv.**, **Scrapping**, or **Pickup** fields.

Warehouse Request of the Returns Type

Definition

This warehouse request defines all product movements that are used to enable goods that were returned by a customer to be processed as an inbound delivery in Extended Warehouse Management (EWM).

Use

The warehouse request of the returns type forms the basis for executing follow-up activities in EWM.

Follow-up activities for returns include:

- Reject the inbound delivery after carrying out a rough inspection

  You first conduct a rough inspection of the inbound delivery. In doing so, if you determine that the forwarding agent does not need to unload the goods, but rather can take the goods straight back because they are damaged, for example, you can immediately reject the inbound delivery.

  On the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Inbound Delivery**
  - > **Maintain Inbound Delivery Notification** and then **Reject**
  - > **Maintain Inbound Delivery** and then **Reject**

    Note

    For more information about rejecting inbound deliveries, see Goods Receipt .
- Putaway

- Scrapping
- Transferring
- Delivery to internal remanufacturer
- Delivery (to customer or external remanufacturer)

> **ℹ Note**
> You cannot change returns deliveries that SAP CRM sends to EWM, for example. The external system controls the returns process and determines which items in which quantities the customer can send back. In particular, you cannot do the following:
>
> - Create new delivery items
> - Delete delivery items
> - Split delivery items

Structure

The structure of a warehouse request of the returns type differs from the generic structure of a warehouse request, as follows:

- Different item types are used
- Item category RET is used
- A different configuration of the item types and the item category is used:
  - EWM uses a status profile specifically for the warehouse request of the returns type:
    - /SCDL/INB_PRD_RET for an inbound delivery
    - /SCDL/INB_REQ_RET for an inbound delivery notification

      Thus, EWM can carry out the following:
    - Create the inspection documents automatically (inspection planning)
    - Adjust the status type **Q-Inspection** automatically when entering the inspection results

      The status type **Q-Inspection** specifies the start and end of the quality inspection.
    - Control the follow-up activities, such as putaway, based on the inspection results

      SAP CRM determines whether SAP CRM or EWM is to determine follow-up activities. EWM maps this using the status type **Blocked According to QM** .

      The status type **Blocked According to QM** specifies that following the quality inspection, the returns stock is blocked from further use until an external system, for example, SAP CRM, makes a usage decision.
      - If the status type **Blocked According to QM** is not set in EWM, you can create the follow-up activity in EWM. EWM automatically triggers the corresponding logistics processes, such as putaway or returning goods to the customer.
      - If the status type **Blocked According to QM** is set in EWM, EWM must first send the inspection result to SAP CRM. Then SAP CRM determines the follow-up activity and sends it to EWM. EWM triggers the corresponding logistics processes. You can carry out stock transfers in the warehouse during the whole process.

If you carry out a step-by-step inspection in multiple warehouses, you carry out an individual inspection for each warehouse. If the current warehouse is not the last warehouse in the sequence, EWM sets the status type **Blocked QM f. Putaway** . As a result, you can only put returns away after the last inspection has been carried out. However, you can scrap or send returns back after each individual inspection. The step-by-step inspection does not affect delivery processing.

- EWM adopts the validity dates/times from SAP CRM.

  SAP CRM determines the date and time for the external and internal validity of the returns and sends this data to EWM. EWM maps the data for external and internal validity to the corresponding date/time types **Internal Validity RMA for Returns** and **External Validity RMA for Returns** and saves them in the delivery item. These date/time types determine whether EWM accepts or rejects returns.

  - If both date/time types have expired, EWM rejects the returns.
  - If the external date/time type has expired, but the internal date/time type is still valid, SAP CRM determines the follow-up activities and decides whether or not to reject the returns. EWM adopts this decision.

    ℹ Note

    For the structural description of a warehouse request, see Warehouse Request .

Integration

SAP CRM triggers the returns. It sends a delivery message to EWM. EWM then creates a warehouse request of the returns type.

EWM conducts inspection planning with quantity offsetting, and saves the inspection results in the warehouse request of the returns type. In addition, EWM sends the inspection result to SAP CRM.

Warehouse Request of the Internal Stock Transfer Type

Definition

This warehouse request defines all warehouse-internal product movements in Extended Warehouse Management (EWM).

Use

You use this document to plan your warehouse process for ad hoc movements, replenishments, and rearrangement in EWM. Since you can only use this document within the warehouse, it does not contain any delivery-based predecessor documents in other systems (for example, SAP R/3). Your processes in this case are only visible within the warehouse. After completion of the process, EWM informs the connected systems about the process execution.

You execute the internal stock transfer directly from storage bin **A** to storage bin **B** without an intermediate stop (such as packing station). As a result, there is one picking and putaway bin for every internal stock transfer. You or EWM indicate these storage bins when creating warehouse requests.

For more information about warehouse-internal functions, see Ad Hoc Movement , Replenishment , or Rearrangement .

Structure

For the structural description of a warehouse request, see Warehouse Request .

Integration

EWM sends the posting change that it executes for the internal stock transfer as a goods movement message to SAP R/3.

Internal Stock Transfers

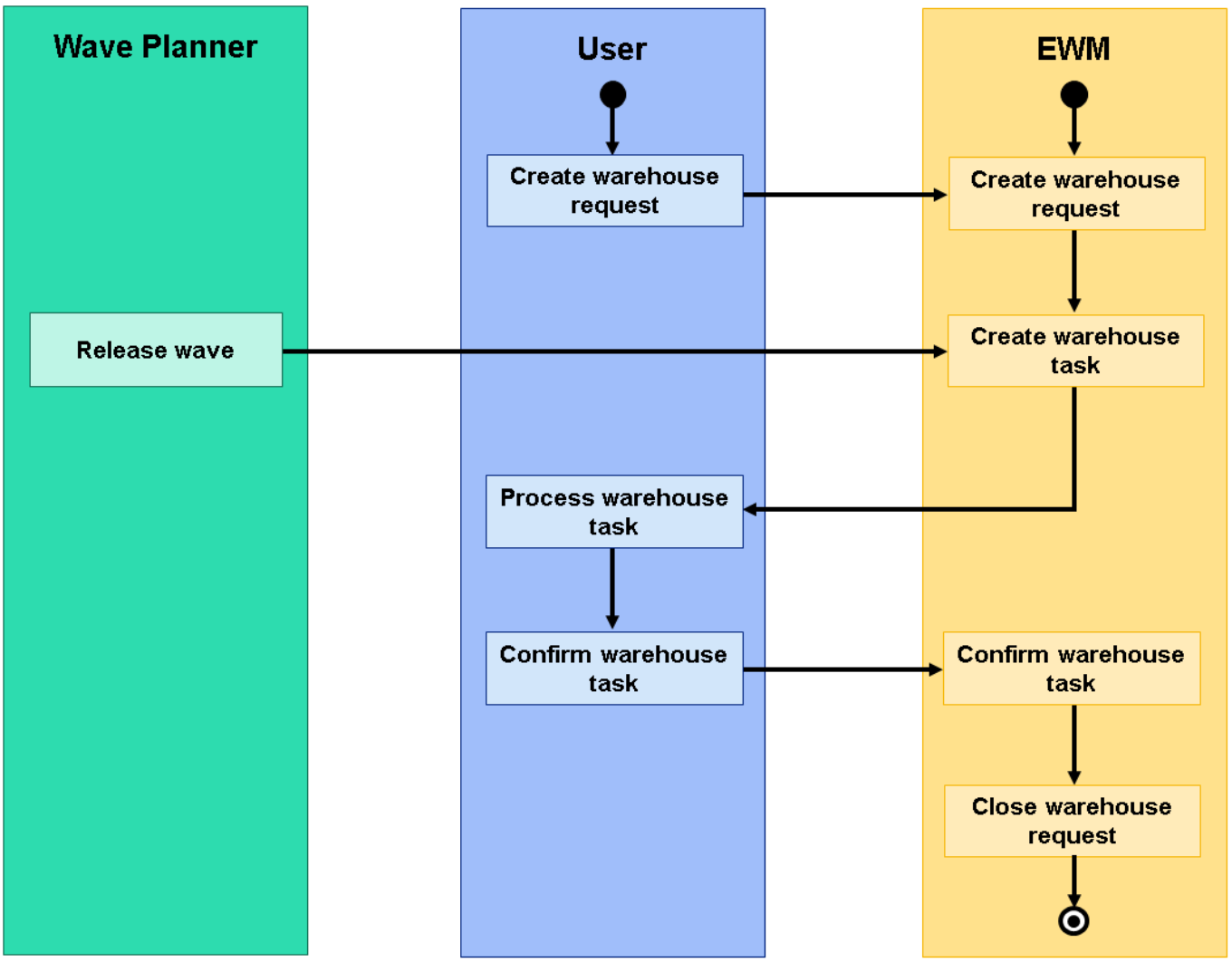
Purpose

With this warehouse process, you can execute the following warehouse-internal product movements in Extended Warehouse Management (EWM):

- Ad hoc movement
- Replenishment
- Warehouse reorganization

Process

The following graphic shows you a sample description for the flow of an internal stock movement.



1. Either you create a warehouse request manually in EWM using the user interface, or EWM does this automatically.
2. When the wave is released, EWM creates the corresponding warehouse tasks (WTs), which reduce the planned open quantity for internal goods movement and block the product.

   When EWM creates WTs, it simultaneously determines from which stock it fulfills the warehouse request. By creating WTs as late as possible, EWM also blocks the necessary stock of the product as late as possible.
3. You process the WTs and confirm them.
4. EWM updates the warehouse request.
5. EWM closes the warehouse request if one of the following options applies:
   - The reduced quantityrepresents the target quantity of the warehouse request.

     The quantity offsetting notifies you at the same time that the open quantity is zero.
   - The quantity reduced by the confirmed WTs does **not** represent the target quantity of the warehouse request. However, you specify process codes for the difference quantities. Quantity offsetting takes these into account.

WTs for warehouse-internal processes move products from one storage bin to another. They therefore contain a putaway and picking step at the same time, whereas the process step **Load** , for example, is omitted.

Result

You have executed an internal stock movement.

EWM has created, updated and closed the corresponding warehouse request.

Executing an Ad Hoc Movement

Use

You can use this function for the following:

- An ad hoc decision for a planned stock movement that triggers the creation of a warehouse request.

  ℹ Note
  In the case of a spontaneous decision for a planned stock movement, you can create a warehouse request.

  You can use this warehouse request in a wave. Once Extended Warehouse Management (EWM) has created the corresponding warehouse task (WT), you can execute the stock movement.
- An ad hoc decision together with an ad hoc movement that triggers the creation of a WT with immediate, automatic confirmation.

  ℹ Note
  In the case of a spontaneous decision for an immediate stock movement, you can immediately create a warehouse task.

  Here you execute the stock movement immediately and without a warehouse request.

Features

You can, for example, transfer a product damaged in the storage bin (for example, water damage) to another storage bin.

To do this, you create a manual warehouse request that sets the picking and putaway storage bin.

On the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Maintain Internal Stock Transfer.

EWM transfers these storage bins during WT creation.

See also: Ad Hoc Movement

Activities

You create warehouse requests for ad hoc movements manually through a user interface.

On the SAP Easy Access screen, choose Extended Warehouse Management > Work Scheduling > Create Warehouse Task Without Reference > Move Product. Then choose Create (Create Warehouse Request) instead of Create (Create Warehouse Task). Alternatively, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Maintain Internal Stock Transfer.

Executing a Replenishment

Use

You can use this function for warehouse-internal replenishment planning. This ensures that you do not fall short of a picking bin's minimum quantity. Replenishment fills picking bins from the reserve storage area.

Furthermore, you can use this warehouse request to schedule several replenishments in waves, and to plan the execution of replenishments separately from the requirements, for example, only execute replenishments at night.

Extended Warehouse Management (EWM) creates this warehouse request with a background job or you can execute the replenishment planning in the dialog.

Features

When creating warehouse requests, EWM specifies possible picking bins and final putaway storage bins. EWM checks the picking bins again before it uses them as final picking bins in the warehouse task.

You can change warehouse requests for the replenishment retroactively.

For more information about executing replenishments, see the relevant report documentation. On the SAP Easy Access screen, choose Extended Warehouse Management > Work Scheduling > Schedule Replenishment. On the Execute Replenishment screen that follows, choose Program Documentation to see the report documentation.

Activities

To retroactively change a warehouse request for replenishment, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Maintain Internal Stock Transfer.

For replenishment planning in the dialog, on the SAP Easy Access screen, choose Extended Warehouse Management > Work Scheduling > Schedule Replenishment.

Executing a Warehouse Reorganization

Use

You use this function to optimize the access paths within your warehouse. Here, Extended Warehouse Management (EWM) uses a warehouse order to reorganize your stock. Since EWM uses a warehouse request for the warehouse reorganization, you can plan your reorganization and start it in a wave.

Features

EWM executes a temporary storage bin determination, which makes no reservation. EWM uses the results of the storage bin determination within the wave determination.

Activities

To create this warehouse request, on the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Work Scheduling** > **Warehouse Reorganization.**

Warehouse Request of the Posting Change Type

Definition

This warehouse request defines the changes in stock key of a product in SAP Extended Warehouse Management (SAP EWM).

Use

The warehouse request of the posting change type is a document that you use in SAP EWM for the following posting changes:

- For unplanned posting changes without reference to a posting change request. In other words, for a process triggered inside the warehouse.
- For planned posting changes with reference to a posting change request. In other words, for a process triggered outside the warehouse.

  SAP ERP triggers this when it sends a delivery message to SAP EWM.

  For more information, see Interfaces for Communication from EWM to SAP ERP.

You can also execute posting changes directly without using a warehouse request.

Structure

For the structural description of a warehouse request, see Warehouse Request.

Integration

You or SAP ERP trigger the posting change externally. In this case, SAP ERP sends a delivery message to SAP EWM. SAP EWM then creates a posting change request.

SAP EWM can execute the posting change directly upon creation of a warehouse request of the posting change type. You can recognize this in the goods movement status of the warehouse request. Planned posting changes, which are based on a warehouse

request, can also have an associated warehouse task for a stock transfer. For example, if you want to execute a posting change from unrestricted-use stock to blocked stock, a physical stock transfer may also be necessary due to the warehouse structure. SAP EWM creates a warehouse task for this stock transfer.

Performing Scrapping

Use

You can use this function in the following scenarios:

- Scrapping within the warehouse, if a product is to be disposed of from the warehouse (for example, defective or obsolete product)
- Scrapping within a returns process

Here you can process warehouse-internal or warehouse-external decisions to scrap a product.

Integration

Only the scrapping triggered outside the warehouse, such as from SAP ERP, has reference to a preceding document (delivery request document). For these types of scrapping, **Extended Warehouse Management** (**EWM**) sends a delivery output to SAP ERP. For scrapping triggered outside the warehouse, EWM sends a delivery message or goods movement message to SAP ERP:

Features

- For warehouse-internal processing of products to be scrapped, you create a warehouse request for the posting change. The system thus posts the products to the blocked stock. From there, you can clear the products through goods issue. EWM can execute the goods issue with or without the corresponding outbound delivery document.
- EWM processes the scrapping in the returns process with a returns delivery, inspection lots, and inspection results. You can trigger the scrapping warehouse-internally in EWM, or warehouse-externally, for example with **SAP Customer Relationship Management** . A warehouse worker in the quality department can use an inspection procedure to carry out the decision to scrap a product. Based on the inspection results, he or she can decide whether a product is still to be used or should be scrapped.

Activities

- You move the products to be scrapped from the different storage bins to a scrapping handling unit (scrapping HU). This represents an internal stock movement that is triggered by the following:
    - Warehouse-internal decision, for example, by a warehouse worker in the quality department
    - Warehouse-external decision when a scrapping message is sent
- At the same time, EWM posts the quantity to be scrapped as blocked stock. In other words, it carries out a posting change. EWM uses a warehouse request for the posting change for this step. You can use this warehouse request for the posting change during wave creation.
- You book the goods issue (GI) for the scrapping HU. **EWM** can execute the GI with the corresponding outbound delivery document. EWM can also execute the GI without a corresponding outbound delivery document. This represents a GI posting without delivery.

Processing, Deleting, and Archiving Warehouse Requests

Use

You can process, delete, or archive warehouse requests.

Features

The following table gives you an overview of the functions you can execute for the individual warehouse requests.

Object for which EWM creates a warehouse request	Create (manually by user, automatically by EWM)	Edit	Delete	Archive
Inbound Delivery Notification	Manually	X	X	X
	Automatically	Activation possible only: restricted processing	------	X
Inbound Delivery	Manually	X	X	X
	Automatically	X	X	X
Outbound Delivery Request	Automatically	-----	-----	X
Outbound Delivery Order	Manually	X	X	X
	Automatically	X	X	X
Outbound Delivery	Manually	X(only header data)	X	X
Posting Change Request	Automatically	-----	-----	X
Posting Change	Manually	X	X	X
Stock Transfer (incl. Scrapping)	Manually	X	X	X

The inbound delivery notification and outbound delivery request created automatically are messages from a sending system, such as SAP ERP. You cannot process or delete these messages.

For an inbound delivery notification or outbound delivery request created automatically, you can only activate or reject the document.

HU Integration in Delivery Processing

Use

Handling unit integration (HU integration) in delivery processing is used to forward HU data between SAP ERP and SAP Extended Warehouse Management (SAP EWM).

The handling unit has a reference to the delivery.

Integration

All HU-relevant processes of the goods receipt process in SAP EWM are based on the correct and complete forwarding of HU data from SAP ERP to SAP EWM. If this forwarding is incorrect or incomplete, errors occur in delivery processing.

SAP EWM simultaneously creates inbound deliveries and HUs in HU management. For this it uses the HU information from the inbound delivery notification. This means that SAP EWM automatically creates HUs that have been defined in SAP ERP.

Prerequisites

The mapping of HU data from SAP ERP to SAP EWM must be defined.

Features

SAP EWM determines the packaged quantity from the handling unit information. Delivery processing sets a status for the packed quantity (for example, unpacked, partially packed, completely packed).

Goods Receipt Process

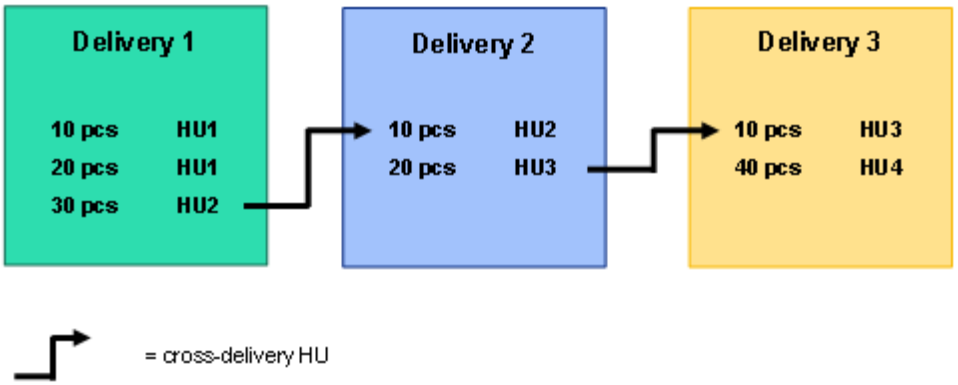
During goods receipt posting (GR posting), delivery processing determines which products exist in which quantities in the handling unit warehouse task. Delivery processing updates these quantities during inbound delivery. You can post the goods receipt (GR) in delivery processing completely or partially.

Goods Issue Process

You can only post the goods issue (GI) on delivery basis completely. A partial GI posting is not possible. You post the GI for every delivery separately. If the Cross-Delivery indicator is set in the data for a cross-delivery handling unit, delivery processing posts the GI automatically for all associated deliveries.

Example

The following figure shows an example GI posting process for cross-delivery handling units.



You have a Delivery1 with HU1 and HU2.

If you post the GI for HU2, delivery processing automatically posts the GI for HU1 as well, since it only posts one GI for the complete delivery.

HU2 is cross-delivery. In other words, it is also relevant for Delivery2. You must either post a GI completely for both deliveries using the selected delivery headers for Delivery1 and 2, or select HU1 and 2 and post the GI. If Delivery2 contains a HU3 that is

also cross-delivery, you will get an error message.

Activities

If you want to view HU data for the inbound delivery, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Inbound Delivery > Process Inbound Delivery.

If you want to view, change, or delete HU data for the outbound delivery, choose one of the following options:

- On the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Outbound Delivery > Process Outbound Delivery Order.
- On the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Outbound Delivery > Process Outbound Delivery.

If you want to view HU data for a warehouse request, open the Process Outbound Delivery Order or Process Outbound Delivery screen and choose Delivery > Follow-On Functions > Pack.

User Interfaces in Delivery Processing

Use

The following user interfaces are available in Extended Warehouse Management (EWM) for the purpose of processing deliveries.

Features

User interfaces for the goods receipt process

- /SCWM/IDN (Maintain Inbound Delivery Notification)
- /SCWM/PRDI (Maintain Inbound Delivery)

User interfaces for the goods issue process

- /SCWM/ODR (Maintain Outbound Delivery Request)
- /SCWM/PRDO (Maintain Outbound Delivery Order)
- /SCWM/FD (Maintain Outbound Delivery)

User interfaces for posting changes

- /SCWM/IM_DR (Maintain Posting Change Request)
- /SCWM/IM_PC (Maintain Posting Change)

User interfaces for internal stock transfers

- /SCWM/IM_ST (Maintain Internal Stock Transfer)

Activities

From the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Inbound Delivery, Outbound Delivery or Posting Change , and then the required user interface.

On the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Maintain Internal Stock Transfer.

Basic Functions in Delivery Processing

Use

Delivery processing provides you with the following basic functions:

- Printing delivery notes
- Text management
- Delivery quantity adjustment
- Validation

Status Management

Use

Status Management controls business transactions (also called activities) by assigning one or several statuses to a delivery document. A status can allow or prohibit a business transaction or it may have no influence at all.

Integration

The status management described here is used in delivery processing exclusively. The field control for delivery processing depends on status management.

Features

A status in delivery processing consists of the following parameters:

- Status type
- Status value

The status typedescribes a partial view or a partial process in the processing of a delivery document (for example, **goods receipt status** , **unloading status** or **putaway status**).

A basic quantity of status types for the document header and the document item is saved in the system profile for status management.

The status types are linked to the document and item types of delivery processing via a status profile. You can change the settings for the status types using this status profile.

The status valuedefines the delivery document status with regard to the status type (for example, **Yes/No** or **Not Started/In Process/Complete** d). It changes along with the business transactions. A status value can disallow a business transaction. On the other hand, a business transaction can set a new status.

The following conditions must be fulfilled before delivery processing can perform a business transaction:

- At least one status value must allow the transaction.
- No status value may disallow the transaction.

SAP delivers predefined status types for the standard processes in delivery processing. Several of these status types are indicated by special attributes which are displayed in the following table. You cannot change these attributes in Customizing.

Attribute	Meaning	Use
Aggregated status	The system calculates the status value of such a status type from the status values of all the partial objects in a lower document level.	An aggregated status at document header level provides compact information on the status of all the items, for example in a display or change transaction.
Projected status	The system calculates the status value of such a status type from the status type with the same name from higher document levels.	A projected status at document item level provides the option of controlling activities and the field control at item level by a status at header level. This attribute has a purely technical meaning.
Overall status	The system calculates the status value of such a status type from the current status values of a predefined quantity of status types from the same partial object (for example, header or item).	An overall status provides compact information on a certain quantity of statuses for the same partial object (for example, header or item).
Transient status	The system only calculates this status type for the runtime. The status type is not saved permanently.	This indicator is only set for derived status types (aggregated status, overall status, for example).
Status value if original status ambiguous	The system sets the derived status value for an aggregated status or an overall status if the status values of the influencing status types are not unique.	
inactive	You use this indicator to activate or deactivate several status types in each status profile. However, there are also status types which always have to be active. You cannot change this indicator for these status types.	

You can increase the number of available status types by adding your own user statuses. As opposed to the status types predefined by SAP, the user statuses can only have two values (**Yes/No**).

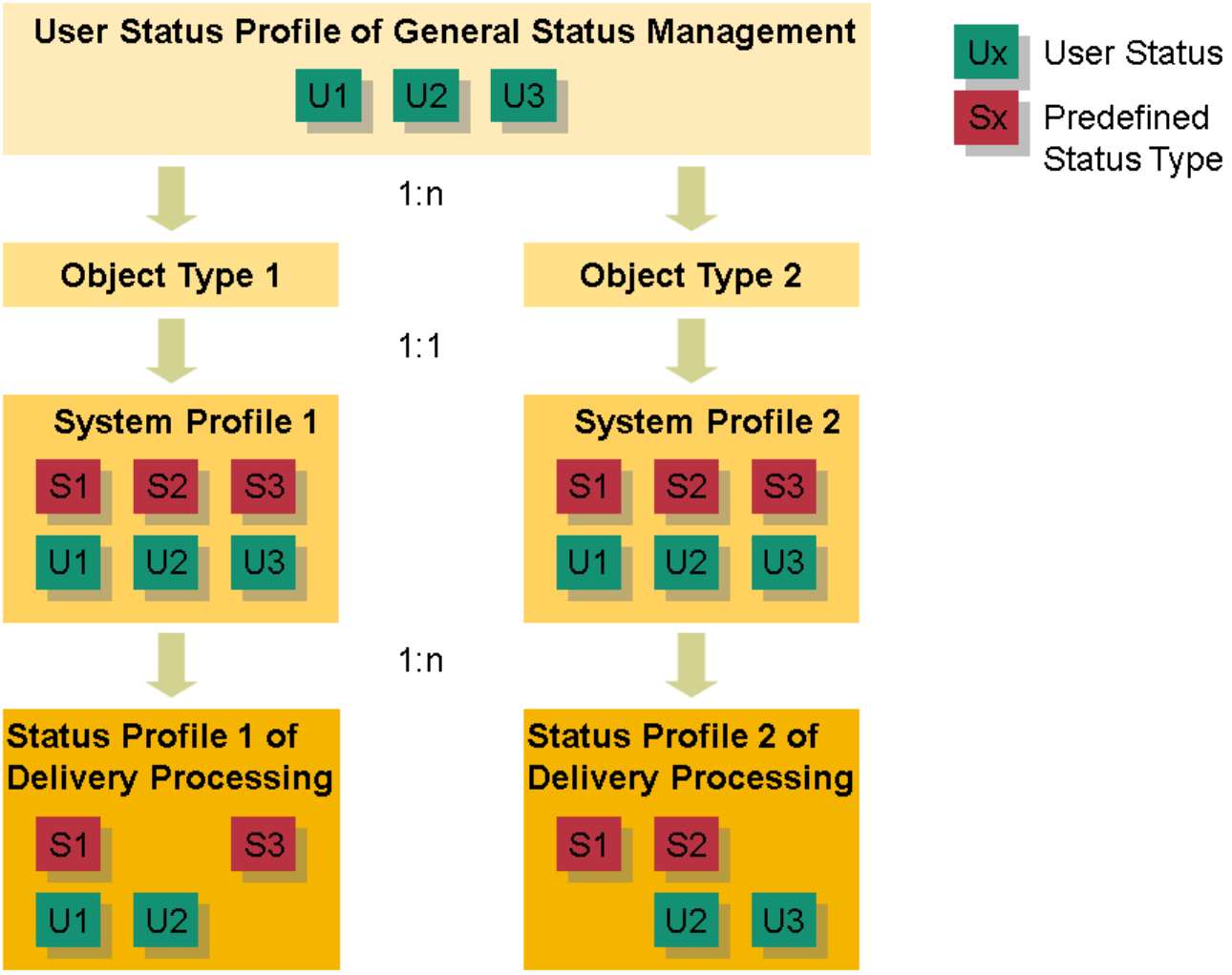
The user statuses are grouped via the user status profile in the general status management. The user status profile in the general status management is linked to the system profile for delivery processing via an object type. Here, a user status profile in the general status management can be assigned to several object types and, therefore, to several system profiles.

⚠ Caution

In the status management for delivery processing, you cannot assign several user status profiles to one object type.

The object type and system profile have a one-to-one relationship. You can derive several status profiles for the configuration of document and item types from one system profile.

The following graphic shows the relationships between the individual objects in status management.



⚠ Caution

The user status profile of the general status management is not identical to the status profile for the document types and item types.

Activities

In Customizing for Delivery Processing , you can execute the following activities:

- Activate or deactivate certain status types in the status profile
- Define status profile
- Define a user status profile of the general status management for the user status
- Define user statuses

Quantity Offsetting

Use

Using the quantity offsetting function, you can offset quantities consistently independently of the application.

In delivery processing, you use the quantity offsetting to calculate the open quantities for a document or to calculate the relevant quantities (requested quantity and reduced quantity) for checking a document execution, for example. This means that you can check whether you actually delivered the agreed quantity on a certain date.

Integration

The quantity offsetting is a cross-application service within Logistics.

Features

The quantity offsetting is executed at item level and at schedule line level of a document. At present, the schedule line level is not supported by the documents in delivery processing.

Quantity offsetting consists of the following features:

- Quantity type
- Quantity calculation
- Quantity offsetting
- Quantity role
- Quantity offsetting profile

Quantity type

The quantity type is a classification of the different quantities for quantity offsetting.

Examples of quantity types include:

- Requested quantity
- Reduced quantity
- Open quantity

Quantity Calculation

The quantity calculation determines the mathematical function (quantity calculation rule) for calculating the relevant data for quantity offsetting. You therefore define how the reduced quantity of a document is determined and which algorithm is to be used to calculate this quantity, for example. The relevant data for the quantity calculation algorithm comes from either the document flow or from a different source (from a service, for example).

If you use the document flow for the quantity calculation algorithm, you have to define which links are relevant in Customizing for delivery processing.

Example

The data in Handling Unit Management (HUM) is used for the packed quantity.

Quantity Offsetting

The quantity offsetting determines the function (quantity offsetting rule) with which the different quantities are calculated.

Depending on how it is used, you can access the quantity offsetting with or without quantity calculation.

Quantity Role

The quantity role is used to group quantity calculation or quantity offsetting rules for certain applications or processes. In this way, you can define the quantity calculation or quantity offsetting rules according to the application or the process.

A quantity role can exist of several quantity types.

Quantity Offsetting Profile

If you want to execute quantity offsetting automatically, a quantity offsetting profile exists in the quantity offsetting. This profile contains the quantity calculation and quantity offsetting rules to be used by the quantity offsetting to calculate the quantities automatically. The quantity offsetting profile is linked to the document type and item type of a document.

SAP delivers predefined quantity offsetting profiles types for the standard processes in delivery processing.

You can find more information in Customizing for Delivery Processing .

Activities

In Customizing for Delivery Processing, you can execute the following activities:

- Define quantity type
- Define quantity role
- Define quantity offsetting profile
- Define quantity calculation rules
- Define quantity offsetting rules

Text Management

Use

With this function, you can do the following to texts belonging to deliveries:

- Display
- Change
- Save
- Delete

In addition, you can create and save new texts for a particular delivery.

Integration

SAP ERP sends a message (inbound delivery notification or outbound delivery request) to SAP Extended Warehouse Management (SAP EWM) to create a delivery. This message also contains the data (including existing texts) for this delivery.

You can create texts in external systems (such as SAP CRM) and send them with the document via SAP ERP to SAP EWM.

Prerequisites

You have made the Customizing settings for text management.

- You have defined text types.

  You have assigned one of the text types to the text objects (such as delivery header, delivery item) delivered by SAP. For example, assign the text type packing text to the delivery header.

  ### i Note

  For more information about defining text types, see the Implementation Guide (IMG) for EWM under Cross-Process Settings > Delivery Processing > Text Management > Define Text Types for Delivery Document Header and Define Text Types for Delivery Document Item.

- You have defined a text profile containing the following:

  - One or more access sequences
  - The assignment of text types to the text profile
  - The assignment of a document or item type to the text profile

    ### i Note

    For more information about defining text profiles, see the IMG for EWM under Cross-Process Settings > Delivery Processing > Text Management > Define Text Profiles with Access Sequences.

  For more information about Customizing the document or item types for the goods receipt process, see the Implementation Guide (IMG) for EWM under Goods Receipt Process > Manual Settings > Inbound Delivery > Define Document Types for Inbound Delivery Process or Define Item Types for Inbound Delivery Process.

  For more information about Customizing the document or item types for the goods issue process, see the Implementation Guide (IMG) for EWM under Goods Issue Process > Outbound Delivery > Manual Settings > Define Document Types for Outbound Delivery Process or Define Item Types for Outbound Delivery Process.

Features

Text management supports texts:

- At header level

  ### Example

  For example, you can include sales texts that indicate that the delivery A is for the key customer Smith and that the product must therefore be delivered as quickly as possible.

- At item level

  ### Example

  For example, you can create a packing text for a product that indicates to the warehouse employee that the product is fragile and that he/she must therefore always pack it in packing material 123.

In SAP EWM, you can display, change, or delete texts belonging to deliveries. Only there can you also create and save new texts directly.

i Note

For more information about the connection between documents and services, such as text management, see Delivery Processing.

Activities

To process existing texts or create new ones in SAP EWM, select one of the following:

- For the goods receipt process, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Inbound Delivery > Maintain Inbound Delivery.
- For the goods issue process, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Outbound Delivery > Maintain Outbound Delivery Order.
- For posting changes, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Posting Change > Maintain Posting Change.

Field Control

Use

A document in Delivery Processing contains a multitude of fields (see Documents in Delivery Processing). The system uses the field control settings to determine which fields are displayed over the course of the delivery process and which fields can be changed.

The field control supports fields at header and item level of a document.

Integration

Field control depends on status management (see Status Management). During the delivery process, the system controls which fields can be displayed and changed through the status of the document and item type.

The field control is accessed by delivery processing during the following events:

- When creating a document or a document item
- When switching between display mode and change mode
- After changing the data in a document

Features

Logical field names

The individual fields of a document are represented in the system as components of structures. Every component in a structure is uniquely identified by the logical field name and can therefore be addressed directly.

Field attributes

In the field attributes of the field control, the system defines the status for which a field is displayed and can be changed for each logical field name.

The following table lists the field attributes in the field control:

Field attribute	Meaning
Status type	Describes a partial view or a partial process in the processing of a delivery document (for example, goods receipt status, unloading status or putaway status).
Status value	Defines the delivery document status with regard to the status type (for example, **Yes**/ **No** or **Not Started**/ **In Process**/ **Completed**).
Display field	If this checkbox is selected, the field is displayed as a non-changeable field depending on the assigned status (status type and status value).

Several statuses can be assigned to one field. One field may have different settings for different statuses. When accessing the field control, the table of field attributes is read sequentially by the system. Here, the system compares the status information in the table with the current status information of the document. Every time this information is consistent, the system assigns the corresponding attributes from the field attributes (display or hide field) to the field. If no consistencies are found, the field remains changeable and is not hidden.

System profile of the field control

The system profile for the field control contains the allowed fields and their field attributes for a certain document or item type. By means of the system profile, you can include new logical field names in the field control. The assignment to the document or item type is carried out by means of the field control profile.

The field attributes saved in the system profile are valid for all assigned field control profiles. You cannot change these field attributes in the system profile nor in the assigned field control profiles.

⚠ Caution

The incompleteness profile also depends on the system profile of the field control. It uses the list of all allowed fields.

Field control profile

A field control profile contains all the allowed fields and their field attributes for a certain document or item type. Every field control profile is assigned to a system profile. When accessing the field control, the system determines the assigned field control profile via the document and item type.

SAP delivers predefined logical field names, system profiles, and field control profiles for the standard processes in delivery processing. You cannot change the field attributes of the system profiles delivered by SAP. However, you can add new field attributes.

Activities

In Customizing (see Menu Paths and Customizing Paths for Delivery Processing, you can execute the following activities for the field control:

- Define a new field control profile
- Maintain field attributes

  You cannot change any of the field attributes delivered by SAP. However, you can add new field attributes.
- Define logical field names for new fields
- Add new logical field names to the system profile of the field control

Validation

Use

Extended Warehouse Management (EWM) works with a validation framework and uses various validation methods. The validation checks whether a delivery document is complete and consistent so that it can be further processed in the system. If a document changes, EWM performs a validation.

Features

The validation comprises the following checks:

- Incompleteness check
- Consistency check with tolerance check
- Checks in the goods receipt process

Activities

You can call validation directly from the user interface.To do so, on the **SAP Easy Access** screen, choose the corresponding user interface, for example, **Extended Warehouse Management** > **Delivery Processing** > **Inbound Delivery** > **Maintain Inbound Delivery** .Select an inbound delivery and choose the function **Check** .

Incompleteness Check

Use

The incompleteness check determines whether all the relevant fields in a delivery document are filled so that it can be further processed in the system.

Integration

The incompleteness check uses the logical field names assigned to the system profile of the field control .

Features

Mandatory fields

Every input and output field that must be filled to be able to further process a document is a mandatory field. A mandatory field is defined by a logical field name in the system. Every logical field name that represents a mandatory field is assigned an action. Every time this action is executed, the corresponding field is automatically checked for completeness.

Incompleteness profile

The settings for the incompleteness check are saved in an incompleteness profile. The incompleteness profile is assigned to the document and item types. When accessing the incompleteness check, the system determines the assigned completeness profile via the document and item types and therefore the corresponding settings for the mandatory fields.

SAP delivers predefined logical field names and incompleteness profiles for the standard processes in delivery processing. You cannot change the field attributes delivered by SAP. However, you can add new field attributes.

Activities

The incompleteness check is executed automatically by the system. If the incompleteness check determines a mandatory field that has not been filled, the action is terminated and an appropriate message is issued.

In Customizing for Delivery Processing , you can execute the following activities for the field control:

- Define a new incompleteness profile
- Maintain field attributes

  You cannot change any of the field attributes delivered by SAP. However, you can add new field attributes.
- Define logical field names for new mandatory fields
- Add new logical field names to the system profile of the field control

Consistency Check

Use

The consistency check determines whether any data inconsistencies exists between the delivery document , Customizing and master data.

Features

The consistency check contains the following checks:

- Check whether the document type, document category, item type and item category correspond to the settings in Customizing for delivery processing.
- Check whether the document contains items.
- Check whether the product data corresponds to the product master (for example, product ID and allowed units of measure).
- Check whether the location data corresponds to the location master.
- Check whether the delivery date lies in the past.
- Check whether the specifications for the Incoterms, process codes, partner roles, reference documents and date types correspond to the settings in Customizing for delivery processing.
- Check whether partial deliveries are allowed.

  Delivery processing determines the number of allowed partial deliveries from the reference document. The consistency check uses these specifications to determine whether partial deliveries are allowed. If no partial deliveries are allowed, the system issues a message and you cannot split the delivery.
- Check whether the retention period is expired.

Activities

The consistency check is executed automatically by the system. You can also trigger the consistency check manually by choosing the delivery document in the maintenance transaction.

If a document is inconsistent, the system issues an appropriate message.

Example

If you want to compare the delivery quantity of the inbound and outbound delivery with the tolerance defined in Customizing, you can perform a Tolerance Check .

Tolerance Check

Use

The tolerance check determines whether the delivery quantity lies within the given tolerances for both inbound and outbound deliveries.

Integration

You define the tolerances for the delivery quantity in the reference document that is used to trigger the delivery process. You cannot change the tolerances in the delivery process.

Features

The tolerance check is split into the underdelivery tolerance and the overdelivery tolerance.

- Underdelivery tolerance

  The minimum allowed delivery quantity. It is specified as a percentage.

- Overdelivery tolerance

  The maximum allowed delivery quantity. It is specified as a percentage.

As soon as the delivery quantity changes in a document, the delivery process checks to see whether the changed delivery quantity still lies within the defined tolerances. If the goods quantity lies below or above the tolerance values, the system issues a warning message. The tolerance check in delivery processing checks the delivery quantities in the following documents:

- Inbound delivery notification
- Inbound delivery
- Outbound delivery Request
- Outbound delivery order

Unlimited Overdelivery

If this indicator is set, overdeliveries of any size are allowed. The system does not issue any message.

Activities

The tolerance check is executed automatically by the system.

Checks in the Goods Receipt Process

Use

When Extended Warehouse Management (EWM) generates an inbound delivery, it performs a series of automatic checks in the background. This enables you to better evaluate whether you want to accept or reject delivered products. For example, you reject a product because the quantity delivered was too large, or the minimum remaining shelf life is below normal.

Integration

For example, you receive a purchase order with a specific best-before date. The checks in the goods receipt process in EWM, for example, the check for the best-before date, provide you with all the necessary data so that you are already able to decide during a goods receipt at the door if the delivered products match the purchase order data in SAP ERP or the delivery or product master data in EWM. If EWM has carried out the checks and the inspection results are positive, that is if EWM has not blocked a delivery item, you can post a partial goods receipt or the complete goods receipt.

EWM executes the checks in delivery processing. For this, the system checks the inbound delivery data with the available delivery or purchase order data. EWM also takes into account retail-specific master data during its checks.

Prerequisites

See under:

- Product Master Data Check
- Checking the Best-Before Date or Expiration Date
- Checking the Overdelivery Tolerance

Features

EWM executes the checks in the background automatically. If the checks result in an error, you receive a message. You can check the corresponding data and correct it, if necessary. If no delivery position is blocked, you can then post the goods receipt.

The following are examples of checks that EWM carries out when it generates an inbound delivery:

- Checking the best-before date or expiration date

  EWM checks the inbound delivery data with the master data in EWM.

- Checking the Overdelivery Tolerance

  EWM checks the data of the inbound delivery with the data from the purchase order in SAP ERP or the delivery data in EWM. The decision whether EWM is to check of the purchase order data in SAP ERP or the delivery data in EWM is made in SAP ERP.

- Checking the product master data during goods receipt

Activities

EWM starts the checks automatically if you carry out one of the following actions on the user interface of the inbound delivery or the physical goods receipt:

- You choose **Check** to first check the data in the inbound delivery or in the physical goods receipt, and to not yet post the goods receipt.
- You choose **Save** to save the inbound delivery.

- You choose Goods receipt on the user interface of the inbound delivery or Post goods receipt on the user interface of the physical goods receipt to post the goods receipt.

Checking the Product Master Data

Use

If you receive an inbound delivery and specify master data, Extended Warehouse Management (EWM) automatically checks in the background if the product master data was already checked. EWM executes this check before you post the goods receipt.

Integration

EWM only performs this check for inbound deliveries.

Prerequisites

You have set the Insp.Prod. indicator in Customizing for EWM so that EWM checks the product master data.For more information, see the Implementation Guide (IMG) for EWM under Goods Receipt Process > Inbound Delivery > Manual Settings > Define Item Types for Inbound Delivery Process .

Features

When EWM generates an inbound delivery, it automatically checks a product in the background to see whether the product master data was already checked, corresponding to Customizing for EWM. You can always post the goods receipt. If the check is negative, that is, if the product master data was not yet checked, you receive a corresponding warning message.

Activities

If required, on the SAP Easy Access screen, choose Extended Warehouse Management > Master Data > Product > Maintain Product. In the group box checked by , choose Set check to confirm the product master data check; for example, for dimension and weight.

Checking the Best-Before Date or Expiration Date

Use

If you receive an inbound delivery and the minimum shelf life expiration date is specified, Extended Warehouse Management (EWM) checks the best-before date or expiration date in the inbound delivery with the minimum remaining shelf life.You can check, reset, or, if necessary, manually overwrite the inspection result.

Integration

EWM only performs this check for inbound deliveries.

The status for the best-before date or expiration date affects the entire status of a delivery item (see "Features").

Prerequisites

In Customizing for EWM, you have defined whether EWM is to include the validation of the best-before date or expiration date in the settings for delivery.You have also defined the settings for checking the best-before date or expiration date in the batch handling.For more information, see the Implementation Guide (IMG) for EWM under **Cross-Process Settings → Batch Management → Make Setting for Delivery.**

In Customizing for EWM in Status Management, you have set the status type **DSL Ck Remaining Shelf** for the delivery item to active. That means you have not set the **Inactive** indicator.For more information, see the IMG for EWMunder **Cross-Process Settings → Delivery Processing → Status Management → Define Status Profiles.**

Features

If EWM generates an inbound delivery, it checks the best-before date or expiration date according to the Customizing settings in EWM.If the check is positive, that is, if EWM does not block any delivery item, you can post the goods receipt, and EWM generates the related warehouse tasks.EWM performs this check in the following cases:

If you change the best-before date or expiration date of a delivery item

If you select **Check** on the user interface of the inbound delivery

If you select the arrow on the right-hand side of the **In Yard** icon and then select **Set Status**

Based on the inspection results, EWM sets the status for the status type **DSL Ck Remaining Shelf** .You can check or reset the inspection result.If the check is negative, you can accept the result despite this before you book the goods receipt.

You can use the Business Add-In (BAdI) /SCWM/EX_DLV_BATCH_SLED_VAL **BAdI: Check Best-Before Date/Expiration Date** to determine in detail within batch management when a check of the best-before date or expiration date is to occur.For more information, see the BAdI documentation.

You can use the BAdI /SCWM/EX_DLV_BT_VAL_DATE_WE **BAdI: Select Comparison Date for BBD Check at GR to change the determination of the date for the check on the best-before date or expiration date.** For more information, see the BAdI documentation.

Checking Best-Before Date or Expiration Date, Goods Receipt Posting and Warehouse Tasks

You check the products that have been delivered directly on site, perform changes for the best-before date or expiration date if necessary, and post the goods receipt.

If EWM sets the status **Not OK** for the check of the best-before date or expiration date, the delivery item is inconsistent.You cannot post a goods receipt for this delivery item.However, you can manually approve the check and accept the best-before date or expiration date.EWM sets the status **OK, manually** for the check. You can post the goods receipt.

If you have mistakenly executed the check on the best-before date or expiration date for an incorrect delivery item, you can manually reset the status of the check for this delivery item.

Checking Best-Before Date or Expiration Date in the Warehouse Management Monitor

You can specify a reference date in the Warehouse Management Monitor and display the stock whose best-before date or expiration date allows no further use in a regular stock removal process.For example, EWM displays the stock whose best-before date or expiration date no longer provides the agreed number of days before reaching this date.This includes critical situations concerning the latest delivery date and the best-before date or expiration date.On the **SAP Easy Access** screen, choose **Extended Warehouse Management → Monitoring → Warehouse Management Monitor , and in the navigation structure, choose Stock and Bin → BBD/Expiration Date Overview.** As a minimum, specify your reference date; that is, the best-before date or expiration date or the latest delivery date.You can also specify additional selection fields; for example, a storage bin or a transportation unit.You can also exclude stock that is assigned to a storage bin, a resource, or a transportation unit.EWM displays the associated stock,

and you can manually assign a new stock type to this to execute, for example, a scrapping.Select the stock and choose the arrow on the right-hand side of the More Methods icon, and then Change Stock Type.

Activities

If you want to manually approve or reset the check, you can, for example, perform one of the following options:

For manually approving or resetting within the inbound delivery, on the SAP Easy Access screen, choose Extended Warehouse Management → Delivery Processing → Inbound Delivery → Maintain Inbound Delivery. Display the inbound delivery in the list view, and choose the arrow on the right-hand side of BBD/Tolerance and Appro ve BBD Violation or Reset BBD Violation.

For manually approving or resetting within the physical goods receipt, on the SAP Easy Access screen, choose Extended Warehouse Management → Delivery Processing → Inbound Delivery → Physical Goods Receipt. Display the inbound delivery in the list view and choose the arrow on the right-hand side of Blocked Status and Appro ve BBD Violation or Reset BBD Violation.

Checking the Overdelivery Tolerance

Use

When you receive an inbound delivery and specify the delivered quantity, Extended Warehouse Management (EWM) checks if the tolerance for an overdelivery was kept.You can check, reset, or, if necessary, manually overwrite the inspection result.

Integration

EWM only performs this check for inbound deliveries.

EWM compares the data of the inbound delivery with the data from the purchase order in SAP ERP or the delivery data in EWM.The decision whether EWM is to check of the purchase order data in SAP ERP or the delivery data in EWM is made in SAP ERP.

EWM copies the quantities and unit of measure from the purchase order or from the order in SAP ERP to an inbound delivery. EWM saves this data as quantity roll PQTOT Total Purchase Order , or PPQTOT Total Order Quantity. For more information, see the Implementation Guide (IMG) for EWM under Cross-Process Settings → Delivery Processing → Quantity Offsetting → Define Quantity Roles and Quantity Types.

Prerequisites

In Customizing for EWM in Status Management, you have set the status type DTL Check Tolerance for the delivery item to active. That means you have not set the Inactive indicator.For more information, see the IMG for EWM under Cross-Process Settings → Delivery Processing → Status Management → Define Status Profiles.

In Customizing for EWM, you have set the PQTOT Total Purchase Order Quantity or PPQTOT Total Order Quantity to active. That means you have not set the Inactive indicator.For more information, see the IMG for EWM under Cross-Process Settings → Delivery Processing → Quantity Offsetting → Define Quantity Offsetting Profiles.

Features

When EWM generates an inbound delivery, it checks the data for a quantity tolerance according to the Customizing settings in EWM.If the check is positive, that is, if EWM does not block any delivery item, you can post the goods receipt, and EWM generates the related warehouse tasks.EWM performs this check in the following cases:

If you change the tolerance for an overdelivery of a delivery item, or the related unit of measure

If you choose Check on the user interface of the inbound delivery

Based on the inspection results, EWM sets the status for the status type DTL Check Tolerance. You can check or reset the inspection result. If the check is negative, you can accept the result despite this before you book the goods receipt.

The tolerance check of the purchase order item can also take into account quantities from other delivery items that refer to the same purchase order.If you, for example, make a quantity change in such a case after EWM has performed the tolerance check for an overdelivery, you can manually trigger the check for the current delivery item.Choose with the quick info text Check .EWM performs the check using the new data.

Checking the Tolerance for Overdelivery, Goods Receipt Posting and Warehouse Tasks

You check the products that have been delivered directly on site, perform changes for the tolerance check if necessary, and post the goods receipt.

If EWM sets the status Not OK for the tolerance check, the delivery item is inconsistent.You cannot post a goods receipt for this delivery item.However, you can manually approve the tolerance check and as a result accept an overdelivery.EWM sets the status OK, manually for the tolerance check. You can post the goods receipt.

If you have mistakenly performed the check on the tolerance check for an incorrect delivery item, you can manually reset the status of the tolerance check for this delivery item.

Activities

If you want to manually approve or reset the check, you can, for example, perform one of the following options:

For manually approving or resetting within the inbound delivery, on the SAP Easy Access screen, choose Extended Warehouse Management → Delivery Processing → Inbound Delivery → Maintain Inbound Delivery. Display the inbound delivery in the list view, and choose the arrow on the right-hand side of BBD/Tolerance and Appro ve BBD Violation or Reset BBD Violation.

For manually approving or resetting within the physical goods receipt, on the SAP Easy Access screen, choose Extended Warehouse Management → Delivery Processing → Inbound Delivery → Physical Goods Receipt. Display the inbound delivery in the list view, and choose the arrow on the right-hand side of Blocked Status and Approve Tolerance Violation or Reset Tolerance Violation.

Process Management and Control

Use

Process management and control manages business transactions using process indicators in a delivery document. A process indicator can allow or prohibit a business transaction or it may have no influence at all.

Integration

The process control described here is used in delivery processing exclusively. Process control can influence field control.

Features

A process indicator has multiple values, for example process indicator **Scrapping** has the value **Not Allowed** and **Required**.

A process indicator describes the entire business transaction represented by the delivery document.

For the document header and document item, SAP delivers a set of process indicators in the system profile and customer profile of process control.

The process indicators are linked to the document and item types of delivery processing via a process profile. You can change the settings for the process indicators in the process profile.

When SAP Extended Warehouse Management (SAP EWM) creates a delivery document, the system sets the process indicator. You can then no longer make changes to these. However, if you have set the value **Allowed** for process indicator **IBGI** (invoice creation before goods issue) in Customizing for **Extended Warehouse Management**under **Cross-Process Settings** > **Delivery Processing** > **Process Management and Control** > **Define Process Profile for Delivery Document**, you can change the value of this process indicator in a direct outbound delivery order. However, you can only do this as long as the outbound delivery has not yet been created.

You can choose from the following process indicators:

Process Indicator	Meaning	Use
Manual creation		Certain deliveries, such as direct outbound delivery orders and posting changes, can be created and processed in SAP EWM directly. They do not require a preceding document in SAP ERP.
Preceding document	Allows a reference to a preceding document (purchase order, sales order, or manufacturing order)	Inbound and outbound deliveries usually have a reference to a preceding document, inbound deliveries for purchase orders, and outbound deliveries for sales orders. Posting changes or outbound deliveries without reference, on the other hand, do not refer to a preceding document.
Production	Allows processes for production, such as kit to stock, goods receipt for production	If inbound and outbound deliveries are used in a production process, you have a reference to a manufacturing order.
Scrapping	Allows subprocesses for scrapping	In the scrapping process, you perform a planned reduction of overstock in two steps: you first post change the stock to locked stock, and then post the locked stock.
Pickup	Allows a process for direct sales	In this process, you only make goods available for the sales location within the warehouse. You do not perform any transportation activities.
Invoice Creation Before Goods Issue	Allows invoice creation before goods issue	You can create the invoice in the early stages of the process flow within the packing or loading goods process step.
Correction delivery	Allows posting differences to the stock of the differences that are to be cleared	If you identify warehouse-external differences, for example, in a stock transfer

Process Indicator	Meaning	Use
		process or for a customer delivery, which require warehouse stock to be corrected, you can post these to the stock of the differences to be cleared using a correction delivery.
Communication to ERP	Prevents replication of inbound deliveries to SAP ERP and allows goods receipt posting that are communicated to SAP ERP via the goods movement interface instead of the delivery interface.	If set to Via Goods Movement Interface, the system communicates with SAP ERP via the goods movement interface only. This setting is used during receipt from production process.

Activities

In Customizing for SAP EWM, you define the process indicators for the delivery document header and for the delivery document item. For more information, see Customizing for Extended Warehouse Management under Cross-Process Settings > Delivery Processing > Process Management and Control > Define Process Profile for Delivery Document Header or Define Process Profile for Delivery Document Item.

Determination of Priority for Inbound Deliveries

Purpose

You deploy Extended Warehouse Management (EWM) priority points to allow an optimal unloading sequence for several trucks at transportation unit level in the yard. At the same time, EWM keeps updating the inbound delivery so long as the Status Warehouse Activities is not yet at Completed .

Process

1. SAP ERP receives an advanced shipping notification and creates an inbound delivery for it.
2. SAP ERP notifies SAP Advanced Planning and Optimization (SAP APO) and EWM about this.
3. Based on the notification, the system executes the following:
   a. SAP APO calculates the priority points at item level.
   b. EWM creates an inbound delivery request.

      EWM or you activate the inbound delivery request.

      EWM creates an inbound delivery from the active inbound delivery request.

      The priority points in the inbound delivery are initial (initial value is nil) until SAP APO sends the results of the calculation for an update.
4. SAP APO communicates the calculated priority points to SAP ERP, which then communicates them further to EWM.
5. EWM updates the inbound delivery in relation to the priority points.

   EWM saves the priority points for an inbound delivery at item level. EWM calls the Business Add-In (BAdI) /SCWM/EX_DLV_DET_LOAD to display the inbound delivery. This BAdI calculates an aggregated priority at header level.

6. EWM aggregates the priority points at transportation unit level using the BAdI /SCWM/EX_SR_PRIOP. You can see this on the user interface for unloading.

   You can start the report /SCWM/PRIOP_UPDREQ for the recalculation of priority points from the SAP Easy Access display, under Extended Warehouse Management > Delivery Processing > Inbound Delivery > Update Priority Points .Via SAP ERP, EWM triggers an asynchronous request for the recalculation of priority points in SAP APO.As with the first calculation (Step 4), SAP APO sends the result of this recalculation to EWM via SAP ERP.EWM updates the inbound delivery in relation to the priority points and aggregates the priority points at transportation unit level.

   i Note

   For more information, see the report documentation.

Result

SAP APO has calculated the priority points and has communicated this via SAP ERP to EWM. So long as the status Warehouse Activities is not yet at Completed , EWM has updated the inbound delivery in relation to the priority points.

Printer Determination in Delivery Processing

Purpose

You can use this process in Extended Warehouse Management (EWM) to print various documents on particular printers in the different storage sections, for example, in the goods receipt area or goods issue area.

EWM uses printer determination in the goods receipt process to print unloading instructions and in the goods issue process to print the delivery note and loading instructions.

In the user interface for inbound delivery and outbound delivery order in the warehouse management monitor, you can manually print a document. You can also manually print an outbound delivery document from the user interface in the outbound delivery transaction. For example, you can use this feature if the first print is damaged, or if you require a second print.

Prerequisites

- If appropriate, you have defined a print profile in the default values. To do so, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Outbound Delivery > Pickup. Choose Default Values and enter a print profile.
- If appropriate, you have implemented your own printer determination in Business Add-In (BAdI) /SCDL/PPF_ADD_DATA in method GET_PRINTER. For more information, see the BAdI documentation.
- If appropriate, you have implemented your own print profile determination in BAdI /SCWM/EX_DLV_PRINT_PROFILE. For more information, see the BAdI documentation.
- If appropriate, you have defined one or more print profiles in Customizing for EWM.

  For more information, see the Implementation Guide (IMG) for EWM under Extended Warehouse Management > Goods Receipt Process > Inbound Delivery > Print > Define Print Profiles for Delivery Processing and Goods Issue Process > Outbound Delivery > Print > Define Print Profiles for Delivery Processing .
- If appropriate, you have defined one or more Post Processing Framework (PPF) actions in Customizing for the PPF.

  For more information, see the IMG for EWM under Extended Warehouse Management > Cross-Process Settings > Delivery Processing > Actions > Change Actions and Conditions > Define Action Profiles and Actions and Define

Conditions .

- You have defined print profile determination for the goods receipt process or goods issue process in Customizing for EWM.

  For more information, see the IMG for EWM under Extended Warehouse Management > Goods Receipt Process > Inbound Delivery > Print > DefinePrint Profile Determination for Goods Receipt Process or Goods Issue Process > Outbound Delivery > Print > DefinePrint Profile Determination for Goods Issue Process .

Process

EWM attempts to determine the printer in the following sequence:

1. EWM checks whether you have specified a print profile in the default values on the user interface for pickup. If so, it uses the name of the printer from this print profile.
2. If EWM cannot find a printer in step 1, it checks whether you have implemented your own printer determination in BAdI /SCDL/PPF_ADD_DATA. If so, it uses this printer.
3. If EWM cannot find a printer in step 2, it checks whether you have implemented your own print profile determination in BAdI /SCWM/EX_DLV_PRINT_PROFILE. EWM uses the printer from this print profile.
4. If EWM cannot find a printer in step 3, it checks whether you have defined a print profile determination for the goods receipt process or the good issue process in Customizing for EWM. EWM uses the printer from this print profile.

   You can define an access sequence for print profile determination. As a result, EWM considers one or more combinations of different parameters for the print profile, for example, goods receipt office, ship-to location, and warehouse number.
5. If EWM cannot find a printer in step 4, it checks whether you have specified a printer in the user master data. On the SAP Easy Access screen, choose System > User Profile > Own Data to specify your own output device in the Spool Control group box on the Defaults tab page.

If EWM does not find a printer, it does not schedule the PPF action for printing.

See also:

Print Profile Determination for Invoice Printout

Printing Delivery Notes

Printing Delivery Notes

Use

You use this function in Extended Warehouse Management (EWM) to print delivery notes.

Prerequisites

You have created either an outbound delivery order or an outbound delivery.

Features

EWM prints the delivery note based on the PPF action /SCWM/PRD_OUT_PRINT for an outbound delivery order or /SCWM/FD_OUT_PRINT for an outbound delivery.

- By default, EWM uses the predefined SAP Smart Form /SCWM/DLV_NOTE. However, you can also create your own SAP Smart Form to use when printing the delivery note and for form determination.
- You can also print delivery notes using a PDF-based form. To do this, you must change the assigned processing of the PPF action from Smart Forms Print to External Communication in Customizing for EWM.

  For more information about action conditions, see the Implementation Guide (IMG) for EWM under Extended Warehouse Management > Cross-Process Settings > Delivery Processing > Actions > Change Actions and Conditions > Define Conditions .

EWM automatically prints the delivery note for an outbound delivery as soon as you have posted the goods issue. You can also use a PPF action to manually print a delivery note for an outbound delivery or outbound delivery order.

i Note

For more information about Smart Forms, see the SAP Library under SAP NetWeaverLibrary > SAP NetWeaver by Key Capability > Application Platform by Key Capability > Business Services > Smart Forms .

For more information about form determination, see the IMG for EWM under Extended Warehouse Management > Goods Issue Process > Outbound Delivery > Print > Determine Form Determination for Delivery Note Printing.

You can display a preview of the delivery note before printing it.

EWM issues the dangerous goods data when printing the delivery note. In doing so, EWM uses detailed information about the route and means of transport to determine the relevant dangerous goods master records.

For more information about dangerous goods data, see EH&S Services .

Activities

You have the following options for printing or displaying delivery notes manually:

- Printing or displaying directly using a PPF action
  1. On the SAP Easy Access screen, choose Extended Warehouse Management > Work Scheduling > Print > Process and Display PPF Actions.
  2. Specify the document number in the Application Key field.
  3. Choose Execute .
  4. Select the line of the required PPF action (/SCWM/PRD_OUT_PRINT or /SCWM/FD_OUT_PRINT).
  5. Choose Process .
  6. EWM prints the document.
- Printing or displaying from the user interface of the delivery
  1. On the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Outbound Delivery and then one of the following transactions:
     - Maintain Outbound Delivery Order (/SCWM/PRDO)
     - Maintain Outbound Delivery (/SCWM/FD)
  2. Enter the required document number.
  3. Choose Change to switch to change mode.

4. Choose the PPF Actions tab page and select the required action (Print Outbound Delivery Order or Print Outbound Delivery).

5. Proceed as follows:

   - If you want to display the document, choose Preview.

     EWM displays the document.

   - If you want to print the document, choose Execute Action .

     EWM prints the document.

     i Note

     You can only process a PPF action once from within the user interface of the delivery. PPF actions with the status Unprocessed have not been processed yet. If you have already processed the PPF action once, you can choose Repeat Action to generate and then process a new, unprocessed print action.

Example of Switching from Smart Form Print Forms to PDF-Based Print Forms

1. Choose the action profile Outbound Delivery and the action Print Outbound Delivery FD_OUT i n Customizing for EWM under Extended Warehouse Management > Cross-Process Settings > Delivery Processing > Actions > Change Actions and Conditions > Define Conditions .
2. Switch to change mode and make sure that the setting Default Settings from Action Definition is not selected in the Action Definition group box.
3. In the Processing Selection group box on the Processing Details tab page, select the assigned processing Smart Forms Print and delete it.
4. Under Available Processings , choose External Communication .
5. Save your entry.

Delivery Quantity Adjustment

Use

In SAP Extended Warehouse Management (SAP EWM), you can record physical delivery quantity difference in the inbound delivery for the outbound delivery order or the posting change delivery. Delivery quantity differences can result from damage, loss, or expiration of the shelf life data, for example.

You can adjust the quantity of multiple outbound delivery order items to zero, or the picked quantity in the case of warehouse activities that have already been started for partial quantities, at the same time.

Prerequisites

- If you adjust the delivery quantity manually in the user interface of the delivery, the following prerequisites apply:
  - A delivery exists in SAP EWM, which has a status value at item level for status Completion that allows delivery quantity adjustment. That is, it has either status value Not Started or Partially Completed.
  - If you want to use your own process codes, you have defined them in Customizing for SAP EWM. For more information, see Customizing for Extended Warehouse Management under Cross-Process Settings > Delivery Processing > Process Codes > Define Process Codes .

- If required, you have assigned the process codes to your own process code profiles in Customizing for SAP EWM. For more information, see Customizing for Extended Warehouse Management under Cross-Process Settings > Delivery Processing > Process Codes > Maintain Process Code Profiles.

- If required, you have assigned the process code profiles to the used document types and items types in Customizing for Extended Warehouse Management.

  For more information, see Customizing for Extended Warehouse Management at the following locations:

  - Goods Receipt Process > Inbound Delivery > Manual Settings > Define Document Types for Inbound Delivery Process or Define Item Types for Inbound Delivery Process

  - Goods Issue Process > Outbound Delivery > Manual Settings > Define Document Types for Outbound Delivery Process or Define Item Types for Outbound Delivery Process

  - Internal Warehouse Processes > Delivery Processing > Posting Changes > Define Document Types for the Posting Change Process or Define Item Types for the Posting Change Process

- For the availability check in the direct outbound delivery process, you have not defined any of the following restrictions for the quantity increase in Customizing for SAP EWM. For more information, see Customizing for Extended Warehouse Management under Interfaces > Availability Check > Define Configuration of Availability Check.

  - Under Allowed Quantity Increase, you have not defined A Quantity Increase Not Allowed, meaning that the quantity increase is generally not allowed for the party entitled to dispose, item type, product group type, and product group.

  - Under Allowed Quantity Increase, you have not defined B Special Authorization Required to Increase Quantity, meaning that the user who wants to execute the quantity increase requires authorization for activity 94 (Override) for authorization object `/SCWM/DLV`.

    If you still choose this setting, you must assign authorization for activity 94 to the corresponding user for authorization object `/SCWM/DLV`.

- If you adjust the delivery quantity using an exception code for an automatic delivery update, you must also have defined exception codes in Customizing for SAP EWM and assigned them to the process codes. For more information, refer to Customizing for Extended Warehouse Management under Cross-Process Settings > Exception Handling > Define Exception Codes.

- For both types of delivery quantity adjustment, you have also assigned partner roles to the process codes in Customizing for Extended Warehouse Management under Cross-Process Settings > Delivery Processing > Process Codes > Define Process Codes. If you record quantity differences for stock transfer orders, for example, the partner you define here corresponds to the person who caused the delivery quantity difference (see Quantity Differences for Stock Transfer Orders). SAP EWM saves delivery quantity differences for this partner and sends the information to SAP ERP. The only information SAP EWM does not send to SAP ERP is information about delivery quantity differences for posting change deliveries, since these involve internal warehouse differences.

- For both types of delivery quantity adjustment, you have also defined in Customizing for SAP EWM the partners for which SAP EWM reports exceptions, namely process codes, to SAP ERP. For more information, see Customizing for Extended Warehouse Management under Interfaces > ERP Integration > Delivery Processing > Map Partner Roles from ERP System to EWM.

Features

SAP EWM considers the following quantities for delivery quantity adjustment:

- In the inbound delivery, SAP EWM considers the quantities for goods receipt, packing, and planned putaway, meaning SAP EWM matches the delivery quantity to the largest of these quantities.

- In the outbound delivery order, SAP EWM considers the quantities for loading, packing, and planned picking, meaning SAP EWM matches the delivery quantity to the largest of these quantities.
- In the posting change delivery, SAP EWM considers the quantity for planned internal movements, meaning SAP EWM matches the delivery quantity to this quantity.

In Customizing for the process code, you can define one of the following types for the delivery quantity adjustment:

- SAP EWM does not adjust the delivery quantity, although SAP EWM saves a process code in the delivery that you can also view in the respective user interface of the delivery.
- SAP EWM adjusts the delivery quantity and saves a process code in the delivery.
- SAP EWM adjusts both the delivery quantity and the transferred delivery quantity, meaning SAP EWM performs a delivery split in the goods receipt process or goods issue process. SAP EWM also saves a process code in the delivery.

For more information, see Delivery Split for Inbound Deliveries and Delivery Split for Outbound Deliveries.

Activities

You can adjust delivery quantities in one of the following ways:

- On the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing and then the corresponding user interface:
  - Inbound Delivery > Maintain Inbound Delivery
  - Outbound Delivery > Maintain Outbound Delivery Order
  - Posting Change > Maintain Posting Change

  You adjust the delivery quantity manually in the user interface of the delivery.
- If you are confirming a warehouse task or recording counting, on the SAP Easy Access screen choose Extended Warehouse Management > Execution > Confirm Warehouse Task or Execution > Quality Inspection and Count, to specify an exception code. SAP EWM assigns this exception code to a process code, corresponding to the settings you have made in Customizing. SAP EWM then uses the delivery update to adjust the delivery quantity for each item, corresponding to the settings of the process code.
- If you are repacking delivery stock, on the SAP Easy Access screen choose Extended Warehouse Management > Execution > Packing - General. If you have defined the work center correspondingly in Customizing for SAP EWM, then SAP EWM displays the tab page Differences. For this, you have defined the work center layout in Customizing for Extended Warehouse Management under Master Data > Work Center > Specify Work Center Layout. You record the difference and specify an exception code. SAP EWM maps this exception code to the relevant process code in the background and updates the delivery.
- To adjust the quantity of multiple outbound deliveries, in the warehouse management monitor, choose Outbound > Documents > Outbound Delivery Order > Outbound Delivery Order Item.

  The following monitor methods are available:
  - Adjust Quantity

    Adjusts the item delivery quantity.
  - Adjust Quantity and Create OD

    Adjusts the item delivery quantity and also creates an outbound delivery if the item quantity was set to zero. This automatically sets the item's completion status to complete.

When you start a method, a dialog box appears where you can choose which type of process codes to use for the adjustment. The following possibilities are available:

- **Adjust Quantity**

  This shows only the process codes that adjust the quantity and do not create new deliveries for the difference quantity. The system shows only process codes that are valid for all the items that have been marked in the monitor. If multiple process codes exist for this option, you choose one in the next dialog box.

- **Adjust Quantity (Create new delivery in EWM for difference)**

  This shows only the process codes that adjust the quantity and create new deliveries for the difference quantity. The system shows only process codes that are valid for all the items that have been marked in the monitor. If multiple process codes exist for this option, you choose one in the next dialog box.

- **Adjust Quantity (Use Default Process Code from Profile)**

  This shows process codes that are configured as the default process code for the item types. Note that you cannot specify a specific process code using this option.

To automate the input for the dialog boxes of the monitor methods, see SAP Note 1688949 .

Example

Delivery Split in the Goods Receipt Process

SAP EWM generates an inbound delivery notification for 100 pieces. The document flow also carries a quantity of 100 pieces. SAP EWM generates an inbound delivery with an inbound delivery quantity of 100 pieces. In the user interface of the inbound delivery, you specify that SAP EWM is to adjust the inbound delivery quantity to 90 pieces, since 10 pieces are defective. In Customizing for the process code, you have defined an adjustment for both the delivery quantity and the transferred delivery quantity. SAP EWM adjusts the inbound delivery quantity to 90 pieces. This triggers a quantity adjustment for the transferred delivery quantity in the document flow to 90 pieces, and a delivery split in the goods receipt process, meaning that SAP EWM generates another inbound delivery using the difference quantity of 10 pieces.

Quantity Differences for Stock Transfer Orders

Purpose

In **Extended Warehouse Managemen** t (**EWM**), you can confirm stock transfer orders with a quantity difference. This enables you to map the following processes:

- You receive a quantity shortage or quantity excess in the inbound delivery, which was caused by a delivering warehouse; for example, the delivering warehouse has not loaded a pallet onto the vehicle.
- You receive a quantity shortage or quantity excess in the inbound delivery, which was caused by a carrier; for example, a carrier damaged the goods.
- In cross-docking , you can confirm quantity excesses or shortages that were caused by the first cross-docking warehouse and not the delivering warehouse.

If a source warehouse caused a quantity excess or shortage, SAP ERP sends a correction delivery to **EWM** in the source warehouse. There, if required, you can execute clearing in the Difference Analyzer between the receivable quantities and the difference quantities to be cleared. If you are unable to execute clearing, you post the difference to SAP ERP.

Example

You deliver 10 pieces of product A. The customer only receives 9 pieces. You post a correction delivery for one piece as a receivable quantity to be cleared. In the physical inventory, you find that one piece of product A is still in the goods issue area. In the Difference Analyzer, you can offset the receivable quantity to be cleared with a difference quantity to be cleared.

If a carrier caused a quantity excess or shortage, you perform an adjustment posting directly in SAP ERP.SAP ERP uses the process code from the last partial goods receipt message of the target warehouse to determine who caused the quantity excess or shortage.For more information, see the SAP Help Portal for mySAP ERP under SAP ERP Central Component > Logistics > Logistics - General (LO) > Service Parts Management (LO-SPM) > Transfer and Inventory Management > Stock Transfers > Adjustment of Stock Transport Order Quantity Difference .

If quantity differences occur in connection with kits, you always post these differences at the kit header. If the kit itself is not available in the source warehouse, rather the kit components only, you have to clear this difference by posting differences in the inventory quantity.

Prerequisites

- You are working in a system landscape that includes SAP ERP, central SAP Advanced Planning & Optimization (SAP APO) for the automatic generation of stock transfer orders, and at least one decentralized EWM .
- You are using interfaces between EWM and SAP ERP. For more information, see Communication from EWM to ERP System .
- You have defined tolerance groups for the Difference Analyzer in the Customizing settings for EWM .

  For more information, see the Implementation Guide (IMG) for EWMunderInternal Warehouse Processes > Physical Inventory > Warehouse-Number-Specific Settings > Define Tolerance Groups > Define Tolerance Group for Difference Analyzer.
- You have assigned the user to a tolerance group for the Difference Analyzer in the Customizing settings for EWM .To do so, on the SAP Easy Access screen, choose Extended Warehouse Management > Settings > Physical Inventory > Assign User to Tolerance Group for Difference Analyzer.

EWM in the Source Warehouse

- You have defined document types for the correction deliveries in Customizing for EWM . EWM finds these using the differentiation attribute DIS (discrepancy process) at header level.

  For more information, see the Implementation Guide (IMG) for EWMunder Interfaces > ERP Integration > Delivery Processing > Define ERP Document Types for Differentiation Attribute .

  i Note

  Make sure that the Customizing settings for EWM for your own document types and for the item types used are correct. If you are defining your own profiles, use the delivered document types and item types and their relevant profiles as a basis.

- You have defined the mapping of the document types from SAP ERP in EWM in Customizing for EWM.

  For more information, see the IMG for EWM under Interfaces > ERP Integration > Delivery Processing > Map Document Types from ERP System to EWM System .

EWM in the Target Warehouse

- In Customizing for EWM , you have defined whether EWM is to execute an automatic or manual goods movement.

  For more information, see the IMG for EWM , under Cross-Process Settings > Delivery Processing > Actions > Use Assistant to Create Actions .

- In Customizing for EWM , you have defined process codes and assigned these to your own process code profiles. You have also assigned partner roles to the process codes (see Delivery Quantity Adjustment).
- You have defined the mapping of the exception code to the process code of the delivery processing in Customizing for EWM .

  For more information, refer to the IMG for EWM under Cross-Process Settings > Exception Handling > Define Exception Codes .

Process

Quantity Differences for Stock Transfer Orders

You work with a central SAP APO and SAP ERP and two decentralized warehouses managed by EWM . You transport goods from one warehouse to another. In the following, the two EWM systems are referred to as EWM 1 for the source warehouse and EWM 2 for the target warehouse.

1. You create a stock transfer order in SAP APO and SAP APO informs SAP ERP about it.
2. SAP ERP generates a corresponding stock transfer order and an outbound delivery for this. It informs EWM 1 of the source warehouse about this.
3. EWM 1 generates a relevant outbound delivery.
4. You perform the required goods issue process steps and post the goods issue.
5. EWM1 informs SAP ERP about this.
6. SAP ERP generates a corresponding inbound delivery for EWM 2 of the target warehouse. It informs EWM 2 about this.
7. EWM 2 generates a corresponding inbound delivery.
8. You or a carrier transport the products to EWM2 .
9. You execute the goods receipt process steps up to goods receipt posting. When posting goods receipt, you identify a quantity excess or shortage.
10. You post goods receipt with the quantity excess or shortage and specify a process code.

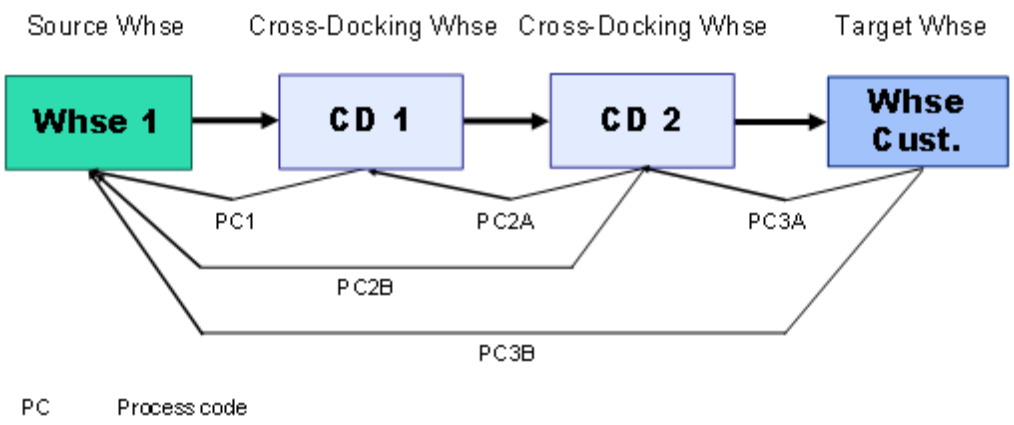
    EWM 2 uses the process code to differentiate who caused the quantity excess or shortage (see Delivery Quantity Adjustment).
11. EWM2 informs SAP ERP about this.
12. SAP ERP posts the quantity differences in the proof of delivery.
13. SAP ERP generates a correction delivery for EWM 1 of the source warehouse. For a quantity shortage, the ERP system generates an inbound delivery for EWM1, and for a quantity excess, an outbound delivery for EWM1 . SAP ERP informs EWM1 about this.
14. EWM 1 generates an inbound or outbound delivery as a correction delivery, based on the notification. It posts the quantity excess or shortage as a receivable quantity to be cleared (external quantity difference). You can reject goods receipt of the inbound delivery or goods issue of the outbound delivery in EWM The unassigned stock remains in SAP ERP. You have to then process it manually, if necessary.

    EWM 1 posts internal quantity differences from the physical inventory or warehouse task creation as difference quantities to be cleared. For example, if you find a pallet in the physical inventory that has 10 pieces of a product, you post this as a difference quantity to be cleared.
15. In EWM 1, you attempt to offset the receivable quantities to be cleared with the difference quantities to be cleared in the Difference Analyzer.

Quantity Differences in Cross-Docking

Unlike the standard process for stock transfer orders, in the cross-docking process (CD process) you work with at least three decentralized warehouses: the source warehouse, the cross-docking warehouse (CD warehouse) and the target warehouse. These warehouses are managed by EWM . You transport goods from the source warehouse to the target warehouse via the CD warehouse.

In the cross-docking process, you record quantity differences in the same way as for stock transfer orders. When you record quantity differences, you can use the process code to decide whether these quantity differences are to be debited from or added to the previous warehouse (if exists) or the source warehouse. The following graphic illustrates these two options. For example, if you want to record a quantity shortage in CD warehouse 2, you can use process code PC2A to debit this quantity difference from CD warehouse 1, or process code PC2B to debit the source warehouse.



For more information about the cross-docking process, see Cross-Docking .

Result

You have posted a quantity difference for a stock transfer order or in the cross-docking process.

Delivery Split for Inbound Deliveries

Use

A delivery split can be necessary in various business processes, for example, if a planned inbound delivery arrives in two trucks instead of one. You can either totally delete a delivery item in the inbound delivery or you can split it by using Adjust Delivery Quantity and choosing a process code.

Prerequisites

- You have defined process codes that adjust both the delivery quantity and the transferred delivery quantity, that is, you have chosen the value 2 Adjustment of Delivery Quantity and Del. Quant. Transferred for the quantity adjustment.For more information, see the Implementation Guide (IMG) under Extended Warehouse Management > Cross-Process Settings > Delivery Processing > Process Codes > Define Process Codes .
- You defined process code profiles.If Extended Warehouse Management (EWM) is to use a default process code (for example, /002) for quantity changes, you have chosen a default process code in the Customizing settings.For more

information, see the IMG under Extended Warehouse Management > Cross-Process Settings > Delivery Processing > Process Codes > Maintain Process Code Profiles .

- You have defined the item types for the inbound deliveries in which you assigned the corresponding process code profiles.For more information, see the IMG under Extended Warehouse Management > Goods Receipt Process > Inbound Delivery > Manual Settings > Define Item Types for Inbound Delivery Process.
- Either you or the system can split delivery items or partial quantities into a new delivery. The prerequisite for this is that you have not performed a warehouse activity for this delivery item or partial quantity. For example, you have not posted the goods receipt (GR) yet.

Features

You can totally delete a delivery item or reduce the delivery quantity of a delivery item. EWM determines a process code according to your Customizing settings for delivery quantity changes. For example, while you are unloading a truck, you discover that delivery item 20 is missing. However, you know that this delivery item will arrive on another truck the next day. Therefore, you only want to post delivery item 20 the next day, which means that you require a delivery split.

i Note

You can book GR for partial quantities, meaning that you have only not yet carried out warehouse activities for the remaining quantity for which you are performing a delivery split.

If you choose the delivery item and manually change the delivery quantity in the user interface of the inbound delivery then, if necessary, EWM determines a default process code (for example, I002 Diff). Adjust Supplier, Document and Quantity .To access the user interface, on the SAP Easy Access Screen choose Extended Warehouse Management > Delivery Processing > Inbound Delivery > Maintain Inbound Delivery. .If necessary, the default process code has the effect that EWM automatically adjusts the delivery quantities.If you want to work without a default process code, choose Adjust Delivery Quantity for the selected delivery item and specify a process code.When you select the corresponding process code, EWM adjusts the delivery quantity and the transferred delivery quantity.EWM performs a delivery split automatically and creates anew inbound delivery that contains delivery items for the partial quantities that are still missing or for the totally deleted delivery item.

i Note

If you perform changes in an inbound delivery using a process code that automatically adjusts to the delivery quantity, you cannot undo these changes.

When EWM adjusts the delivery quantities, it takes the following quantities of the inbound delivery into account:

- the quantities of the open warehouse tasks
- the packed quantity
- the GR posted quantity
- the unloaded quantity

This allows EWM to prevent document inconsistencies. If, for example, a delivery item still has open warehouse tasks, you cannot set the quantity of this delivery item to zero.

You can process the newly created inbound delivery as in the normal GR process . You can then post the GR, for example.

You can check the automatic adjustment of the delivery quantity for a delivery item in the user interface of the inbound delivery in the Quantity field on the Document Flow tab page.

See Also:

Example

Example for a Manual Delivery Split

1. A delivery item with ten pieces is assigned to a warehouse task with seven pieces.
2. EWM adjusts the delivery quantity and the transferred delivery quantity in accordance with your Customizing settings.To do so, on the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Inbound Delivery** > **Maintain Inbound Delivery Notification** and then the **Delivery Quantity** pushbutton.You specify the process code **I002 Diff. Adjust Supplier, Document and Quantity** .EWM adjusts the delivery quantity and performs a delivery split.
    a. Since you have not performed any warehouse activities for the remaining quantity of the delivery item, EWM automatically adjusts the delivery quantity to seven pieces.
    b. You save the delivery quantity change.
    c. During the save, EWM creates a new inbound delivery for the remaining quantity of three that contains three pieces for every ten delivery items.

    In the original inbound delivery, you can check the changed quantity for the original delivery item at item level in the **Quantity** field on the **Document Flow** tab page. EWM displays the quantity of seven pieces. Additionally, on the **Process Codes** tab page, you can see (at item level) that EWM uses the process code **I002** for the difference quantity of three pieces.

Delivery Split for Outbound Deliveries

Use

You can perform a delivery split in various business processes. For example, you have completely picked an outbound delivery order in Extended Warehouse Management (EWM). You find that you cannot load the complete quantity. You can create an outbound delivery for individual delivery items and therefore perform an outbound delivery split. Either you or the system can split an outbound delivery order or an outbound delivery.

Prerequisites

Prerequisites for the Outbound Delivery Order Split Based on the Outbound Delivery Request

- You have performed the Customizing for route determination (see Special Features for Outbound Delivery Objects).

  If EWM finds various routes for different delivery items, it performs an outbound delivery order split.You can program a separate logic for route determination in the Business Add-In (BAdI) **BAdI: Route Determination and Route Check** that causes EWM to perform an outbound delivery order split.For more information, see the Implementation Guide (IMG) for EWM under **Business Add-Ins (BAdIs) for Extended Warehouse Management** > **Cross-Process Settings** > **Delivery Processing** > **Determinations in Delivery** > **BAdI: Route Determination and Route Check.**

Prerequisites for the Outbound Delivery Order Split

- You have defined process codes that adjust both the delivery quantity and the transferred delivery quantity, that is, you have chosen the value **2 Adjustment of Delivery Quantity and Del. Quant. Transferred** for the quantity adjustment.For more information, see the IMG for EWM under **Cross-Process Settings** > **Delivery Processing** > **Process Codes** > **Define Process Codes.**

- You defined process code profiles.If EWM is to use a default process code (for example, 0002) for quantity changes, you have chosen a default process code in the Customizing settings.For more information, see the IMG for EWM under Cross-Process Settings > Delivery Processing > Process Codes > Maintain Process Code Profiles.
- You have defined the item types for the outbound deliveries in which you assigned the corresponding process code profiles.For more information, see IMG for EWM under Goods Issue Process > Outbound Delivery > Manual Settings > Define Item Types for Outbound Delivery Process.
- Either you or the system can split delivery items or partial quantities into a new delivery. The prerequisite for this is that you have not performed a warehouse activity for this delivery item or partial quantity. For example, you have not created a warehouse task for picking yet.

Prerequisites for the Outbound Delivery Split

If the delivery item of the outbound delivery is relevant for picking, you have at least picked the delivery quantity that you want to split into a new subitem. You can only create new subitems for this picked quantity.

Features

Outbound Delivery Order Split Based on Outbound Delivery Request

EWM performs an outbound delivery order split on the basis of route determination. For example, an outbound delivery request consists of delivery items 10, 20, and 30. Delivery items 10 and 20 have route R1 and delivery item 30 has route R2. EWM creates an outbound delivery order for delivery items 10 and 20 and a second outbound delivery order for delivery item 30.

For more information, see Route Determination in the Delivery and Special Features for Outbound Delivery Objects.

Outbound Delivery Order Split

The same functions exist for the outbound delivery order split as for the Delivery Split for Inbound Deliveries .To do this, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Outbound Delivery > Maintain Outbound Delivery Order.

However, when EWM adjusts the delivery quantities, it takes different quantities into account than for the delivery split for inbound deliveries. EWM takes the following quantities of the outbound delivery order into account:

- The quantities of the open warehouse tasks
- The packed quantity
- The loaded quantity
- The aggregated quantity
- The picked quantity
- The number of serial numbers
- The catch weight quantity assignment

Outbound Delivery Split

If you want to perform an outbound delivery split, you split an outbound delivery order item. EWM creates corresponding subitems. You create outbound deliveries for the individual subitems.

You can split delivery items for both packed and unpacked products. EWM displays the subitems that have been created on the Items or HU tab page using + symbols in the Level column. You can also delete an outbound delivery split. To do so, you must first

delete the corresponding outbound delivery and then the subitem. You delete the whole subitem when deleting unpacked products. For packed products, you can delete the parts of a subitem that are packed in an HU.

When you save a newly created subitem, EWM posts the stock of the delivery item to the subitem, meaning that the stock is item based. You can post goods issue for every subitem.

The document flow always displays the quantity of delivery items that have not been split and does not display the corresponding subitems.

Activities

For a description of the activities for the outbound delivery order split, see Route Determination in the Delivery and Special Features for Outbound Delivery Objects . For a description of the activities for the outbound delivery order split, see Delivery Split for Inbound Deliveries.

Activities for the Outbound Delivery Split

On the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Outbound Delivery** > **Maintain Outbound Delivery Order.**

- If you want to split a delivery item for unpacked products, perform the following steps:
  1. Go to the **Items** tab page and select a delivery item.
  2. Choose the arrow on the right-hand side of **Subitem** and then choose **Split U** p Outbound Delivery.
  3. Specify the number of subitems that are to be created.

     EWM creates one or more new subitems for which you can specify the desired delivery quantity.
  4. Choose with the quick info **Save.**

     EWM saves the new subitem, thus splitting the delivery item.
  5. Choose with the quick info **Create Outbound Delivery**.

     EWM creates an outbound delivery for the subitem.
- If you want to split a delivery item for packed products, perform the following steps:
  1. Go to the **HU** tab page and select an HU item.
  2. Choose the arrow on the right-hand side of **Split Item** and then choose **Create**.

     EWM creates a new subitem for the HU, for which you can specify the desired delivery quantity.
  3. Choose with the quick info **Save.**

     EWM saves the new subitem, thus splitting the delivery item.
  4. Choose with the quick info **Create Outbound Delivery**.

     EWM creates an outbound delivery for the subitem.
- If you want to undo a delivery item split, perform the following steps:
  1. Go to the **Items** tab page.
  2. Delete the corresponding outbound delivery first by selecting the outbound delivery and choosing with quick info **Delete**.

- If you want to delete the subitem of an unpacked product, select the subitem and choose with the quick info **Delete**.

  EWM totally deletes the subitem.

- If you want to delete the subitem of a packed product, undo the HU split. Go to the **HU** tab page and select the subitem. Choose the arrow on the right hand side of **Split Item** and then choose **Undo**.

  EWM totally deletes the subitem.

Processing of Non-Picking-Relevant Delivery Items

Purpose

You can use the processing of non-picking-relevant delivery items to remove goods from your warehouse for a warehouse request of the type outbound delivery order . However, in contrast to picking, Extended Warehouse Management (EWM) does not generate any warehouse tasks for the warehouse request or any warehouse orders.

Prerequisites

- The goods movement bin in the warehouse request from which you are booking out the stock for the goods issue must contain a valid storage bin in which a sufficient quantity of the product is available.
- You use an item type that does not contain any of the following status types in the status profile:
  - **Planning Picking**
  - **Picking**

    For more information, see the Implementation Guide (IMG) for EWM under **Cross-Process Settings** > **Delivery Processing** > **Status Management** > **Define Status Profiles.**

Process

1. You or EWM create the warehouse request for an outbound delivery order. Either ERP distributes an outbound delivery to EWM and EWM automatically generates a corresponding warehouse request, or you manually generate the warehouse request in EWM as a direct outbound delivery order .

   You specify all the relevant stock attributes of the product (for example, the stock type) and a goods movement bin in the warehouse request.

   - If you have configured goods movement bin determination, EWM automatically suggests a goods movement bin. On the SAP Easy Accessscreen, choose **Extended Warehouse Management** > **Settings** > **Maintain Goods Movement Bin Determination.** EWM also uses the settings that you specify here to process inbound deliveries that are not relevant to putaway.
   - If you generate the warehouse request manually, choose one of the following options:
     - You specify the items with their stock attributes and quantities on the **Items** tab page.
     - On the **HU** tab page, you choose the **Create** and **Save** pushbuttons and specify a handling unit (HU) and an item type. Then EWM automatically creates the corresponding items with their stock attributes and quantities.
2. You remove the products from your warehouse and perform any other relevant goods issue steps, for example, loading for transportation.

3. You post the goods issue for the outbound delivery order.

Result

EWM has generated a warehouse request for an outbound delivery order. You have removed the products from your warehouse, posted the goods issue (for example in sell from stock) or performed additional goods issue activities, and have then posted goods issue.

Processing of Non-Putaway-Relevant Delivery Items

Purpose

You can use the processing of non-putaway-relevant delivery items to book goods for a warehouse request for an inbound delivery into your warehouse stock. However, in contrast to putaway , Extended Warehouse Management (EWM) does not generate any warehouse tasks for the warehouse request or any warehouse orders.

Prerequisites

- The goods movement bin in the warehouse request into which you are booking the stock for the goods receipt must contain a valid storage bin into which you can book the product.
- You use an item type that does not contain any of the following status types in the status profile:
  - **Planning Putaway**
  - **Putaway**

    For more information, see the Implementation Guide (IMG) for EWM underCross-Process Settings > Delivery Processing > Status Management > Define Status Profiles.

Process

1. You or EWM create the warehouse request of the type inbound delivery . Either SAP ERP distributes an inbound delivery to EWM and EWM automatically generates a corresponding warehouse request, or you manually generate the warehouse request in EWM.

   You specify all the relevant stock attributes of the product (for example, the stock type) and a goods movement bin in the warehouse request.

   If you have configured goods movement bin determination, EWM automatically suggests a goods movement bin. On the SAP Easy Accessscreen, choose **Extended Warehouse Management > Settings > Maintain Goods Movement Bin Determination.** EWM also uses the settings that you specify here to process outbound deliveries that are not relevant to picking.
2. You perform any other relevant goods receipt steps, for example, unloading.
3. You put away the products in your warehouse and you post goods receipt for the inbound delivery.

Result

EWM has generated a warehouse request for an inbound delivery. You have performed any additional goods receipt activities, you have put away the products in your warehouse, and you have posted goods receipt.

Processing of Packaging Items

Purpose

You can process packaging items to put away stock-relevant packaging materials or to remove them from stock. When putting away the packaging material, you use a warehouse request for an inbound delivery. When removing the packaging material from stock, you use a warehouse request for an outbound delivery order (see also Warehouse Req. - Type Inbound Del. or Outbound Del. Order).

If the delivery items are relevant to picking or to putaway, Extended Warehouse Management (EWM) creates warehouse tasks for the warehouse request and the corresponding warehouse orders. If not, EWM creates neither warehouse tasks for the warehouse request nor warehouse orders.

If packaging items already exist in an inbound or outbound delivery in SAP ERP, SAP ERP distributes them to EWM and EWM adds the packaging items to the inbound or outbound delivery.

Prerequisites

- The product of the packaging item is a packaging material. Check the settings for the packaging data in the global view of the product master. To do so, on the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Master Data** > **Product** > **Maintain Product** and then choose the **Global Data** setting in the **View** group box. Choose the **Pkg Data** tab page and display the relevant data.
- In the goods issue process, the goods movement bin in the warehouse request from which you are booking out the stock for the goods issue must contain a valid storage bin in which a sufficient quantity of the product is available.

Process

1. You or EWM create the warehouse request for an inbound delivery or an outbound delivery order. Either SAP ERP distributes an inbound or outbound delivery to EWM and EWM automatically generates a corresponding warehouse request, or you manually generate the warehouse request in EWM.

   You can manually generate packaging items for both the inbound delivery and the outbound delivery. EWM processes these to a large extent in the same way as normal items. Exceptions include the batch split, which is not possible in this case.

   When you manually generate the packaging items, you specify all the relevant stock attributes of the packaging material, for example, the stock type, in the warehouse request.

   If the delivery item is not relevant to picking, you specify a goods movement bin.

   If you have configured goods movement bin determination in Customizing for EWM, EWM automatically suggests a goods movement bin. On the SAP Easy Accessscreen, choose **Extended Warehouse Management** > **Settings** > **Maintain Goods Movement Bin Determination.EWM uses the settings that you specify here to process delivery items that are not relevant to putaway and delivery items that are not relevant to picking.**
2. If the delivery items are relevant to picking or to putaway, you generate the warehouse tasks for the warehouse request and confirm them.
3. You perform any other relevant goods receipt or goods issue steps. For example, you pick products and use the packaging material as pick-handling units.
4. You post goods receipt for the inbound delivery or goods issue for the outbound delivery order.

Result

EWM has generated the warehouse request for an inbound delivery or an outbound delivery order. You have either put away the packaging material in your warehouse and posted goods receipt, or you have removed it from your warehouse and posted goods issue.

Automatic Generation of Packaging Items for Goods Issue

Use

During delivery processing in a goods issue (GI) process, the system can automatically create packaging items and post GI for them. This process is warehouse-dependent.

The system generates the packaging items based on the handling units (HUs) of the delivery items when you create the outbound delivery (OD), as follows:

1. The system collects all delivery items for which an OD should be created. The system does this either directly, by creating the OD for certain items, or indirectly, by creating an OD for a HU or a transportation unit (TU) or by posting GI for a delivery, a HU, or a TU.
2. The system evaluates if the delivery type is relevant for the automatic generation of packaging items, based on the customizing.
3. If the delivery is relevant for automatic generation of packaging items, the system checks if the packaging material type of each individual HU is relevant for the automatic generation of packaging items. If it is relevant, the system counts this packaging material as one quantity with the basis unit of measure of the packaging material.

   The system performs the same tasks for the auxiliary products contained in one HU. If multiple HUs or auxiliary packaging materials with the same packaging material are contained in the items used for the OD creation and are relevant for packaging item creation, the system adds up the quantities.

   If the same packaging material is contained multiple times in the HUs, but with different units of measurement, the system adds the quantities for each unit of measurement separately. For example, if one HU contains the same packaging material in two units of measurement, once as eaches and once in a different unit of measurement, the system creates two packaging items.
4. The system creates a delivery item with the item type and stock type determined in the customizing. If the system cannot determine an item type, it does not create a delivery item.

   These delivery items are normal delivery packaging items. If SAP Extended Warehouse Management (SAP EWM) is integrated with SAP ERP, the integration is identical to manually created packaging items. The same applies to goods movement postings.
5. The system adds the automatically generated packaging items to the OD. In one OD, the system creates a maximum of one delivery item for each combination of packaging material and unit of measure of the HUs.

Prerequisites

- You have created an item type with item category `PAC`, for example, you have activated BC set `/SCWM/DLV_OUTBOUND_PACK`.
- You have made the settings for automatic generation of packaging items in Customizing for **Extended Warehouse Management**, under **Goods Issue Process** > **Outbound Delivery** > **Automatically Generate Packaging Items**.

  It is especially important that you have configured the correct item type and stock type.
- You have configured the following determinations, so that the system uses valid values:

  - Warehouse process type determination for the packaging items. For more information, see Customizing for **Extended Warehouse Management** under **Cross-Process Settings** > **Warehouse Task** > **Determine Warehouse Process Type**.
  - Goods movement bin determination, for example, fixed bin assignment. On the **SAP Easy Access** screen for Extended Warehouse Management, choose **Settings** > **Maintain Goods Movement Bin Determination**.
  - Determination that the GI is possible for the packaging material used for the automatically generated packaging items, for example, sufficient stock is available. For more information, see Goods Issue.
- If SAP EWM is integrated with SAP ERP, you have mapped the SAP ERP item type to the relevant item type in SAP EWM in Customizing for **Extended Warehouse Management** under **Interfaces** > **ERP Integration** > **Delivery Processing** > **Map Item Types from ERP System to EWM**.

Features

- Goods Issue

  If you post GI for the delivery, the system automatically posts GI for the automatically generated packaging items with the OD they belong to.

  Due to the automatic process, no manual steps, such as picking, packing, loading, or TU assignment, are relevant for the automatically generated packaging items.

- Automatic Replication to SAP ERP

  If SAP EWM is integrated with SAP ERP, SAP EWM replicates the automatically generated packaging items to SAP ERP. The automatically generated packaging items behave like manually created packaging items. In SAP ERP, you can use follow-up processes according to your settings in SAP ERP. For example, you can use the new packaging items when you create an invoice for your OD.

  > **ℹ Note**
  > In the case of stock transfer orders, the automatically generated packaging item on the outbound delivery order causes the system to create a corresponding new packaging item in the corresponding stock transfer order. In the follow on process, this new item is created in the inbound delivery. Depending on your inbound process for the inbound delivery, you have to ensure that this packaging item is handled correctly.

- Deletion

  If you delete the OD, any automatically generated packaging items are deleted automatically with the OD.

- TU Processing

  Packaging items generated in this process are not relevant for TU processing. This means that they are not shown on the TU, and they cannot be assigned to the TU manually. The system automatically processes them with the corresponding OD.

- Business Add-In to Prevent Repacking after OD Creation

  After the system generates automatic packaging items, the HUs that the automatically generated packaging items are based on should not be repacked. If they are repacked, the packaging situation and the packaging item information, like packaging material or quantities, might not match. Once the automatically generated packaging items are created, they are not changed, even if the HU is repacked.

  You can use the Business Add-In (BAdI) **Termination of Warehouse Task Creation** to prevent repacking after outbound delivery creation. For more information, see Customizing for **Extended Warehouse Management**, under **Business Add-Ins (BAdIs) for Extended Warehouse Management** > **Cross-Process Settings** > **Warehouse Task** > **Creation of Warehouse Tasks** > **BAdI: Termination of Warehouse Task Creation**.

- Cross Delivery Handling Units

Cross delivery handling units are handling units where items from different deliveries are packed in the same HU. In such cases, the system decides if a packaging item should be created, and if it should, for which delivery and with which quantity it should be created. SAP provides a basis default logic, which is implemented in fallback implementation of the BAdI **Modification of Data for Generated Packaging Items**. For more information, see **Business Add-Ins (BAdIs) for Extended Warehouse Management > Goods Issue Process > Outbound Delivery > Automatically Generated Packaging Items > BAdI: Modification of Data for Generated Packaging Items**.

Direct Outbound Delivery Order

Definition

Document that is triggered in the warehouse directly.

Extended Warehouse Management (EWM) sends the direct outbound delivery order to SAP ERP. The direct outbound delivery order contains the SAP ERP document number. The structure and data of the direct outbound delivery order match those of an outbound delivery order that is triggered in SAP ERP.

Use

Unlike EWM outbound delivery orders, for which SAP ERP generated a reference document previously, you or EWM can also generate direct outbound delivery orders directly and locally in EWM, which EWM then sends to SAP ERP.

You can use direct outbound delivery orders in the following scenarios:

- Pickup
  - Direct sales

    For direct sales, transportation activities do not apply, since the customer asks for the goods and picks them up on-site in the warehouse. You do not need to create a sales order for this. The direct outbound delivery order you generated manually in EWM forms the basis for the delivery in SAP ERP. The delivery in SAP ERP forms the outbound document for billing.

    You can also trigger direct sales in SAP ERP without generating a sales order.
  - Account assignment

    For account assignment, transportation activities also do not apply, since the internal or external customer asks for the goods and picks them up on-site in the warehouse. Unlike for direct sales, you post a goods issue with account assignment for the goods, for the cost center, for example.

    You can also trigger goods issue with account assignment in SAP ERP without generating a sales order.
- Scrapping

  The Direct Outbound Delivery Order for Scrapping forms the last step in the scrapping process. You post the goods to be scrapped with the goods issue from the warehouse stock. Whereas you can also trigger the preparation for scrapping - such as collecting goods in a scrapping container - using a posting change request from SAP ERP, you always trigger this last step locally in the warehouse.
- Immediate Delivery (Push or Pull Method)
- Kit to stock or reverse kitting

  For kit to stock, you use a value-added service orderto create a kit from different components. If you want to reserve the corresponding components, you can use a direct outbound delivery order. When you have created the kit and post goods

issue for the direct outbound delivery order, EWM reduces the stock of the respective components.

You can use the direct outbound delivery order to execute actions that you require for generating and processing an outbound delivery in the warehouse, such as picking. You can carry out the following for a direct outbound delivery order:

- Save
- Validate
- Archive
- Delete

Structure

For the structural description of the direct outbound delivery order, see Outbound Delivery Order .

The structure of a direct outbound delivery order also contains the following, differentiating it from the structure of an outbound delivery order:

- Document categories with different document types

  You can define the document types in Customizing for delivery processing.

  i Note

  For more information about defining document types in EWM, see the Implementation Guide (IMG) for EWM under Goods Issue Process > Outbound Delivery :

  - Define Document Types for Outbound Delivery Process Using Wizard
  - Manual Settings > Define Document Types for Outbound Delivery Process

- An item category with different item types

  You can define the item types in Customizing for delivery processing.

  i Note

  For more information about defining item types in EWM, see the Implementation Guide (IMG) for EWM under Goods Issue Process > Outbound Delivery :

  - Define Item Types for Outbound Delivery Process Using Wizard
  - Manual Settings > Define Item Types for Outbound Delivery Process

Direct outbound delivery orders have the following properties that differentiate them from outbound delivery orders that were generated in EWM based on outbound deliveries created in SAP ERP:

- Necessity of the availability check, as this check is usually triggered in SAP ERP

  To avoid endangering the continual ability to deliver, the planning situation must be adjusted with the availability check.

  i Note

  In SAP SCM 5.1, planning-related reservation of the products only takes place if you are using SAP Advanced Planning and Optimization (SAP APO). No reservation takes place for checks in SAP ERP. In exceptional cases, overconfirmation may occur.

- A lot of organizational and logistically relevant delivery data that is copied to outbound delivery orders from SAP ERP preceding documents must be specified for the direct outbound delivery orders locally in EWM. Either you enter the data directly, such as the product or quantity, or EWM determines the data automatically using configuration data, such as the goods issue bin for items that are not pick-relevant.

  As you start the goods issue process using direct outbound delivery orders in EWM, the underlying document type must display the process correspondingly. EWM uses the process profile to do this. When generating a direct outbound delivery order, you must select the correct document type to ensure the process flow is correct. For more information, see the IMG for EWM under **Cross-Process Settings** > **Delivery Processing** > **Process Management and Control.**

- As no outbound delivery request from SAP ERP exists for direct outbound delivery orders, actions such as splitting direct outbound delivery orders are not possible (see Delivery Split for Outbound Deliveries). Direct outbound delivery orders are also always checked, meaning they have the status type **Locked (Unchecked Item)** and status value **No** , and are immediately ready for processing.

- Direct outbound delivery orders support processing of non-pick-relevant items, for example, or generation of items from handling units.

Integration

- As direct outbound delivery orders are created in EWM directly, no availability check is performed in SAP ERP for the required product. Instead, EWM triggers the availability check. Depending on the configuration, the check can be performed in SAP ERP or in SAP APO. For outbound delivery orders replicated from SAP ERP, however, the availability check is triggered in SAP ERP.

  For more information, see Availability Check .

- EWM can send direct outbound delivery orders for scrapping to SAP ERP. For more information, see the Implementation Guide (IMG) for EWMunder **Interfaces** > **ERP Integration** > **General Settings** > **Set Control Parameters for ERP Version Control** .If you are not using a corresponding Enhancement Package, only the goods movement posting takes place. There is no delivery replication.

- When you generate a direct outbound delivery order, EWM automatically performs a Compliance Check .

  See also:

Processing of Direct Outbound Delivery Orders

Processing of Direct Outbound Delivery Orders

Use

If you want to use direct outbound delivery orders in any of the scenarios listed under Direct Outbound Delivery Order, on the **SAP Easy Access** screen choose **Extended Warehouse Management** > **Delivery Processing** > **Outbound Delivery** > **Maintain Outbound Delivery Order** and choose the function **Create**. Extended Warehouse Management (EWM) starts the standard goods issue process (see Goods Issue) and generates - if required - the necessary documents for billing and shipping.

Prerequisites

- If you are using a corresponding SAP ERP. Only then does EWM give information about direct outbound delivery orders and perform an availability check. For more information, see Customizing for **Extended Warehouse Management** under **Interfaces** > **ERP Integration** > **General Settings** > **Set Control Parameters for ERP Version Control**.

- You have assigned sales organizations. For more information, see Customizing for **Extended Warehouse Management** under **Extended Warehouse Management** > **Interfaces** > **GTS Integration** > **Assign Sales Organization to Warehouse**

Number ▶.

- If you want EWM to perform automatic determination of the goods movement bin for delivery items that are not pick-relevant, you have defined the goods movement bin determination. EWM defines the goods movement bin depending on the following:
  - Document type
  - Item type
  - Warehouse number
  - Combination of the two product master attributes product group type and product group

  On the SAP Easy Access screen, choose ▶Extended Warehouse Management ▶ Settings ▶ Maintain Goods Movement Bin Determination ▶.

Features

For direct outbound delivery orders, all the functions are available to you that you can perform for outbound deliveries. The following list describes the additional functions or functions that vary:

- You can also use direct outbound delivery orders for goods issue in the scrapping process (see Performing Scrapping).
- You can delete a direct outbound delivery order in EWM. EWM only sends a deletion message for direct outbound delivery orders to SAP ERP.
- You can add new delivery items with reference to a reference document to a direct outbound delivery order. EWM sends these added delivery items to SAP ERP, and SAP ERP copies these to the relevant SAP ERP outbound delivery. An availability check is also performed for the added items (see Availability Check).
- If you want to increase the outbound delivery quantity of an item manually, and SAP ERP has already generated an item for this in the SAP ERP outbound delivery, EWM does not perform another availability check for the difference quantity. When you generate a new item or direct outbound delivery order, EWM can perform an availability check for this quantity.
- If you want to reduce the outbound delivery quantity, you can do this directly in the direct outbound delivery order. In this case, no further availability check takes place. Only when SAP ERP copies the quantity change into the SAP ERP outbound delivery does SAP Advanced Planning and Optimization (SAP APO) adjust its planned quantity.
- When you generate a direct outbound delivery order from the user interface of the outbound delivery order, you can manually assign stock to a delivery item that does not have a picking status. Display your direct outbound delivery orders in the form view of the user interface of the outbound delivery order. Choose the input help for **Goods Mvt Bin** and look for a bin in which stock exists.

Constraints in SAP ERP

- You cannot use credit management for direct outbound delivery orders.
- The price determination in SAP ERP does not provide you with the same functions as the price determination in the sales order.
- You cannot use any configurable products in direct outbound delivery orders.
- You cannot perform billing in SAP Customer Relationship Management (SAP CRM) for direct outbound delivery orders.
- You cannot extend bills of materials in direct outbound delivery orders.
- SAP ERP does not send messages for a direct outbound delivery order to EWM.

Constraints in EWM

- If you are adding packing items to a direct outbound delivery order, EWM cannot perform an availability check for these new delivery items.
- The kit-to-order process does not support direct outbound delivery orders.
- Direct outbound delivery orders are not relevant for backorder processing in SAP APO.

Fulfilling Immediate Deliveries with Direct Outbound Delivery Orders

Purpose

You can fulfill immediate deliveries and choose between a push and pull procedure:

In the push procedure, you make a decision for the outbound delivery without a sales order from the current requirements of the warehouse. For example, you want to use your available transportation capacities as optimally as possible by making use of unused space in a vehicle. Or you have to consider the special requirements of the consumer products industry. For example, you want to distribute a very recent delivery containing fresh goods (lettuce) to outbound deliveries going to different stores.

In the pull procedure, the store or customer triggers the process. For example, you only notice later that an article in the last purchase order was not included. You call the warehouse and trigger immediate delivery, so that the goods still delivered on time.

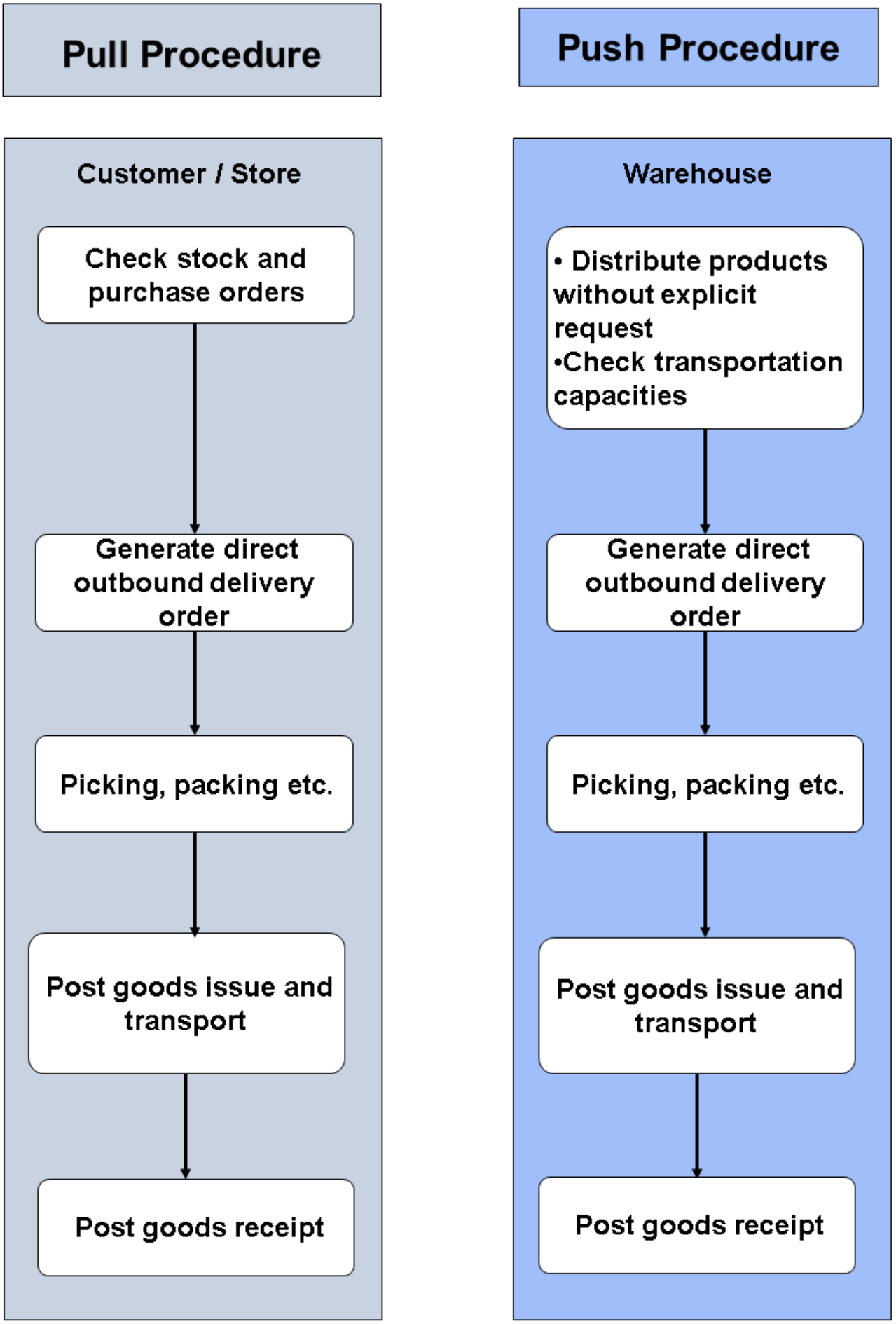
The process flow for the pull procedure is the same as that for direct sales. However, the outbound delivery for the transportation process is relevant here.

Prerequisites

For a description of the prerequisites, see Processing of Direct Outbound Delivery Orders .

Process

The following figure provides an overview of the process flow:


Pull Procedure
Customer / Store
Check stock and purchase orders
Generate direct outbound delivery order
Picking, packing etc.
Post goods issue and transport
Post goods receipt
Push Procedure
Warehouse
• Distribute products without explicit request
•Check transportation capacities
Generate direct outbound delivery order
Picking, packing etc.
Post goods issue and transport
Post goods receipt

The process starts when one of the following steps is triggered:

You want to distribute products to different stores, without an explicit request, or want to make use of available transportation capacities (push procedure).

The customer or store checks the stock or purchase orders and notices that stock or products are missing (pull procedure).

You then generate a direct outbound delivery order.

To do this, on the **SAP Easy Acces** s screen, choose **Extended Warehouse Management → Delivery Processing → Outbound Delivery → Maintain Outbound Delivery Order** and then function **Create** . Specify **ODSH ODO Direct Outbound Delivery** as the document type.

You continue performing the goods issue steps, such as picking and packing .

You post goods issue and transport the products.

The customer or store posts goods receipt .

Result

You have fulfilled an immediate delivery using a direct outbound delivery order.

Scrapping Using a Direct Outbound Delivery Order

Purpose

When scrapping products, you first transport them from a storage bin to the scrapping zone, and then transfer post the products to a scrapping stock type. You then post goods issue for the products that are to be scrapped. For this goods issue posting you use a direct outbound delivery order. Extended Warehouse Management (EWM) generates the direct outbound delivery order corresponding to the handling unit that you are using for the products being scrapped. You do not need to perform picking for this process.

For a descriptions of the prerequisites and of the process, see Scrapping .

Availability Check

Use

If you generate a direct outbound delivery order and have defined an availability check in Customizing, Extended Warehouse Management (EWM) automatically performs an availability check in one of the following ways, corresponding to your Customizing settings:

- Call availability check in SAP Advanced Planning and Optimization (SAP APO)
- Call availability information in SAP ERP
- Call availability check in EWM

Integration

- When SAP APO executes an availability check, SAP APO generates pre-persistent temporary quantity assignments (TQAs), which first reserve the quantities temporarily. As soon as EWM saves the associated outbound delivery order, SAP APO changes these pre-persistent TQAs to persistent TQAs. SAP ERP then later converts these persistent TQAs into reservations for the delivery quantities.
- If SAP APO is not available and is able to perform the availability check, EWM continues with the outbound delivery process. Here, EWM confirms the requested delivery quantity and sends the data to SAP ERP. SAP ERP sends this data to SAP APO as soon as SAP APO is available again. If backorder processing is required due to this current data, SAP APO triggers this.
- If you are using neither the availability check in SAP ERP nor the availability check in SAP APO, you can perform an availability check in EWM. However, this only works on the basis of the availability known in EWM, and only considers stocks that do not refer to a delivery.
- If you are using the availability check in SAP APO and have defined scheduling in Customizing for SAP APO, SAP APO also performs scheduling. EWM copies the results of the scheduling into the direct outbound delivery order. If you are not using SAP APO, EWM sets all relevant dates/times to the availability date/time (see section "Scheduling").

Prerequisites

- If you generate an item manually and want to perform an availability check, you must specify the product, quantity and stock type for the item. If you fail to specify a batch for a product that is subject to batch management, or if EWM is unable to determine one, EWM performs a batch-independent availability check. If you enter the batch at a later time, EWM performs the availability check again, but batch-independently.
- You have performed Customizing for the availability check in EWM. For more information, see the Implementation Guide (IMG) for **EWM** under **Interfaces** > **Availability Check** .
- If you want EWM to set the picking and loading dates, you have defined a corresponding data profile in Customizing for EWM, under **Cross-Process Settings** > **Delivery Processing** > **Dates/Times** . You have assigned this data profile to an item type in Customizing for EWM under **Goods Issue Process** > **Outbound Delivery** > **Manual Settings** > **Define Item Types for Outbound Delivery Process** .

Features

- If SAP APO, SAP ERP, or EWM are able to confirm all quantities and do not change the inspection type automatically, you can check this confirmation by using the status or the additional quantities. In the detailed view of the item, choose the tab page **Status** . For the status type **DAV Availability check** , the status value is **Confirmed** . In the detailed view of the item, choose the tab page **Additional Quantities** . EWM shows you the quantities in the role for the quantity. For example, you have requested 10 pieces of product A for a delivery item. The reduced quantityis 10 pieces, the open quantityis zero, and the requested quantityis 10 pieces.
- If SAP APO, SAP ERP, or EWM can only partially confirm the requested quantities or change the inspection type automatically, EWM displays a dialog box. Here you can find information such as:
    - Check status, for example **Partially Checked**
    - Item
    - Product
    - Stock type
    - Inspection type, for example **Availability Check in EWM**
    - Requested quantity
    - Available quantity
    - Confirmation date/time

For a partial confirmation, EWM also shows you the following:

- Reduced delivery quantity in the relevant item

  For example, you requested 10 pieces for an item but only 8 pieces are confirmed. EWM reduces the delivery quantity to 8 pieces.

- Status value of the checked items as **Confirmed**

  In the detailed view of the item, choose the tab page **Status** .

- Quantities in the role for quantities **Availability Check**

  In the detailed view for the item, choose the tab page **Additional Quantities** . For example, you have requested 10 pieces of product A for a delivery item. However, only 8 pieces were confirmed. The reduced quantity is 10 pieces, the open quantity is zero, and the requested quantity is 10 pieces.

- You can display the available stock for a product. In the view for the item on the user interface for the outbound delivery order, choose **Availability Information** .
- EWM does not perform an availability check in EWM for the following delivery items:
  - Delivery items that belong to a scrapping process
  - Packing items
- For the availability check in SAP APO or SAP ERP, EWM also considers the case where ERP carries a product in a different unit of measure to EWM. For example, you are using Catch Weight in EWM. In Catch Weight, you can work with multiple units of measure. In this case, EWM converts the delivery quantities into the unit of measure used in SAP ERP.

  If the delivery quantity is only partially confirmed, or if you call the availability information, EWM also shows you the delivery quantity in both units of measure in a dialog box.

Scheduling

EWM can set the following dates/times based on the results of the availability check:

- Dates/times at delivery header level:
  - Delivery date/time
- Dates/times at delivery item level:
  - Picking start and finish
  - Loading start and finish
  - Goods issue start and finish

EWM only sets the picking or loading dates/times if you have specified them in the corresponding date profile in Customizing for the item type used.

EWM performs scheduling based on the type of availability check it finds:

- If EWM finds **A Availability Check in SAP APO** as the type of availability check, and you have defined scheduling in Customizing for SAP APO, EWM copies the results of the scheduling into the direct outbound delivery order.
- If EWM finds a different type of availability check, the system sets the dates/times of the direct outbound delivery order - such as the delivery date and goods issue start and finish - to the confirmation date of the availability check.

For more information, see the Implementation Guide (IMG) for **EWM** under **Interfaces** > **Availability Check** .

Maintenance of Proof of Delivery

Use

For monitoring and analysis of outbound delivery items, you can use this function to create or delete proof-of-delivery data, which is imported from SAP ERP and enriched in SAP Extended Warehouse Management by warehouse activity data.

Integration

SAP ERP replicates outbound deliveries indicated as relevant for proof of delivery (PoD) to SAP EWM for further processing.

The BI system extracts the data sets on PoD relevant outbound deliveries for enabling KPI reporting in SAP EWM.

Prerequisites

- You have limited the number of users who are allowed to process the data import by having assigned authorizations for the authorization object `/SCWM/PIMP`. For more information, see Reducing Authorization Checks in Applications.
- You have implemented SAP Note 1625878 in SAP ERP for setting up PoD relevancy, which is also used to support the stock transfer order process.
- You have not archived outbound deliveries in EWM, thus enabling import of PoD entries at item level.
- Items in outbound deliveries are selected as relevant for proof of delivery in SAP EWM.
- Confirmed PoD differences exist in ERP for the corresponding outbound deliveries in ERP.

Features

You can use this function in the background to regularly create proof-of-delivery (PoD) data.

PoD Data Creation

- Selecting outbound delivery items

  The system selects PoD relevant outbound delivery items according to criteria set by the user.

- Importing data

  The system imports PoD differences reported in SAP ERP.

  Before importing data from SAP ERP, you must define a specific RFC destination. In Customizing, see **Assign RFC Destinations to Logical Systems** under **Extended Warehouse Management** > **Interfaces** > **ERP Integration** > **General Settings** > **Assign RFC Destinations to Logical Systems.**.

  By implementing the BAdI `/SCWM/EX_POD_SELECT`, you can replace the standard PoD data import with your own PoD data import.

- Enriching data

  The system determines warehouse activities and tasks performed on selected outbound delivery items and updates the PoD data.

  By implementing the BAdI `/SCWM/EX_POD_ENRICH`, you can replace the standard determination of relevant warehouse tasks with your own determination.

PoD Data Deletion

The system selects and deletes relevant PoD data according to criteria set by the user.

The deletion of PoD data is not linked to delivery archiving because an analysis should be possible even if the delivery is archived.

PoD Data Test

The system creates an application log but does not make any changes to the database.

To activate the application log, on the SAP Easy Access screen choose **SCM Extended Warehouse Management** > **Extended Warehouse Management** > **Settings** > **Activate Application Log.**.

PoD Data in BI

For transfer of PoD transaction data from EWM into BI, use data source `0WM_POD`.

You can use the extracted PoD data with, for example, extracted data on outbound delivery items (data source `0WM_DLVI_OUT`) to build key performance indicators, such as **Ratio Outbound Delivery Errors** and monitor them in the warehouse cockpit. To build the key performance indicator, you must make sure that the PoD data in BI corresponds to the outbound delivery.

See Data Sources for more information.

Activities

To maintain proof-of-delivery data for relevant outbound deliveries, on the **SAP Easy Access** screen choose **Extended Warehouse Management** > **Interfaces** > **ERP Integration** > **Maintain Proof-of-Delivery Data.**

The system processes creation or deletion of proof-of-delivery data based on the selection criteria of at least warehouse number and a goods issue date interval.

After execution, the system displays a message log, which you can use to analyze the changes made. Depending on the application log settings, the message log is also saved to the database. Using transaction code `SLG1`, you can display on the **Analyze Application Log** screen the message log for the object `/SCWM/WME` and sub-object `PoD`.

Example

If you want to use this function to create PoD data in the background on a regular basis, such as weekly, then create a variant with a selection variable for the goods issue data.

For example, give customers 7 days to report incorrect deliveries and if you confirm the PoD based delivery in the same timeframe, you can use the variable current date - xxx, current date + yyy. Enter `14` as xxx value and `7-` as yyy value to import the confirmed PoD data from the previous week.

More Information

For more information, see:

- Proof of Delivery in EWM
- Proof of delivery in SAP ERP, in SAP Library for SAP ERP under **SAP ERP** > **SAP ERP Central Component** > **Sales and Distribution (SD)** > **Shipping (LE-SHP)** > **Proof of Delivery**.

Notification of Expected Goods Receipt

Definition

A document in the inbound delivery process that copies all relevant logistics data from the initiator of the inbound delivery process, such as data of a purchase order or a manufacturing order.

An expected goods receipt is generated out of a notification of expected goods receipt.

Use

You use notification of expected goods receipt to save logistics data transferred from an open purchase order or manufacturing order, and to generate an expected goods receipt from it.

For more information, see Using Expected Goods Receipts.

Integration

The execution of the actions depends on the settings in the post processing framework. For more information, see Post Processing Framework in Delivery Processing.

The following configuration exists in the standard system:

- The delivery process receives a message from a reference document with all the relevant logistics data. SAP Extended Warehouse Management (SAP EWM) automatically generates a notification of expected goods receipt from this message. In the notification, SAP EWM saves the information contained in the message.
- Changing an existing expected goods receipt:

If an inactive notification (inactive version) exists in SAP EWM, you can generate a new expected goods receipt by activating the inactive notification manually.

Expected Goods Receipt

Definition

Document containing the data of an expected goods receipt.

The expected goods receipt contains data of an open purchase order or open manufacturing order. It represents a template that enables you to copy data from a purchase order or production order when creating a delivery manually.

Use

You can copy or delete an expected goods receipt.

For more information, see Using Expected Goods Receipts.

Integration

After SAP Extended Warehouse Management (SAP EWM) has activated the notification of expected goods receipt, it generates the expected goods receipt. In doing so, SAP EWM copies all relevant data from the notification into the expected goods receipt.

For more information about notification of expected goods receipt, see Notification of Expected Goods Receipt.

Using Expected Goods Receipts

Use

You do not want to use an advanced shipping notification in one of the following instances:

- For a purchase order or purchase order item
- For a manufacturing order or manufacturing order item

In this case, you can use the expected goods receipt to copy data from a purchase order, manufacturing order, or corresponding item to an inbound delivery.

You can also copy single delivery items from an expected goods receipt to an existing inbound delivery. For example, a vendor still has space on his vehicle, and sends you more products than he or she has specified in the advanced shipping notification.

Integration

You usually schedule report **Generate and Delete Expected Goods Receipts** `/SCWM/ERP_DLV_DELETE` so that SAP Extended Warehouse Management (SAP EWM) requests new expected goods receipts from SAP ERP. You use these expected goods receipts as templates to generate an inbound delivery in SAP EWM manually.

In exceptional circumstances, you can also use report `/SPE/INB_EGR_CREATE` in SAP ERP to generate an expected goods receipt. SAP ERP informs SAP EWM about this, and SAP EWM generates an expected goods receipt as well.

Prerequisites

- You have scheduled report Generate and Delete Expected Goods Receipts (`/SCWM/ERP_DLV_DELETE`). This report deletes the existing expected goods receipts in SAP EWM and requests new expected goods receipts from SAP ERP for a time period that must be entered.
- If you want to work with expected goods receipts, you have allowed manual generation of single delivery items in Customizing for **Extended Warehouse Management**, under **Goods Receipt Process** > **Inbound Delivery** > **Manual Settings** > **Define Item Types for the Inbound Delivery Process**.
- You have made the required settings for the expected goods receipt in Customizing for **Extended Warehouse Management** under **Goods Receipt Process** > **Expected Goods Receipt**:
  - **Use Wizard to Define Document Types for Expected Goods Receipt**
  - **Use Wizard to Define Item Types for Expected Goods Receipt**
  - **Define Allowed Item Types for Expected Goods Receipt**
  - **Define Document Type Determination for Expected Goods Receipt**
  - **Define Item Type Determination for Expected Goods Receipt**
- In Customizing for **Extended Warehouse Management**, you have defined the document type determination and item type determination for the inbound delivery that SAP EWM is to create under **Goods Receipt Process** > **Inbound Delivery**:
  - **Define Document Type Determination for Inbound Delivery Process**
  - **Define Item Type Determination for Inbound Delivery Process**
- You have defined the required profiles for the document type and item type in Customizing for **Extended Warehouse Management**, under **Cross-Process Settings** > **Delivery Processing**.

- You have defined the integration with SAP ERP for expected goods receipt in Customizing for Extended Warehouse Management under Interfaces > ERP Integration > Delivery Processing >.

  i Note

  You can use BC Set `/SCWM/DLV_EXPGR` Delivery Processing in EWM - Expected Goods Receipt to make the Customizing settings.

- You have authorization for the activity for creating inbound deliveries from an expected goods receipt for authorization object EWM Warehouse Request, or general authorization for creating inbound deliveries manually. On the SAP Easy Access screen, choose Tools > Administration > User Maintenance > Role Administration > Roles >, and check the settings for your role.

Features

If you are not using an advanced shipping notification in the purchase order for your delivered products, the following options are available to you for generating an inbound delivery:

- You generate an inbound delivery in SAP ERP. You send this to SAP EWM, so that you can execute further goods receipt process steps.
- You generate an expected goods receipt as a template document locally in SAP EWM, which you can use to generate an inbound delivery manually.
- You switch to SAP ERP from the user interface in SAP EWM to generate an inbound delivery. You use Business Add-Ins (BAdIs) `/SCWM/EX_DLV_ERP_CALL` and `/SCWM/EX_DLV_ERP_CALL_CTRL` to define whether you generate an inbound delivery in SAP ERP or locally in SAP EWM. On the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Inbound Delivery > Maintain Inbound Delivery >.
- You choose the user interface for the inbound delivery in SAP EWM, and specify the required delivery data without SAP ERP checking this data with the data of the open purchase order.

If you are not using any inbound deliveries in the production for your delivered products, the following options are available to you for generating an inbound delivery:

- You generate an expected goods receipt as a template document locally in SAP EWM, which you can use to generate an inbound delivery manually.
- In SAP EWM, you choose the user interface for the inbound delivery, and specify the required delivery data.

If you are using an expected goods receipt to generate an inbound delivery manually in the purchase order or manufacturing order, this has the following advantages:

- You can execute the goods receipt process steps in one system only, namely SAP EWM.
- If SAP ERP is temporarily unavailable, you can still execute the goods receipt process steps.
- You generate an inbound delivery whose data has been checked by both SAP ERP and SAP EWM. This enables you to make sure that you post goods receipt in SAP ERP as well.
- As the expected goods receipt is based on the data of the open purchase order or open manufacturing order, you can use the expected goods receipt as a preview of the goods receipt.

Actions for Expected Goods Receipts

- You can generate or delete an expected goods receipt.

An expected goods receipt is a temporary document. When SAP EWM no longer requires it, the system deletes the expected goods receipt. You cannot archive the expected goods receipt. When SAP EWM deletes an expected goods receipt, the system makes SAP ERP redefine the expected goods receipts that are still open.

As the expected goods receipt only acts as a template for copying existing data, you cannot change any of this data. In the related inbound delivery, however, you are able to change the data as in an inbound delivery with a preceding document, such as changing the quantity.

If you want to delete one or more expected goods receipts, use report Generate and Delete Expected Goods Receipts (`/SCWM/ERP_DLV_DELETE`).

ℹ Note

SAP EWM does not update the expected goods receipt and does not offset the quantity for the expected goods receipt. In addition, SAP EWM does not show you the expected goods receipt in the document flow of the inbound delivery.

- SAP EWM uses the delivery items of the expected goods receipt to calculate the expected workload (see Goods Receipt Workload).
- In the **Manual** indicator in the header data of the inbound delivery, if the value is set to **Y** for **Generated Manually with Predecessor**, this tells you whether SAP EWM has generated an inbound delivery from an expected goods receipt.

More Information

Generate Inbound Delivery from Expected Goods Receipt

Generate Inbound Delivery Item from Expected Goods Receipt

Generate and Delete Expected Goods Receipts

Use

You can use report **Generate and Delete Expected Goods Receipts** `/SCWM/ERP_DLV_DELETE` to request expected goods receipts for a purchase order or for a manufacturing order, including expected goods receipts for co-products and by-products, from SAP ERP.

You have the following options when executing the report:

- Select possible expected goods receipts in SAP Extended Warehouse Management (SAP EWM)
- Delete expected goods receipts in SAP EWM
- Create expected goods receipts in SAP EWM
- Delete and create expected goods receipts in SAP EWM

Integration

If you want to create or delete expected goods receipts from SAP ERP, you can use report `/SPE/INB_EGR_CREATE` in SAP ERP. This SAP ERP report calls RFC-enabled function modules in SAP EWM, which then delete the expected goods receipts in SAP ERP.

If you want to view the log of the RFC-enabled function modules, choose **Logs in this report in EWM**.

Prerequisites

If you want to use this report to request expected goods receipts from SAP ERP, the following prerequisites must be fulfilled:

- If you are using a corresponding SAP ERP. For more information, see the Customizing for **Extended Warehouse Management** under **Interfaces** > **ERP Integration** > **General Settings** > **Set Control Parameters for ERP Version Control**.
- You are using a logical system that represents the target system, enabling SAP ERP to send the found expected goods receipts to SAP EWM. If SAP EWM is to perform sending using a remote function call (RFC), you have defined an RFC destination for the target system. On the **SAP Easy Access** screen, choose **Tools** > **Administration** > **Administration** > **Network** > **RFC Destinations**.
- You have defined parallel processing in the delivery, in Customizing for **Extended Warehouse Management**, under **Cross-Process Settings** > **Delivery Processing** > **General Settings** > **Parallel Processing in the Delivery**.

Activities

On the **SAP Easy Access** screen, choose **Tools** > **ABAP Workbench** > **Development** > **Function Builder**, and schedule the report as a background job.

Generate Inbound Delivery from Expected Goods Receipt

Use

In Extended Warehouse Management (EWM), you can generate an inbound delivery from an expected goods receipt.

Example

You order products from an external vendor and generate a purchase order in SAP ERP. SAP ERP generates a corresponding expected goods receipt. The vendor's vehicle arrives at the warehouse, and you generate a corresponding inbound delivery in EWM, based on the shipping documents.

Prerequisites

See "Prerequisites" under Using Expected Goods Receipts.

Procedure

1. On the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Inbound Delivery** > **GR Preparation - External Procurement.**
2. Choose the icon with the quick info **Default Values**, and check whether **Select Expected GR** has been set, enabling EWM to search for expected goods receipts.
3. Look for the expected goods receipt that belongs to your inbound delivery, and generate an inbound delivery from an expected goods receipt. To do this, proceed as follows:
    - Group box **Transportation Unit**

      Set **Without TU**. If you are using a transportation unit (TU), enter a valid TU, a carrier, and the Standard Carrier Alpha Code (SCAC) for the partner. If the specified TU is unique, specifying the carrier and the SCAC is optional.
    - Group box **Search Requests**

a. Under **Find**, choose either **Purchase Order** or **Expected Goods Receipt**, and specify the purchase order number or document number of the expected goods receipt.

b. Choose the function **Search**.

EWM displays all the expected goods receipts it finds in the lefthand area of the screen.

c. In the lefthand screen area, set **Flag** for a delivery item in the expected goods receipt.

d. Choose the function **Copy Flagged Item**.

EWM generates a new inbound delivery from the expected goods receipt, and copies all data from the selected delivery item of the expected goods receipt. EWM also displays a dialog box containing header data of the new inbound delivery. If you want to copy multiple delivery items at the same time, you must copy the delivery items of the expected goods receipt in exactly one inbound delivery. You can edit the header data, specifying the advanced shipping notification, for example.

For products with a preferred alternative unit of measure, the quantity for the inbound delivery item is proposed using this UoM, if the value is a whole number (without decimal places).

EWM displays the new delivery items in the righthand screen area.

4. Choose **Save** and save the new inbound delivery.

i Note

On the inbound delivery user interface in SAP ERP, you can check whether EWM has sent the new inbound delivery to SAP ERP.

Result

You have generated an inbound delivery locally in EWM from an expected goods receipt, and can now perform further goods receipt steps on the user interface of the inbound delivery, such as posting goods receipt.

Generate Inbound Delivery Item from Expected Goods Receipt

Use

In Extended Warehouse Management (EWM), you can add additional delivery items to an existing inbound delivery by copying them from the expected goods receipt.

Example

You order products from an external vendor. The vendor sends an advanced shipping notification for these products. However, the vendor also sends other products in the same inbound delivery, which were not included in the advanced shipping notification. The products are requested for the same day, and a corresponding purchase order and expected goods receipt are present. You add further delivery items for the additional products to the existing inbound delivery by copying these delivery items from the expected goods receipt.

Prerequisites

See "Prerequisites" under Using Expected Goods Receipts .

Procedure

1. On the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Inbound Delivery > GR Preparation - External Procurement.

2. Choose with the quick info Default Values , and check whether Select Expected GR and Sel. Inb. Deliveries have been set, enabling EWM to search for expected goods receipts and inbound deliveries.

3. First look for the inbound delivery you require, for example by using the document number of the inbound delivery to which you want to add a delivery item from an expected goods receipt. In a second search, add the expected goods receipts to the result of the first search, for example using the document number of the purchase order. Choose the delivery item that you want to add to your inbound delivery.

   - Group box Transportation Unit

     Set Without TU . If you are using a transportation unit (TU), enter a valid TU, a carrier, and the Standard Carrier Alpha Code (SCAC) for the partner. If the specified TU is unique, specifying the carrier and the SCAC is optional.

   - Group box Search Requests

     Look for the inbound delivery you require.

     a. Select the Inbound Delivery under Find and specify the document number of the inbound delivery.

     b. Choose with the quick info Search .

        EWM displays a dialog box in which you can choose No to cancel the unspecific search query by expected goods receipts, and can continue by searching by inbound delivery. EWM displays all the delivery items it finds in the righthand screen area.

        i Note

        EWM only displays delivery items whose goods receipt status and warehouse activity status are set to Not Started .

        Searching by expected goods receipts:

     c. Select the Purchase Order under Find and specify the document number of the purchase order.

     d. Choose with the q uick info Add Search Result .

        EWM displays the expected goods receipts for the purchase order in the lefthand screen area, and in addition to the delivery items of the first search also displays delivery items of existing inbound deliveries for this purchase order in the righthand screen area.

     e. In the lefthand screen area, set Flag for a delivery item in the expected goods receipt.

     f. Choose to the right of the function with quick info Copy Flagged Item , and choose Add Flagged Item to Inbound Delivery .

        EWM displays a dialog box containing numerous document numbers of the two search queries.

     g. Select the document number you require by choosing Copy .

        EWM copies all data from the selected delivery item to the new inbound delivery item, and also displays it in the righthand screen area along with existing inbound delivery items for this purchase order.

Choose Save and save the new inbound delivery item.

Result

You have generated a delivery item locally in EWM from an expected goods receipt.

Goods Receipt Posting

Use

If you receive products delivered in a goods receipt process, you must post a goods receipt for the inbound delivery. If all necessary data is available in the inbound delivery, for example product, quantity, and supplier, you can post the goods receipt directly in the inbound delivery.

Features

You can either post the goods receipt for the entire inbound delivery, or a partial goods receipt only. If you confirm one or more warehouse tasks, carry out a goods receipt posting for inbound deliveries that are relevant to goods movement. The following procedures are available to you:

- You create and confirm multiple warehouse tasks.
- You confirm a warehouse task in steps, meaning that you post only a portion of the items or quantities.
- You can display individual partial goods receipts in the document flow for the inbound delivery using the relevant number of existing product documents.

  If you work with handling units, you may post goods receipts only for complete HUs. SAP Extended Warehouse Management (SAP EWM) then posts the delivery quantity corresponding to the product quantity specified in the handling unit.

Activities

If you want to post a goods receipt for an inbound delivery, on the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Inbound Delivery** > **Maintain Inbound Delivery**. Search for the inbound delivery and choose **Goods Receipt**.

See also the following topics:

- Goods Receipt Posting from External Procurement
- Goods Receipt Posting from Production

Goods Receipt Posting from External Procurement

Use

If you order products from an external reference source, for example from a supplier, you must execute the good receipt posting from external procurement. The goods receipt process from external procurement can have a different process flow depending on, for example, whether or not there is an advanced shipping notification (ASN).

Prerequisites

If you are working without ASNs but with expected goods receipts, you have carried out the prerequisites for expected goods receipts (see Using Expected Goods Receipts).

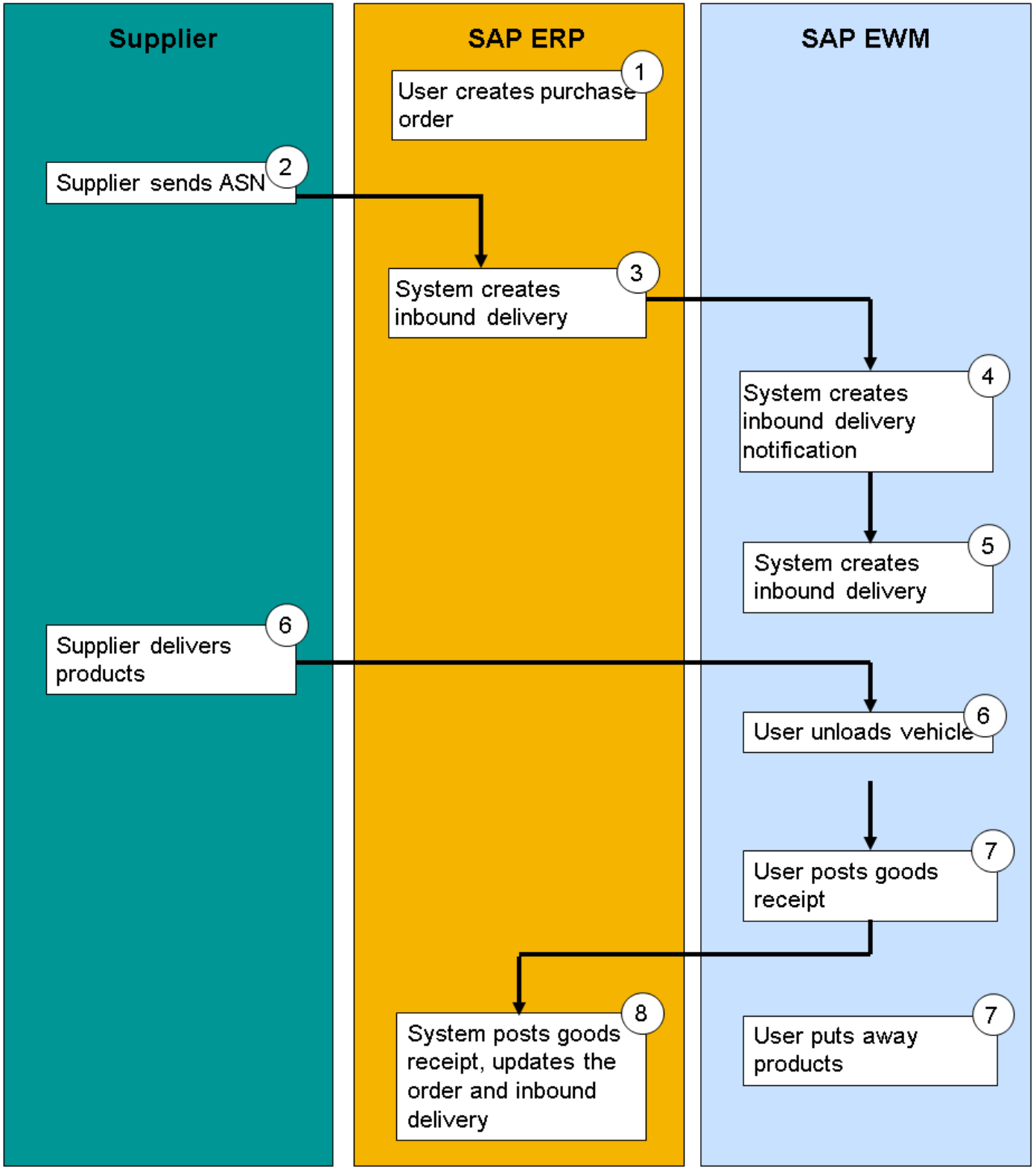
Process

Normally, you receive the ASN from the supplier, and SAP ERP creates the corresponding inbound delivery automatically. Goods receipt posting from external procurement can also take place without an ASN. You can either use an expected goods receipt, or you can create the inbound delivery manually in SAP ERP or in SAP Extended Warehouse Management (SAP EWM).

The following is an example of a goods receipt posting from external procurement in which you use an inbound delivery in SAP ERP, and for which you receive an ASN from the supplier. Afterwards, another example shows a situation in which there is no ASN and no inbound delivery in SAP ERP.

Goods Receipt Posting with Inbound Delivery in SAP ERP and ASN

The following figure provides an overview of the process flow:



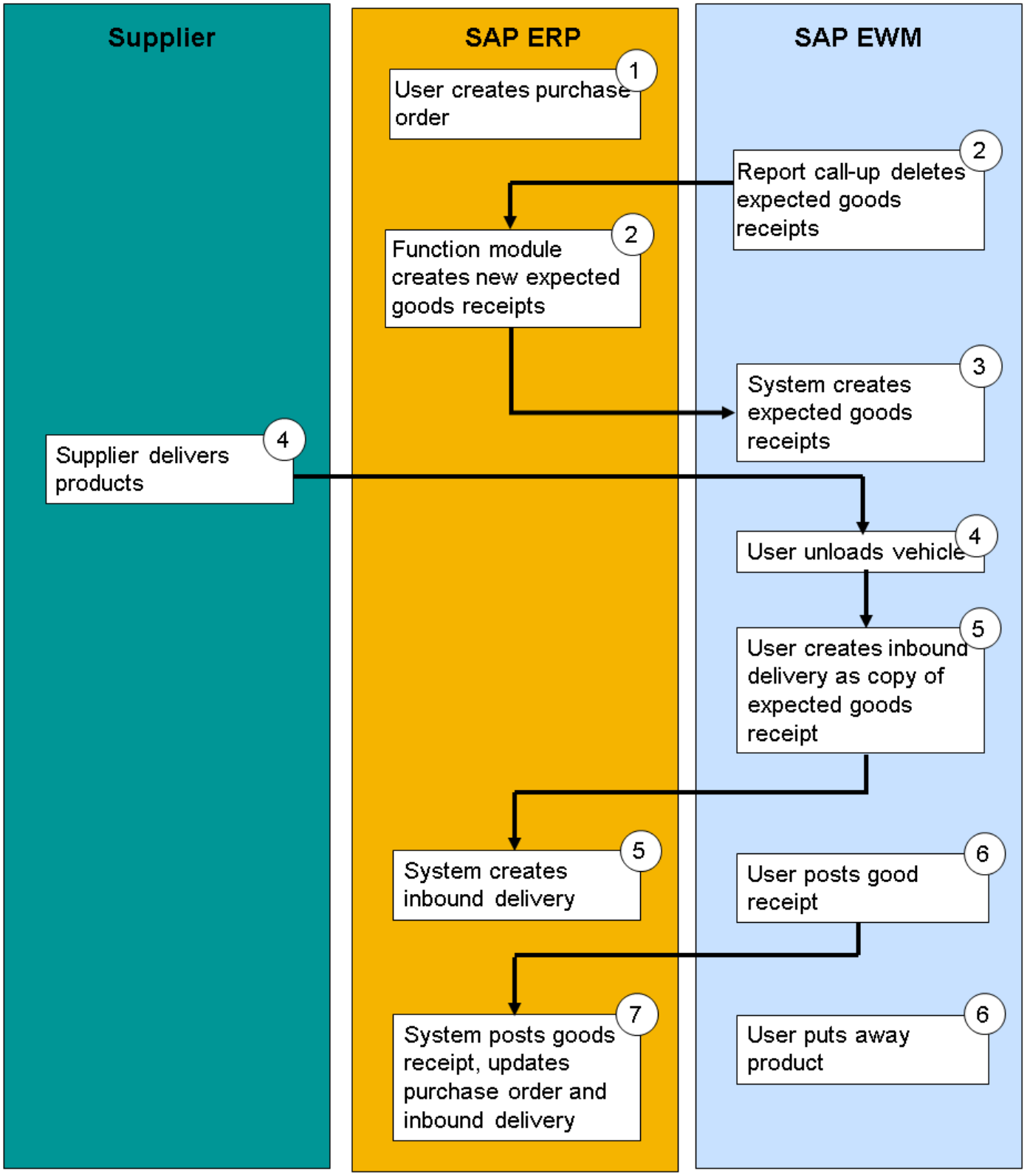
1. You order products from the supplier and create the corresponding purchase order in SAP ERP.
2. The supplier sends an ASN for the purchase order.
3. SAP ERP creates a corresponding inbound delivery and sends it to SAP EWM.
4. SAP EWM generates a corresponding inbound delivery notification.
5. SAP EWM automatically creates an inbound delivery from the active inbound delivery notification.
6. The supplier delivers the products. You unload the vehicle and transport the finished products into the warehouse, for example into the goods receipt area.

7. You post the goods receipt for the entire inbound delivery, and put away the products.

8. SAP EWM transmits the goods receipt posting to SAP ERP. SAP ERP also posts goods receipt for the related inbound delivery. SAP ERP also updates the stock in the purchase order and in the inbound delivery, due to the goods movement.

Goods Receipt Posting Without ASN but with Expected Goods Receipt

The following figure provides an overview of the process flow:



1. You order products from the supplier and create the corresponding purchase order in SAP ERP.
2. The supplier does not send an ASN for the open purchase order. Since SAP ERP has not yet generated an inbound delivery, you work with an expected goods receipt in SAP EWM.

   SAP EWM deletes an existing expected goods receipt in SAP EWM with a report, and calls a function module in SAP ERP. Using this function module, SAP ERP checks whether relevant purchase orders exist, for which you still have to post goods receipt. SAP ERP identifies the relevance from your selection criteria and from the open quantities. If open quantities still exist, the goods receipt indicator is set in SAP ERP but the final delivery indicator is not set.
3. For each relevant purchase order, SAP EWM receives a notification for the expected goods receipt from SAP ERP, and generates a corresponding expected goods receipt in SAP EWM.

   The expected goods receipt contains a link to the purchase order in SAP ERP. You can use the expected goods receipt in SAP EWM for planning purposes. It contains the planned times and quantities, for example.
4. The supplier delivers the products. You unload the vehicle and transport the finished products into the warehouse, for example into the goods receipt area.
5. You can use the number of the purchase order or product to find the expected goods receipt, and generate a corresponding inbound delivery in SAP EWM. For this, you use the expected goods receipt as a template and, if necessary, you can adjust the quantities. For more information, see Generate Inbound Delivery from Expected Goods Receipt.

   SAP EWM transfers the inbound delivery to SAP ERP, which generates a corresponding inbound delivery.

   If the item type allows the use of stock-specific units or measure (UoMs), you can also change the UoM.
6. You post the goods receipt and put away the products.
7. SAP EWM transmits the goods receipt posting to SAP ERP. SAP ERP posts the goods receipt. SAP ERP also updates the stock in the purchase order and in the inbound delivery, due to the goods movement.

More Information

Goods Receipt Optimization

Goods Receipt Posting from Production

Use

In SAP Extended Warehouse Management (SAP EWM), you can post goods receipt for your finished products.

Features

If you want to post goods receipt from production, you have the following options:

- Goods Receipt Posting Triggered by SAP ERP
- Goods Receipt Posting Triggered by EWM

In both instances, SAP EWM generates an inbound delivery for your finished products, for which you post goods receipt.

More Information

Goods Receipt Optimization

Goods Receipt Posting Triggered by SAP ERP

Use

You want to post goods receipt for products from production. For this, you work in SAP Extended Warehouse Management (SAP EWM) with inbound deliveries that SAP ERP generates parallel to production. For more information, see Goods Receipt Posting.

When you confirm the last production step for a product in the manufacturing order, meaning a production or process order, SAP ERP generates an inbound delivery instead of a material document. The inbound delivery provides you with information about the material and quantity currently being produced.

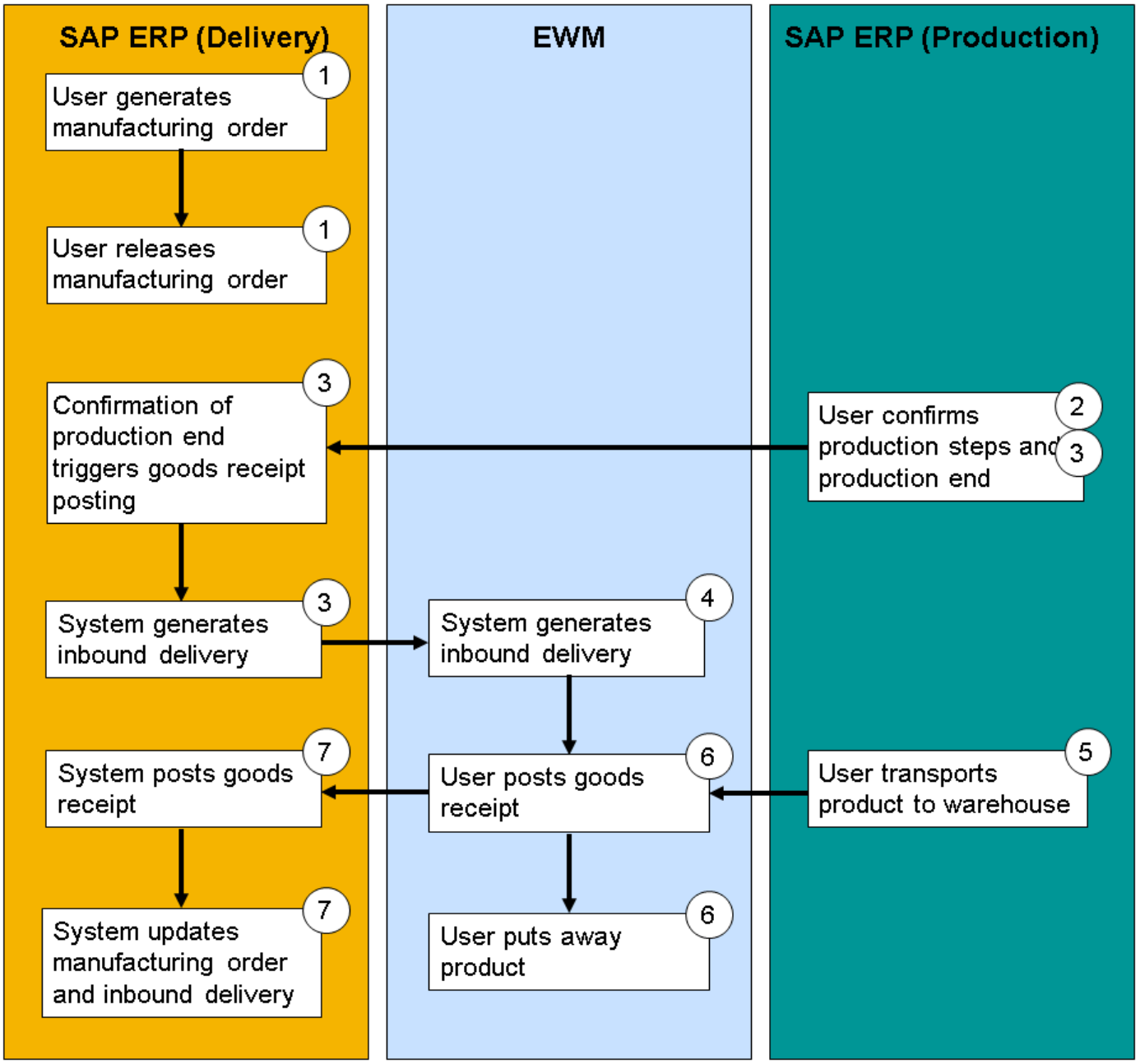
SAP EWM uses document type **IDPD Inbound Delivery from Production** to differentiate that an SAP EWM inbound delivery from production takes place.

Prerequisites

- You are working with SAP SCM and SAP ERP. For more information, see Customizing for **Extended Warehouse Management** under **Interfaces** > **ERP Integration** > **General Settings** > **Set Control Parameters for ERP Version Control**.
- You are using simple transport routes, for example without stock transport orders, so that your goods receipt posting in SAP ERP directly triggers generation of the SAP EWM inbound delivery.

Process

The following figure provides an overview of the process flow:



1. You generate a manufacturing order, namely a production or process order, and release it.

   The manufacturing order contains a plant or storage location combination that refers to a warehouse in which you are using SAP EWM. In other words, SAP EWM manages the plant and storage location combination.

2. You confirm the end of the designated production steps in the manufacturing order. In doing this, SAP ERP updates the finished quantity and marks the completed production steps.

3. When you confirm the last production step for a product in the manufacturing order, SAP ERP generates an SAP ERP inbound delivery and transmits it to SAP EWM. Prerequisite for this is that you have made corresponding settings in Customizing for SAP ERP so that SAP ERP **automatically** executes order confirmation for the goods receipt. This enables the confirmation of the last production step for a product to trigger goods receipt in SAP ERP.

   For more information, see:

   - Customizing for SAP ERP:

     - Production > Production Control > Master Data > Define Production Scheduling Profile

     - Production > Basic Data > Routing > Operation Data > Define Control Key

- SAP Help Portal, under SAP ERP > SAP ERP Central Component > Logistics > Logistics Execution (LE) > Goods Receipt Process for Inbound Deliveries > Goods Receipt > Goods Receipt Posting > Posting Partial Goods Receipts for Inbound Deliveries
- SAP Help Portal, under SAP ERP > SAP ERP Central Component > Logistics > Production Planning and Control (PP) > Production Orders (PP-SFC) > Order Release
- SAP Help Portal, under SAP ERP > SAP ERP Central Component > Logistics > Production Planning and Control (PP) > Production Orders (PP-SFC) > Goods Movement > Automatic Goods Receipt

Once you have released the manufacturing order you can also generate an SAP ERP inbound delivery **manually**. For more information, see Customizing for SAP ERP under Logistics > Materials Management > Purchasing > Purchase Order > Subsequent Functions > MIGO - Goods Receipt.

If SAP EWM does not manage the plant and storage location combination, SAP ERP generates a material document immediately at goods receipt posting. You put away the product, and SAP ERP updates the stock.

4. SAP EWM receives a corresponding inbound delivery notification from SAP ERP, and generates an SAP EWM inbound delivery corresponding to this. The inbound delivery contains a link to the manufacturing order in SAP ERP.
5. You transport the finished product into the warehouse, for example to the goods receiving area.
6. You can use the number of the manufacturing order to find the corresponding SAP EWM inbound delivery, post the goods receipt, and put away the product.
7. SAP EWM transmits the goods receipt posting to SAP ERP. SAP ERP also posts goods receipt for the related SAP ERP inbound delivery. SAP ERP also updates the stock in the manufacturing order and in the SAP ERP inbound delivery, due to the goods movement.

Result

You have posted goods receipt for your products from production, and updated the stock in SAP ERP.

Goods Receipt Posting Triggered by SAP EWM

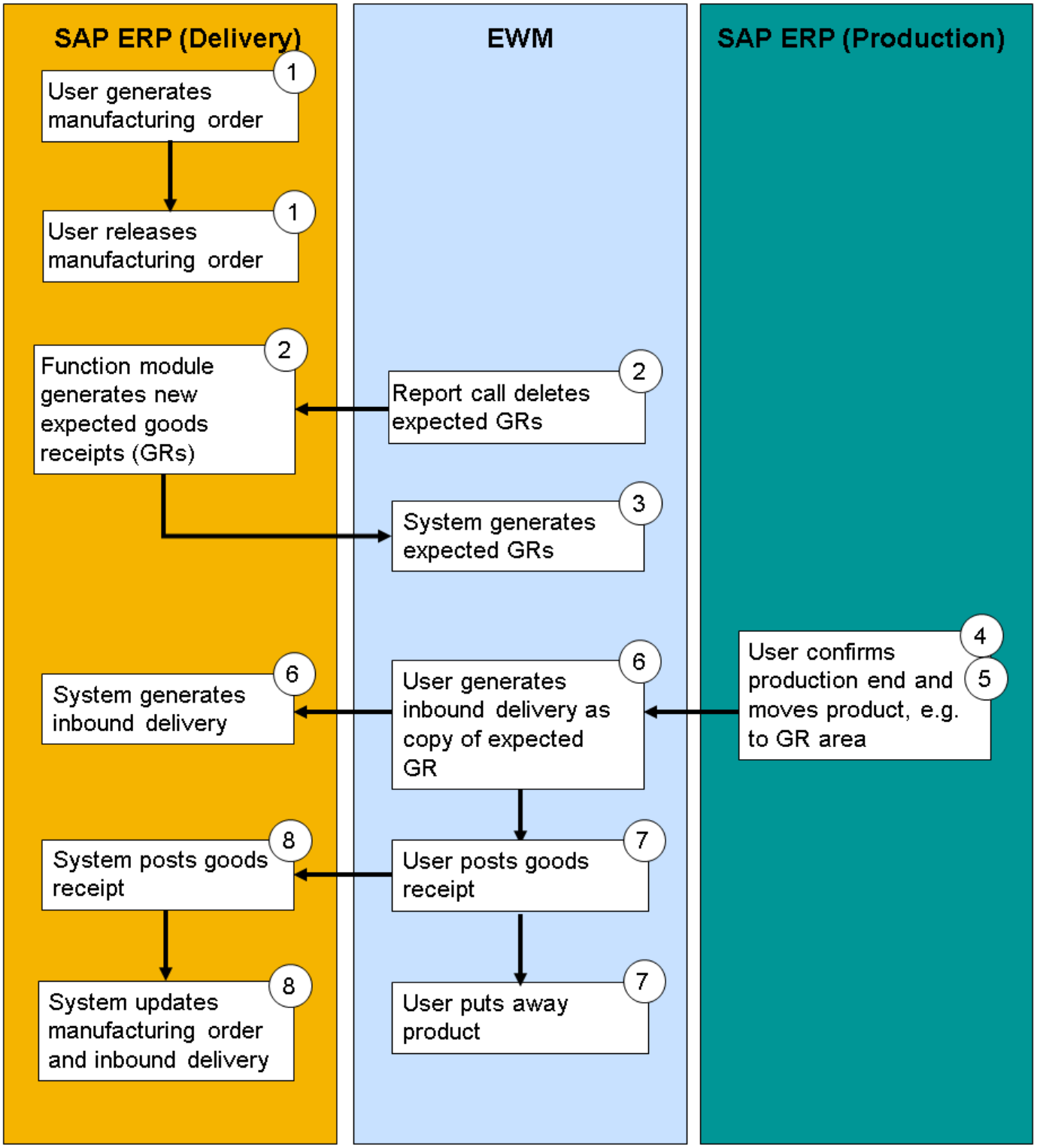
Use

You receive finished products, in the goods receiving area for example, for which you want to post goods receipt. For this, you are working with expected goods receipts in SAP Extended Warehouse Management (SAP EWM). In exceptional circumstances, you generate an SAP EWM inbound delivery manually without a template, meaning without an expected goods receipt. In this case you must make sure that the data in the inbound delivery is validated. You can use the standard implementation of Business Add-In (BAdI) `/SCWM/EX_ERP_PROD`.

For more information about posting goods receipt, see Goods Receipt Posting.

Process

The following figure provides an overview of the process flow:



1. You generate a manufacturing order and release it.

   The manufacturing order contains a plant or storage location combination that refers to a warehouse in which you are using SAP EWM. In other words, SAP EWM manages the plant and storage location combination.

   If SAP EWM does not manage the plant and storage location combination, SAP ERP generates a material document immediately at goods receipt posting. You put away the product, and SAP ERP updates the stock. SAP ERP does not trigger generation of expected goods receipts.

2. If you execute report **Generate and Delete Expected Goods Receipts** (`/SCWM/ERP_DLV_DELETE`), SAP EWM deletes any existing expected goods receipts in SAP EWM, and calls SAP ERP function module `/SPE/INB_EGR_CREATE_PROD`. SAP ERP checks whether relevant manufacturing orders still exist for which you still have to post goods receipt. SAP ERP

identifies the relevance from your selection criteria and from the open quantities. If open quantities still exist, then the goods receipt indicator is set in SAP ERP but the final delivery indicator is not set.

For more information, see Generate and Delete Expected Goods Receipts and Generate Inbound Delivery from Expected Goods Receipt.

3. For each relevant manufacturing order, SAP EWM receives a notification for the expected goods receipt from SAP ERP, and generates a corresponding expected goods receipt in SAP EWM.

   The expected goods receipt contains a link to the manufacturing order in SAP ERP. You can use the expected goods receipt in SAP EWM for planning purposes. It contains the planned times and quantities, for example.

4. You confirm the end of the designated production steps in the manufacturing order. In doing this, SAP ERP updates the finished quantity and marks the completed production steps.

5. You transport the finished products into the warehouse, for example to the goods receiving area.

6. You can use the number of the manufacturing order to find the expected goods receipt, and generate a corresponding SAP EWM inbound delivery. Here you use the expected goods receipt as a template for the SAP EWM inbound delivery, and can adjust the quantities if required.

7. You post the goods receipt and put away the products.

8. SAP EWM transmits the goods receipt posting to SAP ERP. SAP ERP also posts goods receipt for the related SAP ERP inbound delivery. SAP ERP also updates the stock in the manufacturing order and in the SAP ERP inbound delivery, due to the goods movement.

Result

You have posted goods receipt for your products from production, and updated the stock in SAP ERP.

Connection of Delivery Processing to Additional Functions

Use

Delivery processing uses the following additional functions:

- Connection to Yard Management
- Connection to Transportation Management
- Connection to Route Determination
- Connection to Serial Numbers
- Connection to SAP Global Trade Services (SAP GTS)
- Connection to Environment, Health and Safety (EH&S)
- Connection to the Post Processing Framework (PPF)
- Connection to Value-Added Service Order (VAS Order)
- Kit in the Warehouse Request for the Outbound Delivery Order
- Connection to Quality Management
- Connection to Batch Management
- Connection to Pickup and Account Assignment

Connection of Delivery Processing to Yard Management

Use

If you use yard management in **Extended Warehouse Management** (**EWM**), this communicates with delivery processing to exchange relevant information such as status changes. This enables delivery processing to allow follow-up actions, such as status-dependent goods receipt posting.

Yard management and delivery processing exchange the following information changes:

- Status changes in the delivery, such as from **In transit** to **In yard**
- Information about transportation units (TUs)

If you do not use yard management, delivery processing obtains the information about status changes from the user interface of the delivery.

Prerequisites

You use yard management in **EWM** .

See:

The "Prerequisites" section under Yard Management .

Features

As a result of the connection to yard management, **EWM** is able to send modified TU data from yard management to delivery processing.

EWM writes this TU data to delivery processing and the handling unit in the following way:

- If shipping and receiving is activated, delivery processing changes the status value for the **Assign Transportation Unit** status of the delivery document item from **Not relevant** to **Not started.**
- **EWM** updates the TU data with the assignment of transportation unit to a delivery header, delivery document item, or at handling unit level.

    1. Yard management informs delivery processing about the assignment.
    2. For shipping and receiving, delivery processing changes the status value for the **Assign Transportation Unit** status of the delivery document item from **Not relevant** (default) to **Completed** .

       If yard management has not assigned a transportation unit to a delivery document item, it sets the **Assign Transportation Unit** status of the delivery document item to **Not started** .
    3. Delivery processing aggregates the status data of the delivery document items.

       If all delivery document items have the same status value, namely **Not relevant** , **Not started** , or **Completed** , delivery processing also sets this status value for the delivery document header for shipping and receiving.

       If one delivery document item has a different status value to the others, for example, **Completed** rather than **Not started** , delivery processing sets the status value for the delivery document header for shipping and receiving to **In process** .

As long as the status value of the delivery document header for shipping and receiving has not yet reached **In processing** or **Completed** , you can still change the following fields in the delivery document:

- Delivery document header status **In yard** or **Outside yard**
- Vehicle
- Door

4. As soon as the **Assign Transportation Unit** status of the delivery document header has the value **In processing** or **Completed** , delivery processing locks the delivery document for assigning transportation units. It then informs yard management about this lock and the status changes that have occurred. This lock prevents data inconsistencies occurring between delivery processing and yard management. Now you can only change the assignment of transportation units in shipping and receiving or in yard management.

- Using an action for the transportation unit

  If you execute actions for transportation units, such as **Loading Begin** , **EWM** simultaneously updates the delivery documents and handling units that are assigned.

  In delivery processing, you can only execute the actions **In Yard** or **Leave Yard** for transportation units that are assigned to the respective delivery documents or vehicles.

  To keep the data consistent, **EWM** distributes the assignment of transportation units to doors using the delivery documents and handling units that are assigned.

See also:

Assignment of Deliveries to Vehicles and TUs

Connection of Delivery Processing to Transportation Management

Use

EWM sends a shipment output to transportation management when generating a warehouse request of the type inbound delivery or outbound delivery order .

Features

For more information about the features of transportation management, see Transportation Management in EWM .

Route Determination in the Delivery

Use

Delivery processing calls the routing guide for automatic route determination. This determines the route and sends it back to delivery processing. You can also manually enter or change a route on the user interface of the delivery. For more information about route determination, see Route Determination.

Delivery processing executes route checks and, in doing so, it determines whether the route found in the routing guide or the route that was manually entered in the delivery is valid for delivery processing.

When working with automatic route determination, you can use Extended Warehouse Management (EWM) to automatically find and check the route and transportation information in the outbound delivery request, the outbound delivery order, and the outbound delivery (see Special Features for Outbound Delivery Objects).

Integration

For actual route determination, EWM uses the routing guide.

Prerequisites

- You made the settings for route determination.
- The following delivery data has been specified:
  - The supply chain unit of the warehouse at the delivery header

    Route determination uses this supply chain unit as a source location.
  - At least one item with an entry for product, a quantity, weight and volume
  - At least one item with a planned goods issue date (for forwards scheduling) or a planned delivery date at header level (for backwards scheduling)
  - If you are using your own document types, the route status must be activated at header level so that route determination can set this status.
  - If the following delivery data is available, route determination uses it as an additional constraint during route determination:
    - The carrier, if the determination indicator is not initial or the carrier has been specified by EWM.
    - The means of transport, if the determination indicator is not initial or the means of transport has been specified by EWM.
    - The route itself, if the determination indicator is not initial or the route has been specified by EWM.
    - The shipping conditions of all items, if all the items have the same shipping conditions.

Features

Route Determination

Route determination attempts to find a route in delivery processing that corresponds to the delivery data, for example, the product, quantity, or date/time. For this, route determination calls the routing guide, either with a predefined route or it does not supply a route for the routing guide. You either enter the predefined route manually or an external system, such as SAP Customer Relationship Management (SAP CRM) assigns a route to EWM. Route determination in the delivery does not overwrite any routes manually entered. You define the route determination status in Customizing for EWM:

- **Route Determination is Always Active (*<blank>*)**.
- **Automatic Route Determination Only in Case of Initial Route**
- **Automatic Route Determination Inactive**
- **Route Determination Inactive**

For more information, see the Implementation Guide (IMG) for **EWM** under **Goods Issue Process** > **Outbound Delivery** > **Route Determination** > **Activate or Deactivate Route Determination**.

You can directly enter constraints for route determination in the outbound delivery.

Simulation of Route Determination

You can simulate route determination, that is, check which routes EWM finds for specific data or which routes EWM approves as valid after the route check. For more information, see Simulation of Route Determination.

Priority

If, while using the data of the relevant items (see Execution of Automatic Route Determination), route determination finds one or more valid routes for the start date/time and the supply chain unit, it uses the route with the highest priority. The routing guide sends the priority together with the route to delivery processing. Die Priorität wird ausgehend vom verwendeten Kostenprofil gesetzt.

Data Use

If route determination automatically finds a valid route, you specify a valid route, or an external system transfers a valid route to EWM, route determination copies certain data to the delivery and, if necessary, sets further delivery data automatically. The data automatically set can be data that route determination also determines, for example, delivery date/time or means of transport. For more information about the data that delivery processing uses or copies in route determination, see Data Use in Route Determination.

Automatic Route Determination

If you have not yet entered a route (that is, the field for the route is empty) or you have changed data that is route-dependent (for example, item quantities), route determination executes automatic route determination. For more information, see Execution of Automatic Route Determination.

Route Status

Route determination sets a route status at header level depending on the route determination. You can view this on the user interface of the outbound delivery order or the outbound delivery in the detail view on the **Status** tab page under the **Route Determination** status type.

Outbound Delivery Objects

In addition to the features mentioned here, route determination handles the various outbound delivery objects differently. For more information, see Special Features for Outbound Delivery Objects.

Route Determination in Transportation Cross-Docking

In a transportation cross-docking process (TCD process), route determination uses the supply chain unit of the final ship-to party as the destination location instead of the supply chain unit of the ship-to party. If the route contains a TCD location, route determination enters this as the ship-to party in the delivery. The previous ship-to party becomes the final ship-to party. Route determination proceeds in the same way with the delivery date, which means that route determination works with a delivery date and a final delivery date. Route determination enters the route to the next TCD location as the route for the delivery. This also applies to a route that is specified manually. Furthermore, route determination also specifies the goods issue date/time at the TCD location in the delivery header.

If route determination cannot identify a TCD location during a new route determination, it deletes the TCD-specific data from the delivery.

For more information, see Route Determination as Part of Cross-Docking.

Route Check

EWM checks in delivery processing whether a valid route exists for the delivery data. If you specify a route manually or deactivate automatic route determination, EWM then carries out a further check to determine whether the route that you have specified manually is valid for the delivery data.

Activities

If you want to manually trigger route determination, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Outbound Delivery > Maintain Outbound Delivery Order or Maintain Outbound Delivery and then the Route pushbutton.

For more information about the individual settings for the user interface, see Route Determination on the User Interface.

Making Settings for Route Determination

To use route determination, you must make the following settings:

Customizing

- You have activated the route determination of transportation management in Customizing in Extended Warehouse Management (EWM).

  For more information, see the Implementation Guide (IMG) for EWM under Goods Issue Process > Transportation Management > Route Determination > General Settings for Route Determination.

- You have activated the route determination for the delivery or the warehouse that is being used.

  For more information, see the IMG for EWM under Goods Issue Process > Outbound Delivery > Route Determination > Activate or Deactivate Route Determination.

- You also have the option of making additional settings for scheduling. For more information, see the IMG for EWM under Goods Issue Process > Outbound Delivery > Route Determination > Define Scheduling Within Route Determination.

Master Data

- You have created a route and a leg from the supply chain unit of the sending warehouse (source location) to the supply chain unit of the recipient (ship-to location). Here, the destination location may also be a zone.To do this, on the SAP Easy Access screen, you have chosen Extended Warehouse Management > Master Data > Shipping and Receiving > Route Determination.

- You have made sure that the goods recipient in the delivery has been assigned to a supply chain unit that is being used for route determination.

  As a business partner or address is entered in the delivery as a goods recipient, an assignment still has to be made with which the destination location used in a route is determined.

  This is done in the following way:

  - If a document address is entered in the delivery for the goods recipient, the system attempts to find a supply chain unit zone based on the address data. This zone is then used as the destination location for route determination.

  - If no document address is used, a corresponding supply chain unit is found in the following way:

    If you are working without an address or without a zone as a destination location in the route, you have created an individual supply chain unit for the ship-to party in the supply chain unit master data. In the Bus.Partner field on the

General tab page, you assign this supply chain unit to the business partner (ship-to party) that has been entered in the delivery.

> ⚠ **Caution**
> If the goods recipient is a business partner or customer whose master data has been transferred from SAP ERP, the system usually enters the assignment of the business partner to the supply chain unit, and creates a supply chain unit (for example, for the customer). In this case, you are not allowed to enter a business partner in the supply chain unit, and not allowed to change this entry. You are not allowed to change a business partner that has already been entered in a supply chain unit, as it is also used by other processes.

Avoid multiple assignments, that is, only assign one supply chain unit of one type to a business partner.

> ⚠ **Caution**
> If you assign more than one supply chain unit of one type to a business partner, EWM cannot clearly determine the supply chain unit from the business partner.

For master data sent that was sent from SAP ERP, and for the case where you have created multiple assignments, EWM uses the following method when attempting to find the supply chain unit for the business partner:

EWM checks whether an entry with the identification type R/3 Plant (CRM011) exists for the business partner in the business partner master data on the Identification tab page, and whether a supply chain unit exists for this entry.

- If EWM finds **one** distinct entry, it uses this supply chain unit.
- If EWM does not find an assignment, it checks whether an entry with the identification type R/3 Customer Number (CRM002) exists for the business partner on the Identification tab page, and whether a supply chain unit exists for this entry.

If EWM does not find a distinct assignment, it cannot accurately find the routes and issues a message to this effect.

- You also determine the following for the supply chain unit master data:
  - On the Bus.Char. (business character) tab page, you assign the business character to the supply chain unit as a planning location.
  - You have created a ship-to party or an address for a ship-to party. If you do not use addresses in the delivery, you assign the business partner of the ship-to party in the supply chain unit master data of the supply chain unit.

    If you want to create a ship-to party or an address for a ship-to party, in the SAP Easy Access screen choose Extended Warehouse Management > Delivery Processing > Outbound Delivery > Maintain Outbound Delivery Order or Maintain Outbound Delivery. Specify the ship-to party as the partner role on the Partner tab page or use the Ordering Address pushbutton to specify the ship-to party's address.
  - You specify the geocoordinates and the country key for the supply chain unit of the source location and the ship-to location in the supply chain master data on the General tab page.

Executing Automatic Route Determination

Use

Depending on your Customizing settings and various other settings, Extended Warehouse Management (EWM) may be able to execute automatic route determination. This means that you do not need to enter a route manually and route determination determines a new route when data is changed.

Prerequisites

You made the settings for route determination .

Features

If either you or the system change one of the following fields in the delivery header or a delivery item in the delivery, EWM automatically executes new route determination:

Delivery Header

- Changing the route or setting the determination indicator for the route to **Initial** or **Value Determined by System** .
- Changing the means of transport or setting the determination indicator for the means of transport to **Initial** or **Value Determined by System** .
- Change to Delivery Date
- Changing the final delivery date/time.
- Changing the warehouse (of the source location).
- Changing the ship-to party.
- Changing the final ship-to party.
- Changing the freight forwarder.

Delivery item

i Note

Delivery processing only executes new route determination for those items that are relevant for the route. This means that delivery processing does not execute new route determination for changes to text items, kit items, or items that are not warehouse-relevant.

In addition, delivery processing only executes route determination or new route determination for those items that are not yet assigned to an outbound delivery order.

- Changing the product.
- Changing the delivery quantity.
- Changing the shipping conditions.
- Changing the weight or volume.
- Changing the planned goods issue date/time.
- Creating a new delivery item.

If you release a route in the delivery manually, EWM does not execute new route determination. EWM also does not check the validity of the route. EWM only executes route checking again when you change a route.

If EWM has already scheduled a document for picking, meaning that the status type **Picking** has a status value in the delivery other than Not Yet Started, EWM does not execute new route determination or set dependent data. It only checks the validity of the route. As soon as you change a route, EWM sets the dependent data once again.

Data Use in Route Determination

Route determination uses various data from the delivery, a change in which also triggers new route determination.

- From the delivery header, route determination uses the supply chain units of the warehouse and the ship-to party, for example.
- Route determination uses certain delivery-relevant items (see Executing Automatic Route Determination).

  Non delivery-relevant items are text or kit items, or items that are not warehouse-relevant.
- Route determination uses the latest planned goods issue date/time of all the relevant items of the delivery as the start date/time for the forward scheduling of route determination. The planned delivery date is used for backward scheduling.
- If you manually specify the carrier, the route or the means of transport, or if an external system specifies them, route determination uses them as additional constraints. For example, you manually specify the carrier. Route determination determines routes for this carrier only as valid. Using the determination indicator, route determination identifies whether it automatically determined the field or whether you or an external system specified the field.

When route determination determines a valid route, it copies the following data to the delivery:

- Route at header level
- Means of transport at header level
- Carrier at header level on the **Partner** tab page (if route determination automatically determined the carrier)
- Date/time **Departure from Yard** at header level
- Delivery date/time at header level
- Sequence number at header level
- Export relevance at header level

Special Features for Outbound Delivery Objects

Use

Depending on the outbound delivery object, route determination works differently to the function described under Route Determination in the Delivery . The differences for the individual outbound delivery objects are described below.

Features

Outbound Delivery Request

With route determination in the outbound delivery request, Extended Warehouse Management (EWM) can execute a delivery split in different outbound delivery orders.

If you have activated route determination in Customizing, EWM attempts to determine a common route for all delivery items.

- If route determination determines a common route, the system does not execute a delivery split.
- If route determination cannot determine a common route, EWM reexecutes route determination. This time, route determination determines one route for every route-relevant delivery item (see Execution of Automatic Route Determination). If route determination determines varying routes for the delivery items, delivery processing creates an

outbound delivery order for each route. In this way, you can, for example, transport heavy and also smaller products in two outbound delivery orders via separate routes within the same outbound delivery request. However, EWM always processes logically corresponding delivery items together (for example, because of Delivery Groups).

If you have deactivated route determination in Customizing, or if an external system manually specifies the route, route determination no longer determines the route. Route determination specifies the same route for all delivery items.

Outbound Delivery Order

Route determination in the delivery only finds a route for an outbound delivery order if you do not enter a route manually. If you manually enter a route, delivery processing only checks if the manually entered route is valid. The only exception is route determination in transportation cross-docking (see Route Determination in the Delivery , section **Route Determination in Transportation Cross-Docking**).

Route determination for an outbound delivery order does not take into account any delivery items that are already assigned to an outbound delivery.

Outbound Delivery

Delivery processing copies the route and the route status from the outbound delivery order to the outbound delivery, meaning that it does not execute route determination, nor does it check if the route is valid.

As soon as you change delivery item data or delivery header data that is route-relevant, delivery processing checks if the route is valid. If, however, you manually approve the route, delivery processing does not check the route (even with the data change described above).

If you change a route in the outbound delivery, route determination only adjusts the dependent data to the outbound delivery header. The dependent data of the delivery items remains unchanged.

Route Determination on the User Interface

Use

In addition to the functions described in Route Determination in the Delivery and Special Features for Outbound Delivery Objects (outbound delivery), further functions are available on the user interface for the outbound delivery order and the outbound delivery.

Features

Redetermination of Routes

You can execute route determination for one or more deliveries with the **Redetermine** pushbutton. For example, you can use an earlier or a later route. If you change your route master data, you can also execute route determination with the new master data using this function.

- With the **Departure Date/Time**, **Lead Time**, and **Next Trip** fields, you have the option of determining that route determination carries out forward scheduling from a start date/time that is different from the default start date/time.

  If you use the **Departure Date/Time** field, route determination attempts to find a route or trip with this date and time.

  If you use the **Lead Time** field, route determination attempts to find a route or trip that occurs after the current **Departure from Yard** date/time plus the lead time. This is specified in hours.

If you use the **Next Trip** field, route determination attempts to find a route or trip that occurs (in relation to route or trip) directly after the current **Departure from Yard** date/time.

> **ℹ Note**
>
> Note that you can only ever use one of these fields.
>
> Delivery processing saves the information (for Extended Warehouse Management (EWM) to use a changed departure date/time) to the delivery in the planned goods issue date/time of the items. A route redetermination also uses this date/time.

- You can use the **Aggregation** field to include the total weight of the selected deliveries in route determination. This means that route determination also determines routes that it does not automatically find, for example, because the weight of the individual deliveries is too low.
- You can use the **Set Route to Determined Automatically** indicator if route determination is to address routes specified manually or routes specified by an external system in the same way as it addresses routes that it automatically determines. If delivery processing executes a route redetermination, route determination uses the route that was found automatically.

  If you set this indicator, route determination also sets the determination indicator of the route to **Determined Automatically**.

Assigning Routes

With the **Assign** pushbutton, you can assign a route to one or more deliveries. Route determination sets the determination indicator of the route to **Manual**.

As described above, you can use the **Departure Date/Time**, **Lead Time**, **Next Trip**, and **Aggregation** fields, and the **Set Route to Determined Automatically** indicator.

Checking Routes

You can execute a route redetermination with the **Check** or **Check (Aggregated)** pushbutton. With this, you check whether route determination has found a result for the delivery. You also check whether the route determination data agrees with the data in the delivery. If you have made changes to the route master data, you can check, for example, whether the routes are still valid.

Delivery processing issues a warning message if the route that was found deviates from the route contained in the delivery. In the transportation cross-docking process (TCD process), delivery processing also issues a warning message if the route that was found contains TCD information that the delivery does not contain.

For the route check, route determination uses the total weight and total volume of all selected deliveries in the aggregated check.

Manually Releasing Routes

You can manually release a route with the **Release** pushbutton.You can also manually approve a route with the status **Found Invalid**. With this, you can work with deliveries that contain an invalid route, for example, if, in an exceptional situation, you want to use a route that is actually invalid for a delivery. Delivery processing blocks a delivery with an invalid route. Then you cannot post the goods issue for this delivery, for example.

If you manually approve a route, route determination changes the status of the route to **Manually Released**.

After a manual approval, delivery processing does not execute automatic route determination, since the invalid route would lead to another blocked delivery.

You can cancel a manual route approval. You either set the determination indicator to **Initial** or **Value Calculated by System**, or you use the **Redetermine** pushbutton for route redetermination.

ℹ Note

You can only manually approve a route if the route status is **Found Invalid**. If you want to enter a route manually, the determination indicator must not be **Value Calculated by System**.

If you cannot use a route for a delivery because of dangerous goods data, you can force delivery processing to use this invalid route for a delivery using manual route approval.You can use the BAdI default implementation of the Business Add-In (BAdI) `/SCWM/EX_DLV_ACT_ROUTE_RELEASE` to execute a general check as to whether a user is authorized to manually approve routes. You can implement further checks in the BAdI.

Route determination automatically uses dependent data from a valid route determination (see Data Use in Route Determination). If you manually enter a route that is invalid for the delivery or has no master record, route determination can no longer automatically determine this data. Route determination sets this data to initial. You can manually set this data or use the BAdI `/SCWM/EX_DLV_ACT_ROUTE_RELEASE`.

Connection of Delivery Processing to Serial Numbers

Use

You can use serial numbers in the delivery so that you can document them for the delivery and flag them in goods receipt.

Features

For more information about using serial numbers in the delivery, see Using Serial Numbers in the Delivery and Integration of Serial Number in the Delivery .

Connection to SAP Global Trade Services (SAP GTS)

Use

You can use the following functions from SAP Global Trade Services (SAP GTS) in Extended Warehouse Management (EWM):

Connection of Delivery Processing to Transit Procedure

Connection of Delivery Processing to Safekeeping

Connection of Delivery Processing to Customs Warehousing Procedure

Connection of Delivery Processing to Checks in SAP GTS

ℹ Note

For more information about SAP GTS, see the SAP Library under **SAP xAPPs > SAP Global Trade Services.**

Connection of Delivery Processing to Transit Procedure

In SAP Extended Warehouse Management (SAP EWM), you can execute a transit procedure for all transportations that are relevant to transit procedures. Here, SAP EWM differentiates between inbound deliveries and outbound deliveries during the processing. SAP EWM saves the previous document number together with its previous document type as the reference document number and

category in the delivery document. Delivery processing uses these to recognize that an inbound delivery is relevant to transit procedures or to customs blocked stock.

For more information, see Transit Procedure for Inbound Deliveries and Transit Procedure for Outbound Deliveries.

Transit Procedure for Inbound Deliveries

Use

In SAP Extended Warehouse Management (SAP EWM), you can subject complete inbound deliveries that are transit-procedure relevant to a transit procedure. In doing so, you can ensure that the customs authority issues an unloading permission for the vehicles with the goods to be delivered.

Prerequisites

- You have defined reference document categories in Customizing for **Extended Warehouse Management** under **Cross-Process Settings** > **Delivery Processing** > **Reference Documents** > **Define Reference Document Categories**.
- You have assigned these reference document categories that correspond to the previous document types you use to a transit procedure and have set the **TrsPrcRel** (transit procedure relevance) indicator. You do this in Customizing for **Extended Warehouse Management** under **Interfaces** > **GTS Integration** > **Define Previous Document Types**.

Process

1. In the case of inbound deliveries that are relevant to transit procedures, you find the previous document number and previous document type in the transportation papers for the vehicle. You enter these manually in the inbound delivery as the reference document number and reference document category.

   SAP ERP can also send an advanced shipping notification to SAP EWM that already contains the previous document number and previous document type. In this case, you can change the previous document number and previous document type in the user interface.

2. Based on the Customizing settings for the previous document type, SAP EWM determines that the reference document number is a previous document number for which SAP EWM must perform a transit procedure.

3. SAP EWM locks the inbound delivery to prevent additional goods receipt process steps from being performed, for example, posting goods receipt.

   For more information about goods receipt, see Goods Receipt.

4. If the status type **Transit Status** does not have the status value **Registered in Yard**, you set this status manually. By doing so, you ensure that SAP EWM informs SAP Global Trade Services (SAP GTS) about the inbound delivery. To do so, on the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Maintain Inbound Delivery** and then the **In Yard** button. In the dropdown menu, choose **Set Status**.

   For more information about SAP GTS, see Connection to SAP Global Trade Services (SAP GTS).

5. SAP EWM informs SAP GTS that an inbound delivery exists for which it must perform a transit procedure.

6. Based on this message, SAP GTS performs an EDI communication with the customs authority. SAP GTS asks the customs authority whether it is permitted to unload the vehicle. The customs authority sends a corresponding message to SAP GTS.

   - If the customs authority allows the vehicle to be unloaded, SAP GTS sends a notification of the unloading permission to SAP EWM. SAP EWM sets the status type **Transit Procedure** of the inbound delivery to the status value **Released** and unlocks the inbound delivery for additional goods receipt process steps.

- If the customs authority does not allow the vehicle to be unloaded because it is performing a customs inspection, unloading is delayed.

  After the customs inspection, SAP GTS sends a notification with the unloading permission to SAP EWM. SAP EWM sets the status type **Transit Procedure** of the inbound delivery to **Released** and unlocks the inbound delivery for additional goods receipt process steps.

ℹ Note

You can only unload the vehicle and post goods receipt if the inbound delivery has the status value **Not Relevant** or **Released** for the status type **Transit Procedure**.

7. SAP EWM informs SAP ERP about the previous document number that you might have changed (see step 1).

Result

You have performed a transit procedure for your inbound delivery. You unload the vehicle and perform additional goods receipt process steps, for example, you post the goods receipt.

More Information

Loading and Unloading

Transit Procedure for Outbound Deliveries

Use

In SAP Extended Warehouse Management (SAP EWM), you can subject complete outbound deliveries that are transit-procedure relevant to a transit procedure. In doing so, you can ensure that the customs authority issues a loading permission for the vehicles with the goods to be delivered to the goods recipient.

Prerequisites

You have implemented the enhancement spot **Check GTS Release Authority** (`/SCWM/ES_GTS_INTEGRATION`), as required, to determine the transit-procedure relevance.

Process

1. Before you load a vehicle for an outbound delivery, you define whether this outbound delivery is relevant to a transit procedure.

   As the default value, SAP EWM uses the status type **Transit Procedure** with the status value **Not Relevant**.

   You can use the Business Add-In (BAdI) **Set Transit Procedure Relevance** (`/SCWM/SHP_PROC_REL_SET`) to define, for example, that all deliveries for a certain country are relevant to a transit procedure. For more information, see the BAdI documentation.

2. You set the status type **Transit Procedure** of the outbound delivery to the status value **For Checking**. To do so, on the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Outbound Delivery** > **Maintain Outbound Delivery Order** and then the **Transit Procedure** button. In the dropdown menu, choose **Set Status to Check**.

3. SAP EWM locks the outbound delivery to prevent additional goods issue process steps from being performed, for example, posting the goods issue. For more information about goods issue, see Goods Issue.

4. You ask the customs authority for loading permission.

   - If the customs authority allows the vehicle to be loaded, you set the status type **Transit Procedure** of the outbound delivery in SAP EWM to the status value **Released**. To do so, on the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Outbound Delivery** > **Maintain Outbound Delivery** and then the **Transit Procedure** button. In the dropdown menu, choose **Set Status to "Release"**. SAP EWM releases the locked outbound delivery for additional goods issue process steps.

   - If the customs authority does not allow the vehicle to be loaded because it is performing a customs inspection, loading is delayed. You can set the status type **Transit Procedure** of the outbound delivery to the status value **Customs Insp**. To do so, on the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Outbound Delivery** > **Maintain Outbound Delivery** and then the **Transit Procedure** button. In the dropdown menu, choose **Set Status to "Blocked for Customs"**.

   After the customs inspection, you set the status type **Transit Procedure** of the outbound delivery in SAP EWM to the status value **Released**. To do so, on the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Outbound Delivery** > **Maintain Outbound Delivery** and then the **Transit Procedure** button. In the dropdown menu, choose **Set Status to "Release"**. SAP EWM releases the outbound delivery for additional goods issue process steps.

ℹ Note

You can load the vehicle and post the goods issue only if the outbound delivery has the status value **Not Relevant** or **Released** for the status type **Transit Procedure**.

Result

You have performed a transit procedure for your outbound delivery. You load the vehicle and perform additional goods issue process steps, for example, you post the goods issue.

More Information

Loading and Unloading

Connection of Delivery Processing to Safekeeping

Use

You can use safekeeping for inbound deliveries in delivery processing.

For more information about safekeeping, see Safekeeping.

ℹ Note

Delivery processing decides whether a delivery item is relevant to customs procedures on a product-independent basis. The most important aspect is whether the combination of previous document number and previous document type is relevant to customs procedures. This is not dependent on whether the product itself is managed in the customs warehouse.

Prerequisites

- You have performed Customizing for the transit procedure in SAP Extended Warehouse Management (SAP EWM).

  For more information, see Transit Procedure for Inbound Deliveries.

- You have set the **Customs Block** indicator in Customizing for **Extended Warehouse Management** under **Interfaces** > **GTS Integration** > **Define Previous Document Types**. As a result, you have defined that the previous document type is relevant to customs blocked stock.
- If you want to define a customs block for certain previous document types, you have defined stock types with the role **Customs Blocked Stock** in Customizing for **Extended Warehouse Management** under **Master Data** > **Product** > **Define Stock Types**.

Features

SAP EWM copies the previous document number and previous document type from SAP ERP before you post goods receipt for an inbound delivery. If there is no previous document number or previous document type in SAP ERP, you can specify both directly in the delivery document in SAP EWM.

If the previous document type is relevant to customs blocked stock, SAP EWM posts the stock to a stock type for customs blocked stock. Otherwise, SAP EWM posts the stock to a different stock type, for example, **F1 Unrestricted Use in Putaway**. You can put away stock in a normal way in spite of a customs block.

i Note

- If you change the previous document number or previous document type again before posting goods receipt, SAP EWM automatically changes the stock type.
- If you have posted the first (partial) goods receipt or SAP EWM has created a warehouse task (unloading warehouse task or putaway warehouse task), you cannot change the previous document number or previous document type in the delivery.
- If a product is subject to a batch management requirement and is relevant to customs warehousing, you must specify a country of origin in the delivery for goods receipt posting.

Connection of Deliv. Proc. to the Customs Warehousing Procedure

Use

If you use the customs warehousing procedure in SAP Extended Warehouse Management (SAP EWM), the only effect that you have on delivery processing is in the scrapping area. If you work with warehouse requests of the posting change type for scrapping in the scrapping process in the customs warehouse, SAP Global Trade Services (SAP GTS) can apply to the customs authority for permission to scrap.

For more information about customs warehousing procedure, see Customs Warehousing Procedure.

For more information about warehouse requests of the posting change type, see Warehouse Request of the Posting Change Type.

For more information about scrapping process in the customs warehouse, see Scrapping as Part of the Customs Warehousing Procedure.

Prerequisites

For a description of the prerequisites for using the customs warehousing procedure in SAP EWM, see Customs Warehousing Procedure.

You have defined a reference document category SCR Reference Number - Scrap in Customizing for Extended Warehouse Management under Cross-Process Settings > Delivery Processing > Reference Documents > Define Reference Document Categories.

Features

When SAP ERP sends a posting change delivery for a scrapping to delivery processing, delivery processing automatically creates a posting change request for scrapping, or you do this manually. Delivery processing also checks whether the product in this posting change request is relevant to customs warehousing.

- If the product is relevant to customs warehousing, delivery processing requests a scrapping reference number from SAP GTS.
    - If SAP GTS sends this number immediately, delivery processing continues to process the posting change request for scrapping.
    - If SAP GTS does not send this number immediately, delivery processing locks the posting change request to prevent further processing, for example, warehouse task creation. SAP GTS asks the customs authority for permission for the scrapping. As soon as SAP GTS gets this permission, it sends the scrapping reference number to delivery processing. Delivery processing unlocks the posting change request and continues with the processing.
- If the product is not relevant to customs warehousing, delivery processing immediately continues to process the posting change request.

More Information

Performing Scrapping

Connection of Delivery Processing to Checks in SAP GTS

Purpose

Extended Warehouse Management (EWM) uses the compliance check and letter of credit check from SAP Global Trade Services (SAP GTS) to check whether an outbound delivery can be delivered to a certain goods recipient.

The connection to SAP GTS means you can make use of the following checking criteria for your deliveries:

- Checking criteria for the compliance check:
    - Sanctioned Party List
    - Embargos, for example, no exports from the USA to Cuba
    - Legal control, meaning licenses

      For example, a product is only allowed to be exported in a certain quantity and to a certain goods recipient.
- Letter of credit check: Check for whether letter of credit is still valid

The system performs the checks in the following cases:

- When you or EWM generate an outbound delivery order.
- If EWM changes certain fields in an outbound delivery order or outbound delivery, or you manually change certain fields (for example, address, partner, or delivery quantity) in an outbound delivery, EWM checks the outbound delivery order or the

outbound delivery.

- If you determined in Customizing that the outbound delivery is relevant for the legal control, EWM checks the outbound delivery before goods issue.

Prerequisites

- In Customizing, you can define the item types for which EWM is to perform the checks:
  - Define a status profile for the checks, and **do not** set the Inactive indicator for the corresponding status type DGT GTS check .For more information, see the Implementation Guide (IMG) for EWM under Cross-Process Settings > Delivery Processing > Status Management > Define Status Profile.
  - Assign this status profile to an item type.For more information, see the IMG for EWM under Goods Issue Process > Outbound Delivery > Define Position Types for Outbound Delivery Process Using Wizard .
- If you want the system to check outbound deliveries before goods issue, you must set the LegConGTS (GTS Compliance Check: Document Types Relevant for the Legal Control) indicator.

  To do so, in Customizing for EWM choose Goods Issue Process > Outbound Delivery > Define EWM-Specific Settings for Document Types.
- In the outbound delivery, a letter of credit number from the preceding document exists as a reference document. Only in this instance does EWM perform a letter of credit check.
- If you are working with direct outbound delivery orders , no data from a preceding document, such as an SAP ERP outbound delivery, is available.As EWM requires the sales organization, you can define it in Customizing for EWM for the corresponding checks.To do so, in Customing for EWM choose Interfaces > GTS Integration > Assign Sales Organization to Warehouse Number .You can also specify the sales organization manually on the user interface for the outbound delivery order.

Process

1. EWM prepares the data in the outbound delivery for checks and calls SAP GTS .

   In the standard system, EWM calls the checks for both outbound delivery orders and for outbound deliveries.

   ### i Note

   If you are working with direct outbound delivery orders, no data from a preceding document is available in EWM. For this reason, EWM performs restricted checks only. EWM does not perform a letter of credit check or a check for licenses for direct outbound delivery orders.

   If you are working with kit to stock, EWM does not perform any checks in SAP GTS.

2. SAP GTS performs the checks and returns a status.
   - If SAP GTS checks for the outbound delivery order contain errors, SAP GTS returns the relevant status. EWM sets the status GTS Check/Technical Error or GTS Check/Check, NotOK , and does not create an outbound delivery for this outbound delivery order.
   - If SAP GTS checks for the outbound delivery contain errors, SAP GTS returns the relevant status. EWM sets the status GTS Check/Technical Error or GTS Check/Check, NotOK , and does not permit a goods issue posting for this outbound delivery.

     ### i Note

     If you have the required authorization, you can remove a GTS lock.However, we recommend that you accept the decision from SAP GTS to avoid corrupting any check criteria.If, as an administrator, you want to reset a GTS

lock, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Outbound Delivery > Maintain Outbound Delivery .

For more information about SAP GTS, see the SAP Library under SAP xAPPs > SAP Global Trade Services.

If you use SAP Customer Relationship Management (SAP CRM), you can perform the checks for the corresponding sales order there.

To do so, set the GTS Relevance indicator for the transaction type and the item type in Customizing for CRM .For example, you perform these steps for the business transaction category Sales .For more information, see the IMG under Customer Relationship Management > Transactions > Basic Settings > Define Transaction Type or Define Item Type .

If you require the checks in your scenario, we recommend that you perform it in both systems (EWM and CRM), or you do not perform it in either system.

Post Processing Framework in Delivery Processing

Use

The Post Processing Framework (PPF)is a SAP basis technology used in Delivery Processing for scheduling and processing actions with reference to a document .

Integration

The PPF provides basic functions that are delivered with pre-defined settings depending on the application. You can adjust the settings as you require. Note that to what extend it is recommended to make changes to the presettings depends on the application.

Features

The PPF provides the following functions:

- Technique for determining actions
- Technique for aggregating found actions
- Medium for processing actions
- User interface for Customizing action templates including determination and aggregation techniques as well as processing mediums
- User interface for Customizing conditions if you use determination via conditions

Action Profile

The actions are saved in an action profile. The action profile is linked to the document type. You assign the actions to a document type via the action profile.

Requirements

In the PPF there are schedule conditions and start conditions. Delivery processing only uses the schedule conditions. You use the schedule conditions to determine when an action is scheduled.

Example

For more information, see Post Processing Framework (BC-SRV-GBT) .

Activities

In Customizing for Delivery Processing , you can execute the following activities:

- Define actions
- Define action profiles
- Define conditions

Example

Examples of actions include:

- Activate inbound delivery notification
- Post goods receipt
- Print delivery note

Configuration of PPF Action Scheduling

Use

You can configure the scheduling of Post Processing Framework actions (PPF actions) in the PPF application /SCDL/DELIVERY. To do this, you define schedule conditions that use the condition technique. The condition technique works with condition records that control the scheduling of the PPF actions Creating Condition Records).

Prerequisites

- You have defined a schedule condition for the PPF action that contains the call of method /SCWM/CL_DLV_PPF_CONF=>EVALUATE. For more information, see the method documentation. For more information about the PPF, see the SAP Help Portal help.sap.comunder SAP NetWeaver > SAP NetWeaver Library > SAP NetWeaver by Key Capability > Application Platform by Key Capability > Business Services > Business Communication Services > Generic Business Tools for Application Developers (BC-SRV-GBT) > Post Processing Framework (BC-SRV-GBT) .
- You have made the following settings for the condition technique in Customizing for Extended Warehouse Management (EWM):
    1. Create field catalog
    2. Create condition tables
    3. Create access sequences
    4. Create condition types
    5. Maintain determination procedure
    6. Assign determination procedure

7. Create condition maintenance group
8. Register condition maintenance group

   For more information, see the Implementation Guide (IMG) for EWM under Cross-Process Settings > Delivery Processing > Actions > Configure Action Scheduling.

Features

Within the PPF application /SCWM/DELIVERY, you can schedule the PPF action with the Business Add-In (BAdI) EVAL_SCHEDCOND_PPF PPF: Configure Exit for Evaluating Schedule Condition. In this action, you can define conditions that use certain delivery data. For example, you can define that EWM is only to schedule a PPF action if warehouse 001 or partner ABC is affected or if the delivery status has the value Finished . When PPF checks the PPF action to be scheduled, the system uses this configuration.

You call method /SCWM/CL_DLV_PPF_CONF=>EVALUATE in your BAdI implementation for the schedule condition for the PPF action. The method returns the return code RV_RC to the schedule condition. The value zero means that a result was found. Other values indicate various error messages. The schedule condition also returns a return code to the PPF. The value zero means that the schedule condition is fulfilled and that the PPF has scheduled the PPF action.

Condition record determination in delivery processing only returns the result of whether a condition record was found or not. You cannot specify any values that condition record determination then returns as a results record together with the condition record.

SAP provides a default field catalog that contains fields from the delivery document header only and that defines delivery data that you can use in the schedule condition. An exception is the Door field. EWM aggregates this field on header level. If the value in this field is the same for all items, EWM copies the value to the header level too. If the value in this field differs for at least one item, the value for the header field remains empty. You can either use the fields from the default field catalog or you can add your own fields to the field catalog. To fill these additional fields at runtime, you can use BAdI /SCWM/EX_DLV_PPF_CONF BAdI: Determine Delivery Data for Customer-Defined Condition Record Fields. You can also add item fields to the default field catalog. For more information about special fields in the field catalog for the application DPP, see the corresponding field help, for example, for the field /SCWM/DE_DPP_ITEM_REL.

i Note

Due to performance reasons, you cannot use placeholders for the values in the condition records. If you want to define a schedule condition for a status value that has not yet been defined in the system, you must define a condition record for each possible combination of status change, for example, Not Started -> In Process .

Activities

- If you want to define the configuration of scheduling of PPF actions, perform the IMG activities in the sequence specified under Prerequisites in the Implementation Guide (IMG) for EWM under Cross-Process Settings > Delivery Processing > Actions > Configure Action Scheduling.
- If you want to view or edit schedule conditions for PPF actions, in the IMG for EWM, choose Cross-Process Settings > Delivery Processing > Actions > Change Actions and Conditions > Define Conditions.
- If you want to edit condition records for the schedule condition of a PPF action, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Actions > Maintain Condition Records for PPF Schedule Conditions.
- If you want to view and edit PPF actions, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Actions > Process and Display PPF Actions.

i Note

SAP delivers PPF actions whose schedule condition already contains the call of the configuration. The system therefore behaves as follows:

If you do not assign a determination procedure in Customizing for SAP EM, the system behaves as in SAP SCM 5.0 with regard to these actions.

If you assign a determination procedure that matches a delivery, you must also perform all the Customizing settings described under **Prerequisites** and you must create a suitable condition record. Only if you do this are the schedule conditions fulfilled for the actions for this delivery.

Creating Condition Records

Use

You can use condition records to control how the system schedules Post Processing Framework actions (PPF actions) for delivery processing. The following section describes how you can create condition records to influence the planning of PPF actions in the delivery process.

Prerequisites

See Configuration of PPF Action Scheduling .

Procedure

1. On the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Actions** > **Maintain Condition Records for PPF Schedule Conditions (/SCWM/DLVPPFC).**
2. Specify the application **DPP** and the maintenance group that you have defined in Customizing. Choose with the quic k info text **Execute** .
3. In the **Condition Type** field, use the input help to select a condition type.
4. Complete the fields in the condition record and save.

   Here, one table row corresponds to one condition record.

   i Note

   For more information about special fields in the field catalog for the application DPP, see the long text of the corresponding field help, for example, for the field /SCWM/DE_DPP_ITEM_REL.

Result

You have created a new condition record.

i Note

If a PPF action for the delivery contains the call of method /SCWM/CL_DLV_PPF_CONF=>EVALUATE in the schedule condition, it has the following effect on the scheduling of the PPF action:

If the system finds a suitable condition record for a PPF action, it schedules the PPF action since the delivery corresponds to all the delivery data in the condition record.

If the system does not find a suitable condition record for a PPF action, it does not schedule the PPF action since the delivery does not correspond to all the delivery data in the condition record. Moreover, the system does not display the PPF action on the user interface for the delivery.

Connection to Value-Added Service Order (VAS Order)

Purpose

You can document the value-added services that you are providing in your warehouse using a value-added service order(VAS order).

Process

You can either generate a VAS order manually , or have **Extended Warehouse Management** (**EWM**) generate one automatically (see also Generating a VAS Order for the Warehouse Request Item Automatically)

Autom. Generation of a VAS Order for the Warehouse Request Item

Use

You can generate value-added service orders (VAS orders) automatically in Extended Warehouse Management (EWM) to document and monitor these value-added services.

Prerequisites

You have defined in your system that you want EWM to create VAS orders automatically. For more information, see the Customizing for EWM **Cross-Process Settings** > **Value-Added Services** > **Define Relevance for VAS Order** .

You have defined a packaging specification (see also packaging specification).

Process

1. You get a delivery from SAP ERP or create a delivery manually.
2. EWM checks whether it has to generate a VAS order automatically, and does so if this is the case.

   In the goods receipt process, you can configure automatic generation of VAS orders in such a way that EWM does not generate the VAS order until the goods receipt is posted.

   If you have defined in Customizing that EWM is not to generate the VAS order automatically, then the **VAS Order** status has the value **Not Relevant**.

   If the warehouse request item is consistent, for example, the **Blocked (Inconsist. Item)** status has the value **No**, EWM creates the VAS order. If the status has the value **Yes**, EWM creates the VAS order once the warehouse request is saved as consistent, or at the latest when a transfer order is created for the warehouse request item.

   If there is no packaging specification, EWM does not create a VAS order. You have to create or activate an existing packaging specification. EWM creates the VAS order when the warehouse request is changed and saved afterwards, or at the latest when a transfer order is created for the warehouse request item.

3. EWM saves the number of the VAS order as a document flow for the warehouse request item.
4. If you have activated the **VAS Order** in the status profile of the item type for the warehouse request item, EWM sets the **VAS Order** status of the warehouse request item to **Generated**.

Result

EWM has automatically generated the VAS order. You can use it for the confirmation of services (see also Processing a VAS Order).

When you edit the VAS order, EWM changes the status **VAS order of the warehouse request item** (see also: Updating the Warehouse Request Using the VAS Order).

Updating the Warehouse Request Using the VAS Order

Purpose

When you edit a value-added service order (VAS order) in **Extended Warehouse Management** (**EWM**), EWM updates the warehouse request item where required. For example, if the status of the VAS order changes, **EWM** updates the warehouse request item.

Prerequisites

In the Implementation Guide (IMG) for **EWM under Cross-Process Settings → Delivery Processing → Status Management → Define Status Profiles** , you have activated the status **DWV** in the respective status profiles, that is, for the status profiles of the item types for which you want to update the warehouse request. In the standard SAP system for example, these are profiles /SCWM/OUT_PRD_DLV_STANDARD and /SCWM/INB_PRD_DLV_STANDARD.

Process

Changing a VAS Order

You edit the VAS order and with that change the status of the VAS order .

EWM automatically adjusts the **VAS Order** status of the warehouse request item:

If you set the status of the VAS order to **In Process EWM** also changes the **VAS Order** status to **In Process**

If you set the status of the VAS order to **Completed** or **Completed with Quantity Variance EWM** sets the **VAS Order status to Completed**

Deleting a VAS Order

You delete the VAS order.

EWM deletes the document flow for the warehouse request item, and sets the **VAS Order** status to **Relevant** . If required, you can generate a new VAS order manually.

Updating the VAS Order Using the Warehouse Request

Purpose

When you change a warehouse request item in Extended Warehouse Management (EWM), EWM also changes the value-added service order(VAS order) correspondingly, if required.

Prerequisites

- A VAS order exists for the warehouse request item.
- If you change a warehouse request, and you want EWM to update the status of the relevant VAS order, the VAS order must be completed and the VAS Order status of the delivery item must have the value Completed .

Process

Changing the Warehouse Request

If you increase the delivery quantity of a warehouse request item, EWM updates the VAS order. If the VAS Order status currently has the value Completed , EWM changes the status of the VAS order from Completed back to In Process , and changes the VAS Order status of the warehouse request item back to In Process .

If you reduce the delivery quantity of a warehouse request item, EWM does nothing.

Deleting the Warehouse Request

1. You delete an item of a warehouse request for which a VAS order exists.

   Note

   You can only delete a warehouse request item if the VAS order is not yet in process.

2. If the VAS order is not yet in process, EWM deletes the VAS order.

Kit in the Warehouse Request for the Outbound Delivery Order

Definition

Describes a piece list that you always deliver in a complete and assembled form.

Use

Extended Warehouse Management (EWM) uses kits in Kitting . In EWM , the kit information in the outbound delivery forms the basis for kitting.

EWM can only receive information about kits in outbound deliveries.

EWM does not support nested kits, that is, kits in kits.

Structure

The outbound delivery contains the following items as kit information, corresponding to the levels of a kit:

Kit Header Item

Kit Component Item

The kit header item is a direct predecessor of this item.

Substituted kit component item

The kit header item is a direct predecessor of this item, and the kit component item is in turn a subitem of this item.

SAPdelivers preconfigured item types as standard for these kit items. You can use these to individually control the kit items in an outbound delivery.

If you want to change the settings of the preconfigured item types, you must note the following regarding the profiles. We recommend that you use the standard profiles shipped bySAP

You must make sure that the status profile for the item type of the kit header and of the substituted kit components prevents them from being considered in warehouse processing. For example, you must therefore deactivate the picking relevance and the goods movement relevance.

You must configure the quantity offsetting profile of the kit header and of the substituted kit components so that for example EWM does not calculate any open quantities for picking or goods issue. This means that you must set the corresponding quantity roles in this quantity offsetting profile to inactive.

When EWM generates a warehouse request with kit items, it checks the settings of the item types of the kit header items and of the substituted kit component items. If you follow the settings for the status profile and the quantity offsetting profile, EWM generates these items without error status and considers these items in warehouse processing.

For more information about settings for kits in the delivery, see the Implementation Guide (IMG) for EWM under Goods Issue Process → Make Settings for Kitting , section Requirements .

If you change the quantities of a kit component item in EWM through a pick denial, for example, EWM recalculates the quantities of the other kit component items and of the kit header item (quantity correlation).

Make sure thatSAPERP always sends information about entire kits to EWM. That is, the quantities of the kit components must be enough to create entire kits, otherwise the delivery quantity correlation returns incorrect results.

A kit header item and its related subitems can be part of a delivery group together with other delivery items. This enables you to be sure that EWM only ships the kit together with the other delivery items.

Integration

Integration withSAPERP

EWM receives information about the kit as part of the outbound delivery fromSAPERP.SAPERP also saves the kit as items in the outbound delivery (see also: Kit to Order).

Integration with value-added service orders (VAS orders)

You can create a VAS order for the kit header item. EWM copies the kit component items into the VAS order as items (see also VAS Order for the Kit Header Item of the Warehouse Request).

Integration with warehouse processing

Warehouse processing in EWM handles kit component items the same as outbound delivery items, meaning you can pick, pack and load them.

Outbound Delivery Split for Kits

Purpose

You can split kit header items in warehouse requests for outbound delivery orders to then copy the entire kit to an outbound delivery. You perform an item split, for example, if you want to ship finished kits before the remaining kits have been completed.

The process flow is the same as for standard outbound delivery items. However, Extended Warehouse Management (EWM) supports you, for example, by calculating the kit component quantities automatically (see Delivery Quantity Correlation for Kits).

i Note

You can only split entire kits. You cannot split kit component items or replaced kit component items.

You cannot perform a batch split for the kit header item or a replaced kit component item.

Prerequisites

- An outbound delivery with kit header items and kit component items exists.
- The quantity of the kit header item is greater than one.
- All kit component items are completely picked.
- If the system is to automatically generate an HU item for the kit header in the pick-handling unit (pick-HU), set the Create Kit Header Item indicator in Customizing . For more information about the packing profile, see the Implementation Guide (IMG) for Extended Warehouse Management (EWM) under Goods Issue Process > Outbound Delivery > Define EWM-Specific Settings for Item Types.

Process

Item Split for Unpacked Kits

1. You select the warehouse request for the outbound delivery order. To do this, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Outbound Delivery > Maintain Outbound Delivery Order.
2. You select the kit header item and split the warehouse request for the outbound delivery order. To do this, first choose Subitem and the Split Up Outbound Delivery .

3. You specify the number of kit header items that are to be generated.
4. EWM copies the kit, that is, it generates new kit header items and kit component items that are the same as the original kit in EWM .
5. You specify the delivery quantity in the kit header items and confirm your entries by choosing ENTER.
6. EWM calculates the kit component quantities automatically (see Delivery Quantity Correlation for Kits).
7. You can copy each of the split kits to one separate outbound delivery.

Item Split for Packed Kits

1. You select the warehouse request for the outbound delivery order. To do this, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Outbound Delivery > Maintain Outbound Delivery Order.
2. You choose the HU tab page and select all the HUs that you want to ship in advance.
3. You create new split items. To do this, first choose Split Item and then Create .
4. EWM aggregates the quantities of the packed kits using the HU dummy item.
5. EWM copies the kit, that is, it generates new kit header items and kit component items matching the original kit in EWM and automatically sets the quantities in these new kit items .
6. You can now copy the split kit to one separate outbound delivery.

Results

You have performed an outbound delivery split and generated split kits. You can ship these kits by generating an outbound delivery for each of the new kit header items.

Delivery Quantity Correlation for Kits

Use

If you or the system make changes to the delivery quantity, Extended Warehouse Management (EWM) recalculates the delivery quantity changes for the following items:

- Kit header items
- Kit component items
- Batch subitems
- Subitems occurring through substitution

EWM uses the delivery quantities to calculate the additionally required delivery quantities, for example, for creating warehouse tasks belonging to the kit. EWM derives open delivery quantities for these additional delivery quantities.

Integration

When calculating the delivery quantity ratios, EWM always bases its calculation on the current delivery quantity ratios between kit header item and kit component items. The original "delivery quantity structure", that is, the delivery quantities with their hierarchical dependencies, are already saved in the delivery in SAP ERP. SAP ERP sends this data to EWM. Either you or the system

can execute a batch split for kit component items. If you or the system execute this batch split in SAP ERP already, then the system also sends the resulting batch subitems to EWM.

Prerequisites

For EWM to be able to process kits correctly with their hierarchy, you must have configured the item types for the relevant delivery items appropriately in Customizing. For example, in Customizing, you define the field control profile in such a way that the field cannot be modified for the delivery quantity. For more information, see the Implementation Guide (IMG) for EWM under Cross-Process Settings > Delivery Processing > Field Control > Define Field Control Profiles.

Features

- When EWM performs a split, or when you manually change a delivery quantity for a kit header item, EWM recalculates the delivery quantities of all delivery items belonging to the kit.

  EWM calculates the delivery quantities for new and changed kit header items and kit component items.

- When EWM performs a pick denial, or when you manually change a delivery quantity for a kit component item, EWM recalculates the delivery quantities from bottom to top through the hierarchy levels.

  i Note

  When EWM recalculates the delivery quantities for kit header items, it may be the case that EWM also recalculates the delivery quantities for the related kit component items that neither you nor EWM changed.

  You can use Business Add-In (BAdI) `/SCWM/EX_DLV_DET_HIER_CORR` to influence the results of the delivery quantity correlation at a later time. For example, you can adjust items with fixed quantities or discount in kind on a quantity basis, depending on other items.

- EWM considers the process code in the delivery quantity correlation.

  Example

  You have a kit component item 10 for product A with a delivery quantity of 10 pieces. This delivery item 10 consists of batch subitem 15 for batch 1 with 6 pieces, and batch subitem 17 for batch 2 with 4 pieces. During picking, you notice that batch 2 only contains 2 pieces. You perform a pick denial and specify the process code QTY-2. Since the delivery quantity for batch 2 cannot be fulfilled, EWM also adjusts the delivery quantity of the kit component item, and sets the corresponding process code for the missing delivery quantity.

Delivery Quantity Recalculation for the Delivery Split

When you or the system perform a delivery split for a kit header item, EWM calculates and distributes the kit quantities contained in this 'original' kit proportionately across the new kits that have been split up. When you perform the delivery split manually on the user interface, you only have to enter the delivery quantity of the split kit header items. The delivery quantity correlation then automatically calculates the delivery quantities of all other subitems.

Delivery Quantity Recalculation for the Batch Split

When you perform a batch split manually on the user interface prior to picking, or when EWM performs a batch split automatically, EWM performs a corresponding delivery quantity correlation. Here, EWM ensures that the total of the batch subitem delivery quantities is not greater than the delivery quantity of the split kit component item.

- If the total is the same in the batch split, the delivery quantity is not changed.
- If the total is less in the batch split, the delivery quantity is again not changed. However, EWM does not distribute the delivery quantity across all batch subitems. This enables you or the system to add additional batch subitems at a later time.

- If the total is greater in the batch split, the delivery quantity of the batch subitems is changed corresponding to the delivery quantity.

  Example

  You have a kit component item 10 with batch subitems 65 and 67. The requested delivery quantity of the kit component item is 5. Batch subitem 65 contains a delivery quantity of 2 and batch subitem 67 contains a delivery quantity of 3. You enter a delivery quantity of 4 for both batch subitems. EWM leaves the delivery quantity for batch subitem 65 as 4, but changes the delivery quantity of batch subitem 67 to 1. This fulfils the requested delivery quantity of 5.

When you perform a pick denial at batch item level, EWM first recalculates at the level of the kit component item. The correlation function then calculates the delivery quantity of the kit header item. EWM then recalculates the delivery quantities for all kit component items across the hierarchy levels, from top to bottom.

When you reduce the delivery quantity of a kit header item, EWM also reduces the delivery quantities of the related kit component items. If these items also contain batch subitems, EWM also recalculates the delivery quantities (such as open pick delivery quantities) for these across all hierarchy levels, from top to bottom.

Recalculation for Delivery Quantity Changes from External Systems

For requests for delivery quantity changes from external systems, EWM expects the delivery quantity correlation to have already been performed in the external system. For this reason, EWM does not recalculate the delivery quantities.

Activities

You can trigger delivery quantity changes that lead to a delivery quantity correlation by:

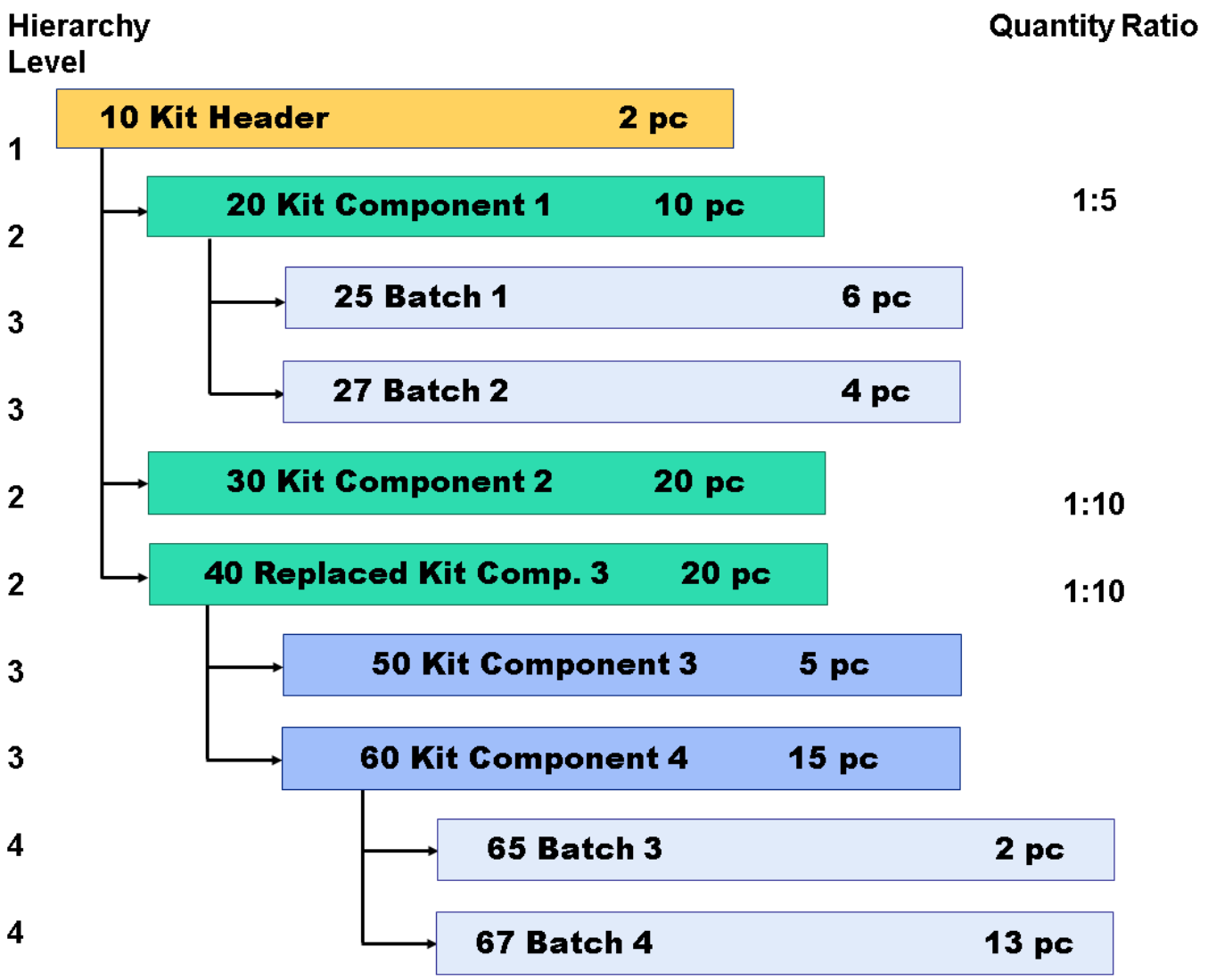
- Directly changing the delivery quantity field on the user interface:
  - On the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Inbound Delivery** > **Maintain Inbound Delivery** and then choose the **Quantity** column on the **Items** tab page. Change the delivery quantity.
  - On the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Outbound Delivery** > **Maintain Outbound Delivery Order** and then choose the **Quantity** column on the **Items** tab page. Change the delivery quantity.
  - On the **SAP Easy Access** screen, choose **Extended Warehouse Management** > **Delivery Processing** > **Outbound Delivery** > **Maintain Outbound Delivery** and then choose the **Quantity** column on the **Items** tab page. Change the delivery quantity.
- Changing the delivery quantity in the pick denial
- Changing the delivery quantity using the delivery quantity adjustment function

  Either you choose **Adjust Delivery Quantity** on the user interface, or EWM changes the delivery quantities when the system updates a delivery in the background.
- Performing delivery splits

  Either you perform a delivery split manually on the user interface, or EWM changes the delivery quantities when the system updates a delivery in the background.

Example

The following graphic shows you the hierarchy for a kit in SAP ERP together with the respective delivery quantities.



You have a delivery item 10 for the kit header with a delivery quantity of 2 pieces.

Delivery item 10 consists of kit component items 20 and 30 and delivery item 40 for the substituted kit component. Kit component item 20 for kit component 1 has a delivery quantity of 10 pieces, kit component item 30 for kit component 2 has a delivery quantity of 20 pieces, and delivery item 40 for the substituted kit component has a delivery quantity of 20 pieces.

Kit component item 20 consists of batch subitems 25 and 27. Batch subitem 25 for batch 1 has a delivery quantity of 6 pieces, and batch subitem 27 for batch 2 has a delivery quantity of 4 pieces.

Substituted kit component item 40 consists of delivery items 50 and 60. Kit component item 50 for kit component 3 has a delivery quantity of 5 pieces, and kit component item 60 for kit component 4 has a delivery quantity of 15 pieces.

Kit component item 60 consists of batch subitems 65 and 67. Batch subitem 65 for batch 3 has a delivery quantity of 2 pieces, and batch subitem 67 for batch 4 has a delivery quantity of 13 pieces.

1. SAP ERP sends the data for the kit to EWM.
2. In EWM, you reduce the delivery quantity of kit component item 30 to 12 pieces, and the delivery quantity of substituted kit component 40 to 10 pieces, due to a pick denial, for example.
3. EWM uses the correlation function to calculate the new delivery quantities.

EWM changes the delivery quantities across the hierarchy levels, from bottom to top. In the case of batch subitems, EWM always changes the delivery quantities "backwards" in terms of the sequence of the items.

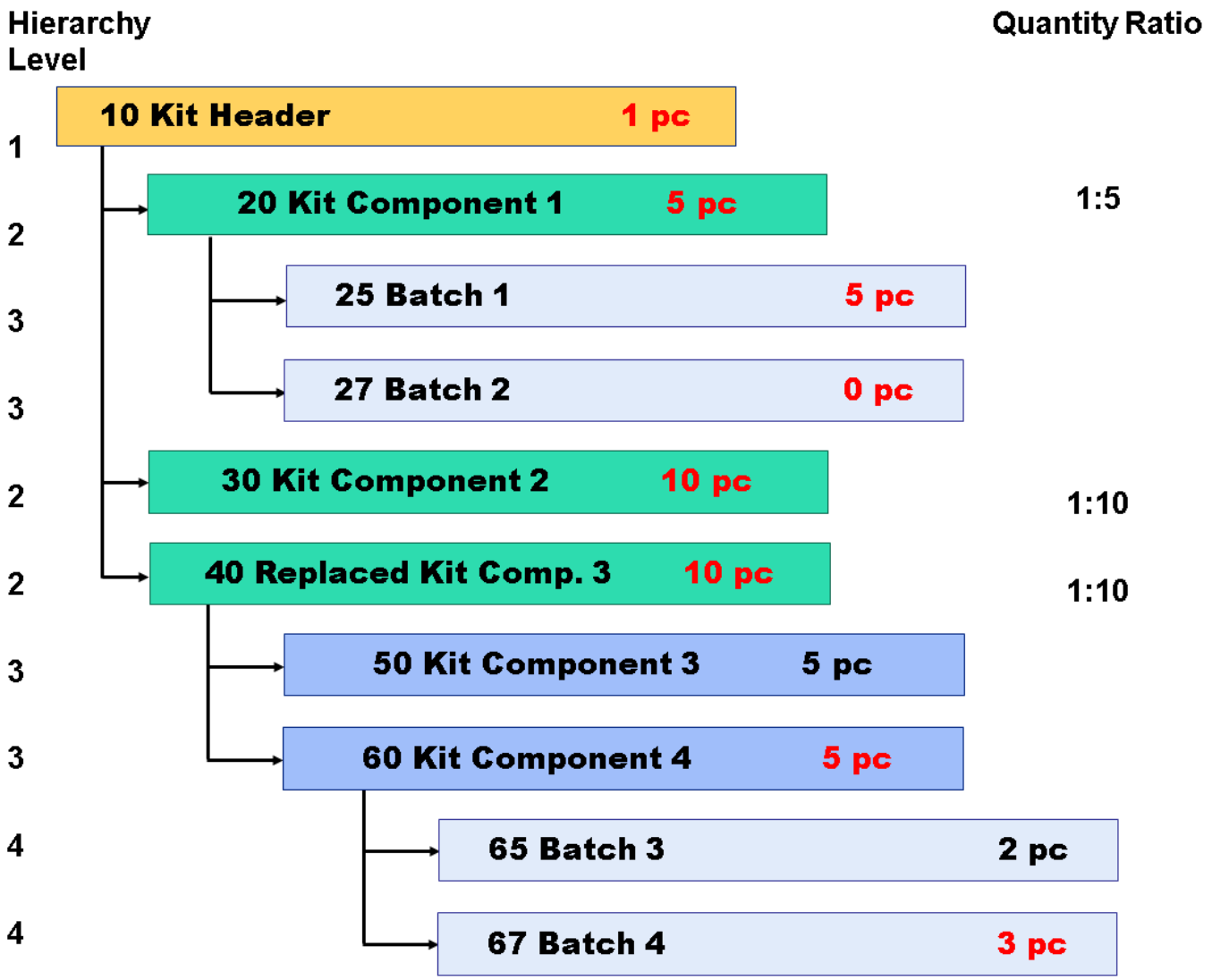
a. EWM starts by executing the following delivery quantity changes across the hierarchy levels from bottom to top:

- Batch subitem 67 for batch 4: reduce to 3 pieces
- Batch subitem 65 for batch 3: no change, remains at 2 pieces
- Kit component item 60: reduce to 5 pieces
- Kit component item 50: no change, remains at 5 pieces
- Kit component item 40: reduce to 10 pieces
- Kit component item 30: reduce to 10 pieces, although you only reduced the delivery quantity by 8 pieces. However, since a fixed ratio of 1:10 is defined, EWM reduces the delivery quantity of this item by 10 pieces.
- Kit header item 10: reduce to 1 piece

b. EWM then executes the following delivery quantity change across the hierarchy levels from top to bottom:

Kit component item 20: reduce to 5 pieces, since a fixed ratio of 1:5 is defined.

The following figure shows you the hierarchy of the kit in EWM following the delivery quantity correlation, together with the respective delivery quantities.



Connection of Delivery Processing to Quality Management

Use

SAP Extended Warehouse Management (SAP EWM) groups together all the necessary data for quality management (see Quality Management) to automatically create inspection documents in the background in accordance with the Customizing for quality management. Here, SAP EWM copies the inspection document number to the inbound delivery. In addition, SAP EWM updates the document flow, the status for inspection planning, and, if necessary, the stock type in the inbound delivery.

Integration

See Creation of Inspection Document.

Prerequisites

- You have carried out the Customizing for creating inspection documents (see Creation of Inspection Document).
- You have defined inspection rules (see Master Data).
- In the delivery, you have specified all the data necessary for the creation of an inspection document.

  If all of the necessary data does not exist in the delivery by the time SAP EWM is to create the inspection documents, SAP EWM creates the inspection documents later. For example, SAP EWM is to create an inspection document when the inbound delivery is activated. The batch number is required for creation. SAP EWM only creates the inspection document once you have specified the batch number.

Features

In the following cases, SAP EWM creates inspection documents in the background, regardless of Customizing:

- If you or SAP EWM activate an inbound delivery
- If SAP EWM sets the status **In Yard** for an inbound delivery
- Before SAP EWM creates a warehouse task
- Before you post the goods receipt

When SAP EWM creates an inspection document, it simultaneously updates the inbound delivery. SAP EWM sets the **QM Check Planning** status to **Completed** and the **QM Counting Planning** status to **Completed**. Regardless of whether you carry out a quality inspection or not, SAP EWM always updates the status data. For example, the result of quality inspection planning could be that SAP EWM is not to carry out a quality inspection. SAP EWM also sets the status data in this case.

Once SAP EWM has created the inspection documents, it updates the document flow for the reference document types **Q01** and **Q03**. Thus, you can view all of the inspection documents for an inbound delivery in the document flow.

If necessary, SAP EWM updates the stock type. For example, SAP EWM changes the stock type in the inbound delivery from unrestricted-use stock to stock in quality inspection.

For more information, see Quality Inspection.

Activities

If you want to display the QM Inspection Planning and QM Counting Planning statuses, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Maintain Inbound Delivery . Go to the Status tab page.

Connection of Delivery Processing to Batch Management

Use

You can work with batch numbers in both inbound and outbound deliveries.This means that you work continuously with homogenous partial quantities that are, for example, required by law or because of hazardous substance directives.

Prerequisites

If you want to create new batch numbers, you have defined number ranges for batches in Customizing for Extended Warehouse Management (EWM).For more information, see the Implementation Guide (IMG) for EWM under Master Data → Product → Define Number Range for Batch

Features

A batch number can be specified via a system (SAP ERP, for example) or you specify it directly in EWM.Using a determination indicator, EWM determines where the batch information originates from and whether you can still change the batch number.The following options exist:

A different system specifies a batch number, for example in a purchase order in SAP ERP.You must use this batch number, that is, you cannot change the batch number.

A different system proposes a batch.For example, you have entered the batch in the inbound delivery of SAP ERP.You can change this batch.

You specify a known batch number directly in EWM, that is, it does not originate from another system.To do this, on the SAP Easy Access screen, choose Extended Warehouse Management → Delivery Processing → Inbound Delivery → Maintain Inbound Delivery Order. Go to the form view and enter a batch number in the Batch field.EWM can also create new batch numbers for inbound deliveries.For this, leave the Batch field empty and choose the function Batch .As soon as you save the inbound delivery, EWM creates a new batch number.

You can also directly specify the existing batch number for outbound deliveries.To do so, on the SAP Easy Access screen , choose Delivery Processing → Outbound Delivery → Maintain Outbound Delivery Order Go to the form view and specify the batch number.

You can only change the batch number before saving.

You can also specify the batch in the corresponding warehouse task.EWM copies the batch to the delivery.Here, EWM creates a batch subitem with the confirmed quantity and batch.

When you are processing the inbound or outbound delivery, EWM copies the batch characteristics to the delivery, for example, the country of origin, minimum shelf life, or the expiry date.

You can split a batch in the inbound delivery.To do this, on the SAP Easy Access screen, choose Extended Warehouse Management → Delivery Processing → Inbound Delivery → Maintain Inbound Delivery Order. Select a batch and choose the function Batch Subitem. EWM creates a corresponding subitem for the batch.

Connection of Delivery Processing to Pickup & Account Assignmt.

Use

Delivery processing supports goods issue for pickup.

Features

For more information, see Pickup and Account Assignment .

Communication from EWM to SAP ERP

Use

Communication from Extended Warehouse Management (EWM) to SAP ERP uses defined interfaces, and enables distribution, changes to, and confirmation of delivery-relevant data.

In SAP ERP, the interfaces between EWM and SAP ERP are based on deliveries. In EWM, they are based on the following:

- Inbound delivery notifications and inbound deliveries in the inbound delivery process
- Outbound delivery orders and outbound deliveries in the outbound delivery process

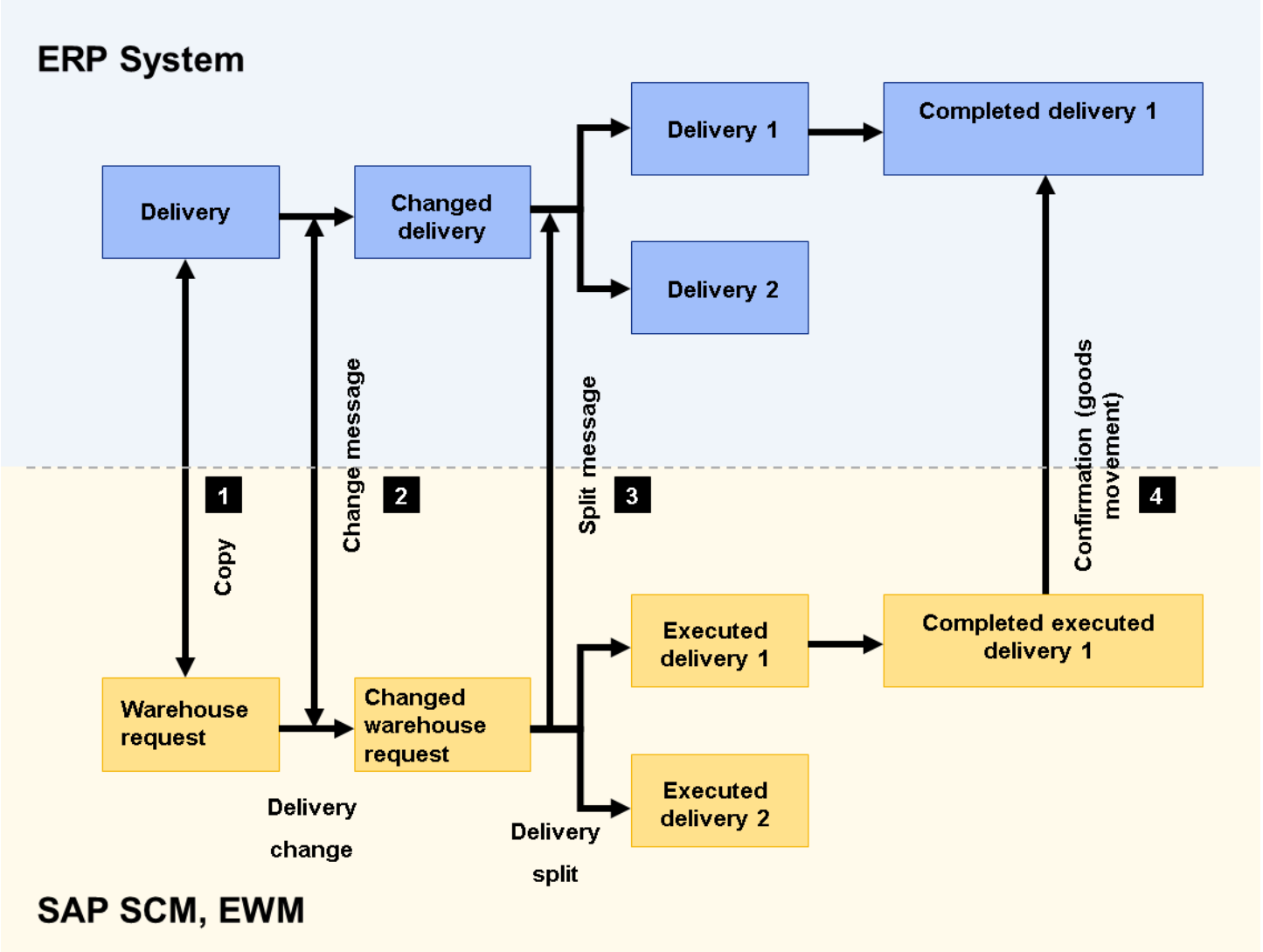
Where necessary, SAP ERP triggers subsequent communication steps for specific processes.

For more information, see Warehouse Req. - Type Inbound Del. or Outbound Del. Order , Generating a Warehouse Request of the Inbound Delivery Type , Generating a Warehouse Request of Type Outbound Delivery Order .

Communication is asynchronous and uses queued Remote Function Call (qRFC) technology. This technology guarantees serialization of the messages.

Integration

The following figure gives you an overview of the functions supported by the interfaces between EWM and SAP ERP:



1. Distribution of deliveries
    - Send deliveries from SAP ERP to EWM
    - Send deliveries from EWM to SAP ERP (for example, for manual inbound deliveries)
2. Changes to deliveries
    - Send changes from EWM to SAP ERP (for example, confirmation of a pick denial from EWM to SAP ERP)
    - Send changes from SAP ERP to EWM (for example, changed ASN from vendor)
3. Confirmation of an inbound and outbound delivery split from EWM to SAP ERP
4. Send confirmation of deliveries from EWM for follow-on functions in SAP ERP (for example, goods receipt, goods issue, confirmation before goods issue for invoicing before goods issue)

Prerequisites

- If you are using SAP ERP. Here, you define for each plant and storage location whether you manage your warehouse in a separate, decentralized EWM. EWM receives the requests for goods movements from SAP ERP.
- Transmit all information to EWM that EWM requires for the goods movements. These include:
    - Partner master data (for example, customers, vendors, transportation service providers)
    - Location master data (for example, customers, vendors, plants)

    - Product master data

- You have made the Customizing settings for integration with SAP ERP.

  In the Implementation Guide (IMG) for **EWM** , choose **Interfaces** > **ERP Integration** .

Features

The following functions are available for EWM and SAP ERP for the purpose of processing inbound and outbound deliveries.

- Processing of Inbound Deliveries
    - Receipt of an inbound delivery from SAP ERP to EWM
    - Receipt of a change request for inbound deliveries from SAP ERP to EWM
    - Notification of changed data before goods receipt from EWM to SAP ERP
    - Confirmation of a goods receipt from EWM to SAP ERP
    - Notification of a reversal or correction of the goods receipt from EWM to SAP ERP
    - Notification of an inbound delivery split from EWM to SAP ERP
- Processing of Outbound Deliveries
    - Receipt of an unchecked outbound delivery from SAP ERP to EWM
    - Receipt of a checked outbound delivery from SAP ERP to EWM
    - Reporting of an outbound delivery change from EWM to SAP ERP
    - Quantity change request from EWM to SAP ERP
    - Confirmation of a goods issue from EWM to SAP ERP
    - Notification of a reversal of the goods issue from EWM to SAP ERP
    - Direct billing or invoicing before goods issue
    - Communication of an outbound delivery split from EWM to SAP ERP

For more information, see Processing of Inbound Deliveries , Processing of Outbound Deliveries , and Interfaces for Communication from EWM to SAP ERP .

Activities

In order to create messages, you must perform the relevant processes using the appropriate user interface .

Processing of Inbound Deliveries

Use

SAP Extended Warehouse Management (SAP EWM) and SAP ERP have access to a range of functions that enable you to distribute, change, and confirm delivery-relevant data.

Integration

To read the integration description, see Communication from EWM to SAP ERP.

Prerequisites

To read the description of the prerequisites, see Communication from EWM to SAP ERP.

For more information about customizing the integration between SAP EWM and SAP ERP, see Customizing for **Extended Warehouse Management** under **Interfaces** > **ERP Integration**.

Features

- Receipt of an inbound delivery notification from SAP ERP in SAP EWM

  Normally, SAP EWM receives an inbound delivery notification as a result of the following process: SAP ERP receives an advanced shipping notification (ASN) and validates this. SAP ERP creates an inbound delivery and sends an inbound delivery notification to SAP EWM. SAP EWM can also receive an inbound delivery notification as a result of inbound deliveries created in SAP ERP. SAP EWM uses the inbound delivery notification to create an inbound delivery.

  If errors occur during ASN validation in SAP ERP , then SAP ERP gives the incorrect delivery items an error status before sending the information to SAP EWM. SAP EWM blocks the incorrect delivery items for execution. Delivery items that do not contain errors are open for execution even if the inbound delivery contains other incorrect delivery items. The following rules apply when solving the errors:

  - You can remove the errors in SAP ERP. SAP ERP sends a change message to SAP EWM. SAP EWM processes this change message as long as the inbound delivery is not yet in processing in SAP EWM.
  - You can remove the errors in SAP EWM. SAP EWM sends the changed data with the goods receipt confirmation to SAP ERP. SAP ERP decides whether it accepts the data in order to correct the errors, or whether you need to manually process the inbound delivery again in SAP ERP before the system processes the goods receipt confirmation. In this case SAP ERP returns the message, which was originally sent by SAP EWM, to the qRFC inbound queue. After you have solved the error in SAP ERP manually, you can start the message again in the qRFC inbound queue in SAP ERP. SAP ERP then continues processing accordingly.

    Inbound delivery notifications that SAP EWM receives can contain handling unit data (HU data). These are taken either from the electronic ASN from the vendor, or were entered during manual packing in SAP ERP. Only SAP EWM supports automatic packing. You cannot use automatic packing in SAP ERP.

    For more information, see Inbound Delivery Notification.

- Receipt of a change request for inbound deliveries from SAP ERP in SAP EWM

  The vendor can send a new ASN or an ASN rejection for an existing inbound delivery. SAP ERP receives these messages and validates them. It forwards them to SAP EWM. SAP EWM checks whether it can replace the existing inbound delivery. If it can replace the inbound delivery, it sends a confirmation message to SAP ERP. If SAP EWM cannot replace the existing inbound delivery, it sends a rejection message to SAP ERP. If the inbound delivery is not yet being processed, in other words, if the status **Warehouse Activity** does not have the value **Not Started**, EWM replaces the inbound delivery.

- Manual receipt of an inbound delivery in SAP EWM

  As an alternative to receiving an inbound delivery from SAP ERP, you can create an inbound delivery directly in SAP EWM. You can either enter all the data manually or use an expected goods receipt (purchase order or manufacturing order) as a reference document. The system sends the inbound delivery to SAP ERP. If you are using advanced production integration to receive HUs from production, SAP EWM creates the inbound deliveries when you receive the HUs. You can configure the system so that it does not send these inbound deliveries to SAP ERP. In this case, the system sends only the goods movements posted with reference to the delivery to SAP ERP. For more information see Receipt from Production in Advanced Production Integration, and Customizing for Extended Warehouse Management under **Cross-Process Settings** > **Delivery - Warehouse Request** > **Process Management and Control** > **Define Process Profile for Document Header**.

- Notification of changed data before goods receipt from SAP EWM to SAP ERP

  If you make changes in SAP EWM, such as setting the transit status **In Yard**, or entering new items, then SAP EWM sends these to SAP ERP before confirming the goods receipt. If you only change the transit status in SAP EWM, then SAP EWM only sends the header data for the inbound delivery to SAP ERP. SAP ERP replaces the header data accordingly. If you also replace relevant item data in SAP EWM, then SAP EWM sends the entire document, and SAP ERP replaces the entire document with the data from SAP EWM. SAP ERP does not overwrite any data that cannot be provided by SAP EWM, for example, the document type or company code.

- Confirmation of a goods receipt from SAP EWM to SAP ERP

  For inbound deliveries, you can make either partial (partial goods receipt) or complete goods receipt posting. You can specify whether there should be a confirmation to SAP ERP for each individual partial goods receipt, or for the entire delivery. You can also specify whether SAP EWM should also transfer the current HU data with the goods receipt confirmation to SAP ERP.

- Notification of a reversal or correction of the goods receipt from SAP EWM to SAP ERP

  This enables you to correct or reverse goods movement postings. SAP EWM creates a material document for these postings, which triggers confirmation of the correction or cancellation to SAP ERP.

- Notification of an inbound delivery split from SAP EWM to SAP ERP

  You can split inbound deliveries in SAP EWM. SAP EWM creates new inbound deliveries for this delivery split, and reports them to SAP ERP. SAP EWM determines the document number and item number in SAP ERP. You must have defined the number range intervals in Customizing. These number ranges must not overlap with the number ranges in SAP ERP.

For more information about the interfaces that enable these functions, see Interfaces for Communication from EWM to SAP ERP.

For more information, see Inbound Delivery and Generating a Warehouse Request of the Inbound Delivery Type.

Activities

In order to create messages for processing inbound deliveries, you must perform the relevant processes using the appropriate user interface.

If you want to create inbound deliveries manually in SAP ERP, on the **SAP Easy Access** screen, choose **Logistics** > **Logistics Execution** > **Inbound Process** > **Goods Receipt Automotive** > **Goods Receipt Transaction**.

Processing of Outbound Deliveries

Use

Extended Warehouse Management (EWM) and SAP ERP have access to a range of functions that enable you to distribute, change, and confirm delivery-relevant data.

Integration

To read the integration description, see Communication from EWM to SAP ERP.

Prerequisites

To read the description of the prerequisites, see Communication from EWM to SAP ERP.

Features

- Change of existing outbound delivery in SAP ERP

  You can change an outbound delivery in SAP ERP after it has already been replicated to EWM. If you open an SAP ERP outbound delivery in edit-mode, the system calls EWM and checks which fields you can change without harming the process in EWM.

  You can change fields unrelated to EWM, for example billing block, regardless of the status of the delivery in EWM. These fields are not influenced by EWM. If you need to change EWM-related fields, such as means of transport or delivery priority, these are evaluated field by field to find out what the impact of a change in EWM is, how long the change makes sense from the business point of view, and how long the documents are technically consistent after a change. For quantity changes, you can change the delivery quantity until a warehouse task is created. If you need to change the quantity of the sales order, see **Quantity change requests from SAP ERP** below.

  While you are in edit-mode in the SAP ERP delivery, warehouse processing is not blocked and the outbound delivery order is not locked. The system uses an optimistic locking concept to ensure data consistency. When you save the SAP ERP delivery, the system checks the outbound delivery order in EWM. If there was a change since you started editing, you cannot save the delivery in SAP ERP. If no error message is raised, the changes were successfully transferred to EWM.

  To be able to use this feature, you must be connected to an SAP S/4HANA system that is running SAP S/4HANA 1709 or higher.

  You need to enable delivery changes in order to be able to use this feature. For more information, see Customizing for **Logistics Execution** under **Logistics Execution** > **Extended Warehouse Management Integration** > **Basic Setup of Connectivity** > **Configure SAP EWM-Specific Parameters**.

  The following Business Add-Ins (BAdIs) are available in Customizing for **Extended Warehouse Management**:

  - **Business Add-Ins (BAdIs) for Extended Warehouse Management** > **Interfaces** > **ERP Integration** > **Inbound Messages from ERP System to EWM** > **BAdI: Receipt of Outbound Delivery Change**
  - **Business Add-Ins (BAdIs) for Extended Warehouse Management** > **Interfaces** > **ERP Integration** > **Delivery Processing** > **BAdI: Modify Change Behavior of Original Delivery**

- Receipt of an outbound delivery from SAP ERP

  If SAP ERP converts unchecked outbound deliveries into checked outbound deliveries, then it sends two messages to EWM.

  - A deletion message for deleting the outbound delivery request and the outbound delivery order that EWM created for the unchecked outbound delivery
  - A message for creating an outbound delivery request and an outbound delivery order for the checked outbound delivery

  You can immediately process outbound delivery orders for checked outbound deliveries.

  EWM creates the following documents for checked outbound deliveries:

  - An outbound delivery request
  - An outbound delivery order

  In EWM, you can create an outbound delivery for an outbound delivery order if EWM has created this outbound delivery order from a checked delivery in SAP ERP.

- Receipt of an unchecked outbound delivery from SAP ERP

  SAP ERP can distribute unchecked outbound deliveries to EWM. SAP ERP creates outbound delivery orders for unchecked deliveries. You cannot process these outbound delivery orders in EWM, but you can preview them. EWM creates the following documents for unchecked outbound deliveries:

- An outbound delivery request
- An outbound delivery order

In EWM, you cannot create an outbound delivery for an outbound delivery order if EWM has created this outbound delivery order from an unchecked delivery in SAP ERP.

- Reporting of an outbound delivery change to SAP ERP

  If you create a pick denial in EWM, then EWM reduces the delivery quantity and writes an exception code in the outbound delivery order. This triggers confirmation of the change to SAP ERP. SAP ERP reports the change to SAP Customer Relationship Management (SAP CRM). The pick denial causes a reduction in the delivered quantity. The message to SAP ERP contains an indicator for the quantity change due to a pick denial. If you are using the Business Add-In (BAdI), you can use this indicator to perform another round of sourcing. SAP ERP can create a new checked delivery for the missing quantity and can distribute this to EWM. This message to EWM contains an indicator for delivery creation, based on a confirmation of a pick denial. EWM sets this indicator in the outbound delivery order in the **Wave Priority** field. EWM uses this to take the outbound delivery order immediately into account during wave pick creation.

  i Note

  To use the BAdI, in Customizing choose **Customer Relationship Management** > **Transactions** > **Settings Before Sales Transactions** > **Business Add-Ins** > **Business Add-In for Data Exchange.**

  This BAdI contains the method CRM_DATAEXCH_R3D_MBDOC_FILL, which you can use to set an indicator to trigger a new availability check.

For more information about processing waves, see Wave Management.

- Quantity change requests from SAP ERP

  EWM can execute quantity change requests in the following cases:

  - When you change your sales order in SAP CRM or in SAP ERP.
  - In the case of delivery-based production supply, when you reduce the quantity of a component in a manufacturing order. For more information, see Integration of Production Supply (PP) in Extended Warehouse Management (EWM)).

  If you increase the quantity of an order in SAP ERP, the outbound delivery order in EWM is not updated automatically. If you reduce the quantity of an order item in SAP ERP, including a reduction to zero, the system tries to update the outbound delivery order in EWM.

  If there is no warehouse task yet, the system can reduce the quantity. EWM immediately changes the quantity of the outbound delivery order and updates the SAP ERP delivery.

  If warehouse tasks have been created or confirmed, you might be able to reduce the quantity if you have configured the order reduction function. For more information on what happens if you have activated order reduction, see Order Reduction.

- Communication of a goods issue confirmation to SAP ERP

  EWM uses an outbound delivery in order to report a goods issue posting or reversal to SAP ERP. You can only cancel an outbound delivery if it does not have the status **In Transit** .

- Direct billing or invoicing before goods issue

  You can print the invoice in the warehouse, if necessary, even before the goods issue. EWM uses an outbound delivery order to create an outbound delivery before the goods issue posting. If you have defined the status **Invoicing Before Goods Issue** for a route in Customizing, then EWM sets an indicator for this status that results in direct billing. If an outbound delivery has set this indicator, either SAP ERP or SAP CRM generates the billing document automatically. After the outbound

delivery has been generated, EWM triggers confirmation of the data for direct billing. SAP ERP receives this data and performs billing. If the system performing the billing is SAP CRM, SAP ERP forwards the data to the accounting system in SAP CRM.

- Communication of an outbound delivery split to SAP ERP

  EWM can split an outbound delivery order, for example, if the requested quantity does not fit onto the means of transport. New outbound deliveries are generated by this split, which EWM reports to SAP ERP. EWM determines the document number and item number in SAP ERP. You must have defined the number range intervals in Customizing. These number ranges must not overlap with the number ranges in SAP ERP.

  For more information about Customizing for number range intervals, see Customizing for EWM under **Interfaces > ERP Integration > Delivery Processing > Define Number Ranges for ERP Documents**.

- Replication of outbound delivery orders that were created manually in EWM, to SAP ERP

  If you create and save a direct outbound delivery order manually in EWM without errors, EWM sends this to the ERP system immediately after saving, using an interface for communicating between EWM and SAP ERP.

For more information about the interfaces that enable these functions, see Interfaces for Communication from EWM to SAP ERP.

Activities

In order to create messages for processing outbound deliveries, you must perform the relevant processes using the appropriate user interface.

Related Information

Outbound Delivery Request
Outbound Delivery Order
Outbound Delivery
Generation of Warehouse Request of Outbound Delivery Order Type

Order Reduction

The order reduction process enables you to reduce the quantity of an order in SAP ERP even after warehouse processing has started in EWM.

Use

If warehouse processing has not yet started, the EWM system changes the quantity of the outbound delivery order. If you want to be able to reduce the quantity after warehouse processing has started, you need to activate order reduction as described below.

EWM can then store the information it receives from SAP ERP about the quantity reduction and start reversing the outbound process, for example, it cancels or updates the warehouse task of picking. If stock has already been picked, the stock is updated. Stock marked for order reduction can be identified by the checkbox on stock item or handling unit header level. At the work center, the worker is informed when packing.

It may happen that EWM agrees to reduce the quantity though this is not possible in reality, for example, if the material can no longer physically be excluded from the outbound process. Therefore, if warehouse processing has started, you do not receive a final update of the quantity in SAP ERP until goods issue.

Configuring Order Reduction

You can control whether order reduction after the start of warehouse processing is active on warehouse number level by making the following settings for the item type in Customizing:

- Select the checkbox Order Reduction Allowed in the warehouse process type you use for the item type.

  For more information, see Customizing for Extended Warehouse Management under Cross-Process Settings > Warehouse Task > Define Warehouse Process Type.

- In the Customizing activity Define Quantity Offsetting Profiles, for the quantity offsetting profile used in the item type, activate quantity role Reduce By (`CANCEL`) and Reduced Quantity (`OQ_RED`).

  For more information, see Customizing for Extended Warehouse Management under Cross-Process Settings > Delivery - Warehouse Request > Quantity Offsetting > Define Quantity Offsetting Profiles.

- In the Customizing activity Define Status Profiles, for the status profile used in the item type, activate status type Order Reduction (`DCA`).

  For more information, see Customizing for Extended Warehouse Management under Cross-Process Settings > Delivery - Warehouse Request > Status Management > Define Status Profiles.

- Optional: If you want to record the time when the system received the request for order reduction, in the Customizing activity Define Date Profiles, you can activate date/time type Order Reduction Received (`TCANCORD`).

  For more information, see Customizing for Extended Warehouse Management under Cross-Process Settings > Delivery - Warehouse Request > Dates/Times > Define Date Profiles.

- Optional: If you want your outbound delivery order to be completed after a reduction to zero, in the Customizing activity Define Document Types for Outbound Delivery Process, select the indicator Create Outbound Delivery at ODO Cancellation for your document type.

  For more information, see Customizing for Extended Warehouse Management under Goods Issue Process > Outbound Delivery > Define Document Types for Outbound Delivery Process.

The following Business Add-Ins (BAdIs) are available in Customizing for Extended Warehouse Management:

- Business Add-Ins (BAdIs) for Extended Warehouse Management > Interfaces > ERP Integration > Delivery Processing > BAdI: Prediction Service for Order Reduction
- Business Add-Ins (BAdIs) for Extended Warehouse Management > Goods Issue Process > Outbound Delivery > BAdI: Sorting of Warehouse Tasks for Partial Cancellations

Features

If you want to integrate order reduction in your own warehouse processing UIs, you can use the API `/SCWM/IF_API_CANCEL_PICKING`.

Related Information

Packing at the Work Center
Internal Exceptions

Delivery Processing Administration

Use

The following provides you with technical information for:

- Interfaces for Communication from EWM to SAP ERP
- Functional Enhancement Through Business Add-Ins (BAdIs)
- Data Enhancement for Warehouse Requests

Interfaces for Communication from EWM to SAP ERP

Use

There are a range of interfaces available for communication from Extended Warehouse Management (EWM) to SAP ERP.

Features

Interfaces for Inbound Deliveries

Interface Name	Description	Additional Information
/SPE/INB_DELIVERY_SAVEREPLICA	Replication of inbound deliveries EWM to SAP ERP (EWM > SAP ERP)	For more information, see the function module description for /SPE/INB_DELIVERY_SAVEREPLICA.
/SCWM/INB_DLV_SAVEREPLICA	Replication of manually created inbound deliveries in SAP ERP to EWM (SAP ERP > EWM)	For more information, see the function module description for /SCWM/INB_DLV_SAVEREPLICA.
/SPE/INB_DELIVERY_CONFIRM_DECENTRAL	Confirmation of the goods receipt in EWM (EWM > SAP ERP) Correction or reversal of the goods receipt in EWM (EWM > SAP ERP)	For more information, see the function module description for /SPE/INB_DELIVERY_CONFIRM_DECENTRAL.
/SPE/INB_DELIVERY_REPLACE	Replication of inbound delivery changes before goods receipt (EWM > SAP ERP) Confirmation of deletion of an inbound delivery (EWM > SAP ERP)	For more information, see the function module description for /SPE/INB_DELIVERY_REPLACE.
/SCWM/INB_DELIVERY_REPLACE	Replication of changed inbound deliveries (SAP ERP > EWM)	For more information, see the function module description for /SCWM/INB_DELIVERY_REPLACE.
/SPE/INB_DELIVERY_SPLIT	Reporting of inbound deliveries that were generated by the split in EWM (EWM > SAP ERP)	For more information, see the function module description for /SPE/INB_DELIVERY_SPLIT.
/SPE/INB_DELIVERY_RESPONSE	Reject or confirm replicated inbound delivery changes (EWM > SAP ERP)	For more information, see the function module description for /SPE/INB_DELIVERY_RESPONSE.

Interfaces for Outbound Deliveries

Interface Name	Description	Additional Information
/SCWM/OUTB_DLV_SAVEREPLICA	Replication of unchecked and checked deliveries from SAP ERP to EWM (SAP ERP > EWM)	For more information, see the function module description for /SCWM/OUTB_DLV_SAVEREPLICA.
BAPI_OUTB_DELIVERY_CONFIRM_DECENTRAL	Confirmation of unchecked and checked deliveries for follow-on functions (for example, goods issue, invoicing before goods issue) (EWM > SAP ERP)	For more information, see the function module description for BAPI_OUTB_DELIVERY_CONFIRM_DECENTRAL.
BAPI_OUTB_DELIVERY_CHANGE	Confirmation (at delivery item level) of quantity change request for checked deliveries (EWM > SAP ERP) Change checked deliveries (EWM > SAP ERP) Confirm pick denials (EWM > SAP ERP) Reversal of goods issue (EWM > SAP ERP)	For more information, see the function module description for BAPI_OUTB_DELIVERY_CHANGE.
/SCWM/OUTB_DLV_CHANGE	Delete unchecked deliveries (SAP ERP > EWM)	For more information, see the function module description for /SCWM/OUTB_DLV_CHANGE.
BAPI_OUTB_DELIVERY_SPLIT_DECENTRAL	Confirm outbound delivery split to SAP ERP (EWM > SAP ERP)	For more information, see the function module description for BAPI_OUTB_DELIVERY_SPLIT_DECENTRAL.
/SCWM/OBDL_CHNG_QUAN_MUL	Quantity change requests for checked deliveries from SAP ERP to EWM (SAP ERP > EWM)	For more information, see the function module /SCWM/OBDL_CHNG_QUAN_MUL.
BAPI_OUTB_DELIVERY_REJECT	Reject a quantity change request (EWM > SAP ERP)	For more information, see the function module description for BAPI_OUTB_DELIVERY_REJECT.
/SPE/OUTB_DELIVERY_SAVEREPLICA	Replication of outbound delivery orders created manually in EWM, to SAP ERP (EWM > SAP ERP)	For more information, see the function module description for /SPE/OUTB_DELIVERY_SAVEREPLICA.

Activities

If you want to call up the function module documentation or edit an interface, on the SAP Easy Access screen, choose Tools > ABAP Workbench > Development > Function Builder .

Functional Enhancement Through Business Add-Ins (BAdIs)

Use

You use Business Add-Ins(BAdIs) in delivery processing to execute user-defined functional enhancements in Extended Warehouse Management (EWM).

You can carry out functional enhancements for the following areas:

- Goods receipt process
- Goods issue process
- Warehouse-internal processes
- Cross-process settings
- Interfaces

Features

For a description of the individual BAdIs, see the relevant BAdI documentation in the Implementation Guide (IMG) for EWM .

Activities

- To call the BAdI documentation, see the Implementation Guide (IMG) for EWM under Business Add-Ins (BAdIs) for Extended Warehouse Management .

  Select the desired BAdI and choose Documentation for IMG Activity .

  EWM shows you the BAdI description of the IMG.

- To carry out functional enhancements for delivery processing, see the Implementation Guide (IMG) for EWM under Business Add-Ins (BAdIs) for Extended Warehouse Management and then choose the area for which you want to carry out the functional enhancement.

  For more information about Business Add-Ins (BAdIs), see the SAP Library under SAP NetWeaver > Application Platform (SAP Web Application Server) > ABAP Technology > ABAP Workbench (BC-DWB) > ABAP Workbench: Tools > Further Concepts > Business Add-Ins.

Data Enhancement for Warehouse Requests

Use

You can use the data enhancement in Extended Warehouse Management (EWM) to add your own fields to the warehouse request, and therefore to adapt it to your requirements.

Integration

EWM uses the append technique for the data enhancement. You can use this method to add further structures to tables and structures.

For more information, see the following:

- SAP Library for SAP NetWeaver under SAP NetWeaver Library: Function-Oriented View > Application Server > Application Server ABAP > Application Development on AS ABAP > ABAP Programming Tools > ABAP Dictionary > Tables > Append Structures
- SAP Library for SAP NetWeaver under SAP NetWeaver Library: Function-Oriented View > Application Server > Application Server ABAP > Application Development on AS ABAP > ABAP Workbench - Classic > ABAP Dictionary > Database Tables > Making Changes to Tables > Adding an Append Structure >

Features

For the document categories for the warehouse request, EWM offers extendable structures for the document header and document item. You can extend these structures using append structures.

Document categories for the warehouse request and corresponding structures

Document Category	Structure for Document Header	Structure for Document Item
Inbound Delivery Notification	`/SCDL/INCL_EEW_DR_HEAD_STR`	`/SCDL/INCL_EEW_DR_ITEM_STR`
Inbound Delivery	`/SCDL/INCL_EEW_DLV_HEAD_STR`	`/SCDL/INCL_EEW_DLV_ITEM_STR`
Outbound Delivery Request	`/SCDL/INCL_EEW_DR_HEAD_STR`	`/SCDL/INCL_EEW_DR_ITEM_STR`
Outbound Delivery Order	`/SCDL/INCL_EEW_DLV_HEAD_STR`	`/SCDL/INCL_EEW_DLV_ITEM_STR`
Outbound Delivery	`/SCDL/INCL_EEW_DLV_HEAD_STR`	`/SCDL/INCL_EEW_DLV_ITEM_STR`
Stock Transfer	`/SCDL/INCL_EEW_DLV_HEAD_STR`	`/SCDL/INCL_EEW_DLV_ITEM_STR`
Posting Change Request	`/SCDL/INCL_EEW_DR_HEAD_STR`	`/SCDL/INCL_EEW_DR_ITEM_STR`
Posting Change	`/SCDL/INCL_EEW_DLV_HEAD_STR`	`/SCDL/INCL_EEW_DLV_ITEM_STR`

⚠ Caution

Several document categories have the same structure. This means that an enhancement is simultaneously present in more than one document category.

EWM offers a range of Business Add-Ins (BAdIs) that you can use to fill the new fields with data.

Activities

1. On the **SAP Easy Access** screen, choose **Tools** > **ABAP Workbench** > **Development** > **Dictionary** .

   The **ABAP Dictionary: Initial Screen** screen appears.

2. Choose **Data Type**.

3. In the **Data Type** field, enter the structure that you want to enhance using an append structure.

4. Choose **Change**.

5. Create an append structure and enhance the append structure by adding your own fields.

   For more information, see Extending Tables and Structures.

6. Enter data into the new fields. To do this, in Customizing for **Extended Warehouse Management** under **Business Add-Ins (BAdIs) for Extended Warehouse Management** enter the relevant BAdIs.

   ### Example

   For the data enhancement for messages to the decentralized SAP ERP, choose the following BAdI:

   Extended Warehouse Management > **Business Add-Ins (BAdIs) for Extended Warehouse Management** > **Interfaces** > **ERP Integration** > **Inbound Messages from ERP System to EWM** > **BAdI: Creation of Inbound Deliveries in EWM.**

Menu Paths and Customizing Paths for Delivery Processing

The following section describes where you can find delivery processing functions.

Menu Paths

- For the inbound delivery, start from the SAP Easy Access screen and choose Extended Warehouse Management > Delivery Processing > Inbound Delivery.
- For the outbound delivery, start from the SAP Easy Access screen and choose Extended Warehouse Management > Delivery Processing > Outbound Delivery.
- For the posting change, start from the SAP Easy Access screen and choose Extended Warehouse Management > Delivery Processing > Posting Change.
- For the internal stock transfer, on the SAP Easy Access screen, choose Extended Warehouse Management > Delivery Processing > Maintain Internal Stock Transfer.

Customizing

Processing document and item types:

- For the inbound delivery, in the Implementation Guide (IMG) for Extended Warehouse Management choose Goods Receipt Process > Inbound Delivery.
- For the outbound delivery, in the IMG for EWM choose Goods Issue Process > Outbound Delivery.
- For the posting change or internal stock transfer, in the IMG for EWM choose Internal Warehouse Processes > Delivery Processing > Posting Changes or Stock Transfers.

General Settings:

To define the cross-process settings, such as services, in the IMG for EWM choose Cross-Process Settings > Delivery Processing.

Business Add-Ins (BAdIs):

- For the outbound delivery, in the IMG for EWM choose Business Add-Ins (BAdIs) for Extended Warehouse Management > Goods Issue Process > Outbound Delivery.
- For the cross-process settings, such as services, in the IMG for EWM choose Business Add-Ins (BAdIs) for Extended Warehouse Management > Cross-Process Settings > Delivery Processing.
- For the warehouse-internal processes, such as posting changes or stock transfers, in the IMG for EWM choose Business Add-Ins (BAdIs) for Extended Warehouse Management > Internal Warehouse Processes > Posting Change DeliveryorStock Transfer Delivery.

Delivery Processing with Transportation Integration

Use

Delivery processing offers features to integrate SAP Extended Warehouse Management (SAP EWM) with the Transportation (LE-TRA) component of SAP ERP. With this integration, you can control and monitor the entire transportation process from the planning stage to the goods receipt at the customer location or your own plant.

Features

Delivery processing offers you the following features:

- Outbound deliveries and outbound delivery orders that are relevant for transportation planning in SAP ERP are not relevant for transportation cross-docking in SAP EWM (see Transportation Cross-Docking (TCD)).
- In the case of a late split of an outbound delivery order by means of a process code or item deletion, the system does not automatically take over the transportation unit (TU) assignment from the old outbound delivery order to the new outbound delivery order.
- The following screens display the transportation planning type, which indicates the system that is responsible for the transportation planning:
    - Inbound delivery header
    - Inbound delivery notification header
    - Outbound delivery header
    - Outbound delivery order header
    - Outbound delivery request header
- Upon an update of TU information to delivery, the system starts a new staging area and door determination based on the new transportation information such as carrier, means of transport, handling unit type of the TU packaging material.
- When a door is assigned to an item, the system redetermines the staging area by using the assigned door.
- The **Blocked (Transp. Plan)** status on delivery header and item level blocks SAP EWM execution as long as transportation planning results are missing. Depending on your Customizing, you can manually release a delivery for execution in SAP EWM before the results of the transportation planning are available. For more information, see Customizing for **Extended Warehouse Management** under **Goods Receipt Process** > **Inbound Delivery** > **Integration with Transportation** > **Allow EWM Execution Without Transportation Planning (Inbound)** or **Goods Issue Process** > **Outbound Delivery** > **Integration with Transportation** > **Allow EWM Execution Without Transportation Planning (Outbound)**.

  If you want to process your delivery without waiting for the TU assignment, you do not need to use the status **Blocked (Transp. Plan)**.
- The **Transport Plan** status indicates whether a transportation plan exists for an inbound delivery, similarly to the status of an outbound delivery order.

More Information

Integration of ERP Shipments in Shipping and Receiving